TAMIL NADU VETERINARY AND ANIMAL SCIENCES UNIVERSITY
Veterinary College and Research Institute, Namakkal

UNIT -I - Veterinary public health and food safety(3+1)

Meat science part

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Food Safety: Why?????


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1	Pathological conditions associated with the transport of food animals.
2	Hygiene in abattoirs and meat plants
3	Detection of conditions or diseases and judgements during **ante mortem** inspection.
4	Detection of conditions or diseases and judgements during **post mortem** inspection.
5	Examination of lymph nodes.
6	Meat as a source of disease transmission
7	Sources of contamination of meat and
8	Methods of carcass decontamination.
9	Speciation of meat.
10	Animal welfare and public health issues
11	Classification of low risk and high risk material generated in an abattoir
12	Hygienic disposal of low and high risk material generated in abattoir
13	Inspection of poultry for human consumption
14	Occupational health hazards in abattoir and meat plants
15	Foodborne infections and intoxications associated with foods of animal origin.
16	Toxic residues (pesticides, antibiotics, metals and hormones) in foods and associated health hazards
17	Types of biohazards.
18	Hazard analysis and critical control points (HACCP) system. Importance of ISO 9000 and 14000 series in meat industry
19	Risk analysis, assessment and management, International food safety standards: World Organisation for Animal Health (OIE), World Trade Organization (WTO) agreements and Codex Alimentarius Commission
20	Sanitary and phytosanitary measures in relation to foods of animal origin. Food Safety and Standards Act and Regulations. Role of Food Safety and Standards Authority of India (FSSAI), Bureau of Indian Standards (BIS) and other national agencies

PATHOLOGICAL CONDITIONS ASSOCIATED WITH TRANSPORT

The need to transport food animals occurs essentially in commercial agriculture and to a lesser extent in the rural or subsistence sector. These animals need to be moved for a number of reasons including marketing, slaughter, re-stocking, from drought areas to better grazing and change of ownership. Typically, methods used to move animals are on hoof, by road motor vehicle, by rail, on ship and by air.

Generally the majority of livestock in developing countries are moved by trekking on the hoof, by road and rail. Historically, livestock has been moved on foot, but with increasing urbanisation of the population and commercialisation of animal production, livestock transport by road and rail vehicles has surpassed this.

Transport of livestock is undoubtedly the most stressful and injurious stage in the chain of operations between farm and slaughterhouse and contributes significantly to poor animal welfare and loss of production.

Effects of transport

Poor transportation can have serious deleterious effects on the welfare of livestock and can lead to significant loss of quality and production.

Effects of transport and movement include:

Stress	Leading to DFD beef and PSE pork Conditions are inevitable sequel to transportation and do have a bearing on meat quality.
Bruising and *torn skin*	are noticed due to transportation in most of the species. The instances are particularly high in sheep and pigs. Muscular bleeding may occur especially in pigs. Perhaps the most insidious and significant production waste in the meat industry
Trampling	This occurs when animals go down due to slippery floors or overcrowding
Suffocation	This usually follows on trampling;
Heart failure	Occurs mostly in pigs when overfed prior to loading and transportation;
Heat stroke	Pigs are susceptible to high environment temperatures and humidity;
Sun burn	Exposure to sun affects pigs seriously
Bloat	Restraining ruminants or tying their feet without turning them will cause this;
Poisoning	Animals can die from plant poisoning during trekking on hoof;
Predation	Unguarded animals moving on the hoof may be attacked;
Dehydration	Animals subject to long distance travel without proper watering will suffer weight loss and may die
Exhaustion	May occur for many reasons including heavily pregnant animals or weaklings
Injuries	Broken legs, horns
Fighting	This occurs mostly when a vehicle loaded with pig stops, or amongst horned and polled cattle.
Transit-fever	***Transit-fever or shipping fever*** is a catarrhal disease which affects mainly store cattle in poor condition that get fatigued due to long journey by rail or sea without sufficiency of food. It develops due to Pasteurella and requires

	proper treatment, otherwise virus may act as secondary invader and aggravate the condition. It is common in colder months, on post-mortem lobar pneumonia is noticed, the interlobular septa being some time thickened due to serious infiltration. Acute enteritis is usually present, though spleen appears normal. The affection does not respond well to treatment early slaughter is advisable before the on-set septic lung changes.
Transit-tetany	***Transit-tetany or rail-road-sickness*** occurs under similar circumstances but almost invariably in cows, particularly those in advanced pregnancy and in warmer month of the year. It is a disease, which bears resemblance, to milk fever and the affected animals usually respond to calcium therapy. There is no specific post-mortem lesion
shrinkage	*Loss of weight or shrinkage occur due to dehydration and depletion of muscle glycogen during the period of journey. In general, it ranges from 3 to 10 % depending on the conditions and duration of transport*
Stress and fatigue	*Conditions are inevitable sequel to transportation and do have a bearing on meat quality.*
Death	*may occur during long transportation. Sheep and pigs are particularly susceptible if animals of unequal age and size are loaded in road trucks without proper partitions due to suffocation. Sheep and goats could also die in long distance transportation by ship due to non-inflammatory diarrhoea.*

Pre-loading precautions

There are a number of simple procedures that can be implemented prior to the loading of livestock, which will considerably reduce the risk of injury and stress.

1. Pre-mixing of cattle or pigs leads to greater familiarity and these animals travel better than animals that are strangers. Cattle should be mixed in a pen 24 hours before loading. Victimised or wild animals can be weeded out during this period. Fighting amongst pigs that are strangers is common, resulting in skin damage, wounds and stress. Mix pigs from different pens together before loading, smearing pigs with litter or excreta from the same pen so that they smell similar.

2. Most animals can be fed and watered before transporting. This has a settling effect. However pigs should **not** be fed before transport as the feed ferments and the gas causes pressure on the heart in the thoracic cavity, leading to heart failure and death.

3. Do not mix horned and hornless animals in the vehicles as this causes bruising and injury. Different species should also not be mixed - sheep, goats and calves under 6 months can be mixed and individual animals can be transported in a loose sack tied at the animal's neck. Feet should not be tied, and animals should be turned every 30 minutes or so. Pigs should not travel with other species unless separated by a partition. Bulls should not be carried together with other stock unless separated by a strong partition.

4. Animals that are diseased, injured, emaciated or heavily pregnant should not be transported, and unfit, heavy, penfed animals should not travel far as they cannot stand up to the rigours of transport.

5. Vehicles should be fitted with a portable ramp to facilitate emergency offloading in case of prolonged breakdowns.

Types of vehicles

Any vehicle used for the transport of slaughter livestock should have adequate ventilation, have a non-slip floor with proper drainage and provide protection from the sun and rain, particularly for pigs. The surfaces of the sides should be smooth and there should be no protrusions or sharp edges. No vehicle should be totally enclosed.

Ventilation-Transport vehicles should never be totally enclosed, as lack of ventilation will cause undue stress and even suffocation, particularly if the weather is hot. Poor ventilation may cause accumulation of exhaust fumes in road vehicles with subsequent poisoning. Pigs are particularly susceptible to excessive heat, poor air circulation, high humidity and respiratory stress. Well-ventilated vehicles are necessary . The free flow of air at floor level is important to facilitate removal of ammonia from the urine.

Floors-Non-slip floors in all vehicles are necessary to reduce the risk of animals falling. A grid of cross slating made from wood or metal is suitable. The grid can be removable, so the vehicle can be used for other purposes. Other forms of non-slip surfaces such as grass or sawdust are not suitable. Additional balance for animals is provided by partitioning the interior of the vehicle with either wood or metal poles or solid boards. Broken floors will cause leg and other injuries. Vehicle floors should be level with off-loading platforms otherwise animals will injure themselves climbing off or be manhandled in order to remove them (Fig. 36).

Floor space-Livestock require sufficient floor space so that they can stand comfortably without being overcrowded. Overloading results in injuries or even death of livestock .

Approximate floor space for transporting different classes of animals

Space requirement for Cattle while being transported in commonly sized road vehicles

Vehicle Size Length x Width Square Meter	Floor Area of the vehicle in Square Meter (Sq.mtr.)	Number of Cattle			
		Cattle weighing upto 200 Kg (1 Sq.mtr. space per cattle)	Cattle weighing 200-300 Kg (1.20 Sq.mtr space per cattle)	Cattle weighing 300-400Kg (1.40 Sq.mtr. space per cattle)	Cattle weighing above 400Kg (2.0 Sq.mtr space per cattle)
6.9 x 2.4	16.56	16	14	12	8
5.6 X 2.3	12.88	12	10	8	6
4.16X1.9	7.904	8	6	6	4
2.9X1.89	5.481	5	4	4	2

I) Horses: Space requirement while being transported by road/rail

Horses	Space in Square Meter
Stallion Horses	2.25
Mares (including pregnant)	2
Ponies	1.5
6 Months to 12 Months	1.4
12 Months to 18 Months	1.6
Over 18 Months and upto 2 years	2
Mares with Foal at Foot (upto 6 months)	2.25

In the said rules, in rule 73, after the words "for a sheep", the following shall be substituted namely:-

"and Goat while being transported by rail/road shall be as under:-

Sheep and Goat: Space requirement while being transported by Rail/Road

Approximate weight of animal in Kilogram	Space required in Square Meter	
	Wooled	Shorn
Not more than 20	0.17	0.16
More than 20 but not more than 25	0.19	0.18
More than 25 but not more than 30	0.23	0.22
More than 30 but not more than 40	0.27	0.25
More than 40	0.32	0.29"

Pig

"(a) Maximum Number of Pigs permitted for commonly sized road vehicle

S.No.	Type of Animal	Maximum Number of Pigs Permitted for Road Vehicles			
		Vehicle having size 5.6mx2.35m	Vehicle having size 5.15mx2.18m	Vehicle having size 3.03mx2.18m	Vehicle having size 2.9mx2.0m
1.	Weaner	43	37	22	19
2.	Young	31	26	15	13
3.	Adult	21	18	10	9

Weaner - Piglet which has just been separated from the mother for the purpose of independent rearing and commonly in the weight range of 12kg - 15 kg.
Young - Male of female pig between 0.3 to 0.6 months of age and commonly in the weight range of 15 Kg - 50 Kg.
Adult - A male or female pig above 06 months of age and having weight more than50kg.

Poultry

interfere with ventilation.

(c) all the containers shall be clearly labelled showing the name, address and telephone number of the consignor and the consignee.

(d) the minimum floor space per bird and the dimensions of the containers for transporting poultry shall be as specified in the Table below, namely :

TABLE

S.No.	Kind of Poultry	Minimum Floor space cm²	Dimension Length cm	Width cm	Height cm	Number in a container
i.	Month old chickens	75	60	30	18	24
ii.	Three month old chickens	230	55	50	35	12
iii.	Adult stock (excluding geese and turkeys)	480	115	50	45	12
iv.	Geese and turkeys	900	120	75	75	10
						youngs
		1300	75	35	75	2
						Growing
		1900	55	35	75	1
						grown up
v.	Chicks	-	60	45	12	80
vi.	Poult	-	60	45	12	60

84.Special requirement of containers for chicks and poults - In transport of poultry by road, rail or air-

(a) wire mesh or a net of any material shall not be used as a bottom for the containers.

(b) the container shall be properly secured to avoid pilferage

(c) the following instruction shall be printed on a label and fixed to the lid or printed directly on sides,

Allowances should be made for breed and body size. If the floor area is too large for the number of animals, partitions should be used to avoid animals being thrown about.

Sides-The sides of vehicles should be high enough to prevent animals, particularly pigs, from jumping out and injuring themselves. Insides could also be padded at hip level with, for example, old tyres to reduce bruising of cattle and ostriches. Also there should be no gaps through which a leg might protrude and be broken. Narrow entry doors can lead to considerable bruising of hips. Rail trucks should be fitted with spring coupling to cushion jerky movement.

Roof-A roof is not necessary on a transport vehicle for bovines and small ruminants provided the animals are not exposed for hours in the hot sun. Vehicles for pigs should have roofs unless the pigs are to be transported in the early morning or late evening. Poultry should be protected from sun and rain. Transporting in cages or crates will protect them from physical injury. They should be large enough to allow all the birds to sit down and move their heads freely. Ventilation should be adequate.

At the small-scale level in more primitive conditions animals are often transported under very unsuitable conditions, which may cause a great deal of pain or even death through suffocating, heat stress, dehydration etc.

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Transport operations

A number of factors must be taken into account during the journey in order that the animals do not suffer, become injured or die.

1. Trekking-Only cattle, sheep and goats can be successfully moved on hoof, and here certain risks are involved. The journey should be planned, paying attention to the distance to be travelled, opportunities for grazing, watering and overnight rest. Animals should be walked during the cooler times of the day and, if moving some distance to a railhead, they should arrive with sufficient time to be rested and watered before loading. The maximum distances that these animals should be trekked depend on various factors such as weather, body condition, age etc., but the distance given in Table 4 should not be exceeded when trekked.

TABLE 4. Maximum distances for trekking

Species	**One day journey**	**More than one day**	
		First day	**Subsequent days**
Cattle	30 km	24 km	22 km
Sheep/goats	24 km	24 km	16 km

2. Time of the day-High environment temperatures will increase the risk of heat stress and mortality during transportation. It is important to transport animals in vehicles during the cooler mornings and evenings or even at night. This is particularly important for pigs. A combination of high humidity and high environment temperatures is especially deadly to pigs. Heat can rapidly build up to lethal levels in a stationary vehicle. Wetting pigs with water will help keep them cool.

3. Duration of journey[2]-Where possible, journeys should be short and direct, without any stoppages. If the vehicle stops, pigs will tend to fight. Cattle and sheep/goats should not travel for more than 36 hours and should be offloaded after 24h for feed and water, if the journey is to take longer than that. Pigs should have access to frequent drinks of water during long journeys, particularly in hot and humid conditions.

There are recent moves in developed regions, seeking to limit the duration of livestock transports to 8 hours or less.

4. Driving-Vehicles should be driven smoothly, without jerks or sudden stops. Corners should be taken slowly and gently. The second person should be in attendance to spot downer animals so that the vehicle can be stopped and the animal lifted. Train drivers should avoid "fly shunting" of trucks with livestock.

5. Wind chill-Wind blowing on wet animals being transported in cold weather causes a wind chill factor, where the body temperature is considerably reduced, resulting in severe stress or deaths.

Rest after transport

Rest after transport is desirable as an animal slaughtered without an adequate period of rest shows a reduction in keeping quality of flesh due to (i) incomplete development of acidity of muscle and (ii) early invasion of the system, by putrefactive bacteria from the intestinal tract.

The meat of animals slaughtered while exhausted also appears dark and firm and gives the impression that bleeding of the animal had been incomplete, but this dark colouration must be attributable in some degree to changes of chemical nature which take place in the muscles of fatigued animal, and to the fact that in such animals there is decreased oxygenation of blood, haemoglobin and muscle myoglobin with a consequent darkening of muscle pigments.

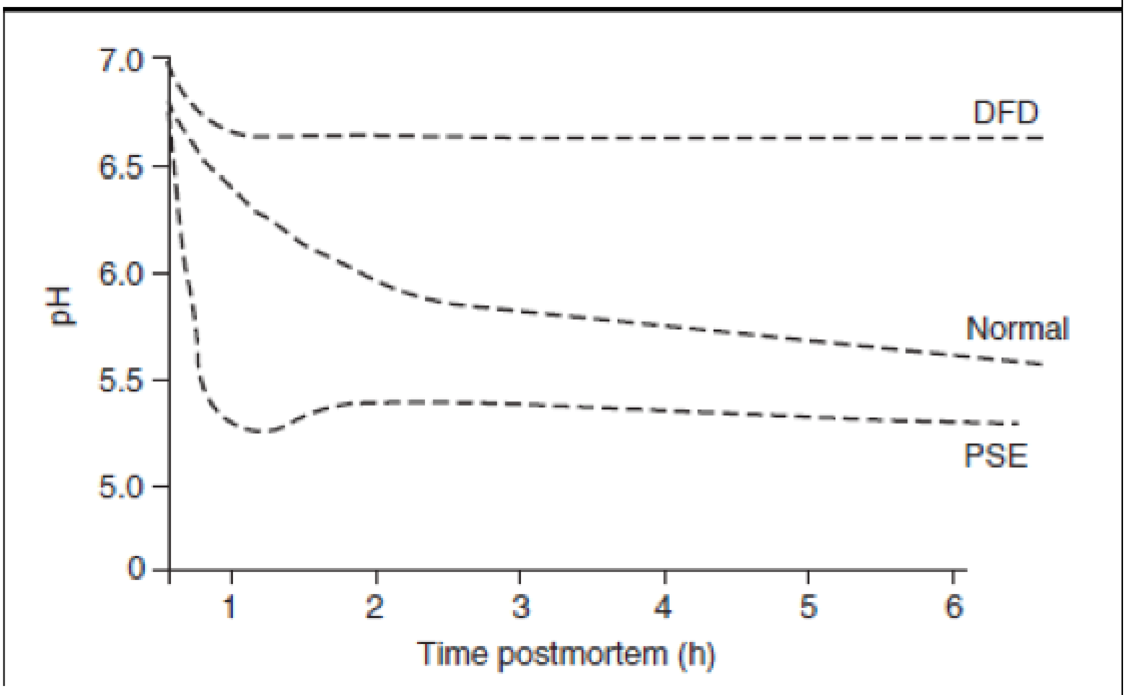
Stress and fatigue lower the quality of meat in several ruminant species due to depletion of glycogen in muscle. Due to low acid production, the ultimate ph of the muscle remains high causing a condition called ***dark cutting meat or dark, firm and dry (DFD) meat.*** Thus, the keeping quality of meat is reduced and it looks dark due to high water content. Such meat is unusually tender on cooking.

In pigs, acute stress or excitement before slaughter causes another abnormal condition wherein low ultimate pH is achieved within 45 minutes due to rapid glycolysis even when the temperature of muscle is quite high. Such ***pale, soft and exudative (PSE)*** pork has higher drip and cooking losses.

Changes in eating quality of meat due to stress is:

Stress during transport is reflected in weight loss and poor meat quality and conditions like Dark, Firm and Dry (DFD) in beef and Pale, Soft and Exudative (PSE) in pork. Weight loss can be regained to some extent by resting, watering and proper feeding after transport.

PSE	DFD
P = Pale due to Myoglobin denaturation S = soft due to breaking of tropomyosin E = Exudative due to low pH, Water Holding Capacity	D- Dark due to low refractive index (partial conversion of Mb in to Oxy Mb) F- Firm due to high pH D= Dry due to more water binding, muscle fibres tightly packed,
Generally in pigs	Generally seen in beef cattle
Due to acute stress prior slaughter	Prolonged stress
Rigor pattern : Short delay short onset\	Long delay long onset
Fast drop in pH to 5.3	Very slow drop in Ph from 6.8 to 6.2
Poor eating quality	Fit for consumption but low in colour and flavour. Low keeping quality due to high pH, which favours bacterial growth.



Porcine Stress Syndrome (PSS) this is a fatal hyperthermia relatively common in Pietrain and Landrace. PSS is also referred as fatal syncope, herztod, and shock heart failure syndrome. The condition is characterized by sudden death; preceded by elevated temperature, skin blanching, erythema and dyspnoea in fat pigs, which have been subjected to stress. Post-mortem examination shows diffuse and severe degeneration of skeletal muscles with excessive pericardial fluids. The halothane test is generally done on pigs between 3-11 weeks to determine their susceptibility to PSS. There may be mortality over 8% in positive reactors.

Halothane sensitive pigs are more susceptible to PSE. This is because of Halothane gene present in pigs as the double recessive genotype, which confers susceptibility to the common anesthetic gas halothane. Susceptible pigs, referred to as halothane positive, exhibit a characteristic response when they breathe the gas through a facemask. Their limbs become extended and rigid and they develop a raised body temperature or hyperthermia.

Two- toning

Because of different types of muscle fibres such as red and white and their relative susceptibility to PSE and DFD there could be both dark and pale coloured areas in the same muscle. This is referred as Two – toning. Red fibres are more prone to PSE than white fibres. This is mostly seen in Ham region of pig.

How to reduce the incidence of PSE and DFD

- Handle the animals gently
- Provide proper ramps for loading and unloading
- Maintain the social groups of the livestock.
- Avoid mixing of the livestock within 24-48 hours before slaughter
- Lairage pens and race arrangements should allow for easy movement of stock
- Water should be provided at all the times in the lairage and food when necessary
- Give proper rest prior slaughter (cattle-24 hours, pig 12 hours poultry 1-4 hours)
- Feeding of electrolytes / sugar in lairage also helps in replenishing glycogen levels.
- The use of sticks or electric goads should be avoided
- Aggressive animals should be separated in lairage
- Use of overhead thin water spray in pigs has been shown to reduce PSE due to cooling effect.

Hygiene in abattoirs and meat plants

Meat Hygiene may be defined as expert supervision of all **meat** products with the object of providing wholesome **meat** for human consumption and preventing danger to public health.

Personal Hygiene
Premises hygiene (area of slaughtering & dressing)
Equipment hygiene
Environmental hygiene

Personal Hygiene

- High level of personal cleanliness while in the production facility
- Everyone must wash and sanitize their hands before entering to process.
- Use hairnets
- Wear washed uniforms
- Do not enter without safety shoes
- All visitors must wear white cleaned Lab coats.
- No outer pockets should be provided in uniforms
- Do not take any food, tea or coffee inside the process
- People should carry only the metal pen without cap inside the process
- Smoking is not allowed anywhere inside the factory.
- Chewing gum, tobacco or any other mouth freshener is not allowed in process area
- No jewelry and watches while in the process .
- Finger nails should be short, clean and without polish
- No body odors, including the use of strong perfumes
- Personnel, having any disease, boils, sores, infections must not work within direct contact of raw materials or finished product
- Any operator, who comes in direct manual contact with raw materials or products, must wear disposable gloves
- Regular Medical Checkups for all the medical Non – Compliances

Premises hygiene

- No access to rodents, birds or insects
- Smooth walls and ceiling
- Easily cleanable floors and drains
- Sloping surfaces
- Externals clean and free of debris
- Doors to be self closing and without gaps
- Windows to be closed and have screens
- There should not be any visible cracks in walls, floors etc.
- Separate change room and eating areas

Environmental hygiene

- Ceiling, walls, overhead pipes & equipment in general must be clean and free of peeling paint, rust, adhering debris, structural damage, dust or other contaminations

- All plumbing and effluent lines as well as ventilation or exhaust system must be clean and properly maintained
- Manufacturing and warehousing facilities must be structurally separated from each other and access must be controlled
- Lighting levels in all the areas of plant must be of at least 500 lux at work level.

Environmental hygiene

Environmental hygiene and its implementation will depend on the area where the slaughterhouse/meat plant is situated. The precautions to be taken will be different if the site is in a town or in the country.

The main principles of environmental hygiene will consist of:

- proper fencing (public, dogs, etc.)
- pest control (rodents, insects)
- liquid and solid waste disposal

1. Solids (meat or skin trimmings, hair, pieces of bones, hooves, etc.) must be screened. This may be done by providing the drains with vertical sieves.
2. Effluents from slaughterhouses always contain small amounts of fat (melted fat or small pieces of fatty tissues). Grease traps should be installed in the drains. The fat solidifies, rises to the surface and can be removed regularly.

The final effluent disposal will depend on local conditions and legislation. Disposal of the effluents into a lake or permanent river should not be allowed because it will contaminate the stream.

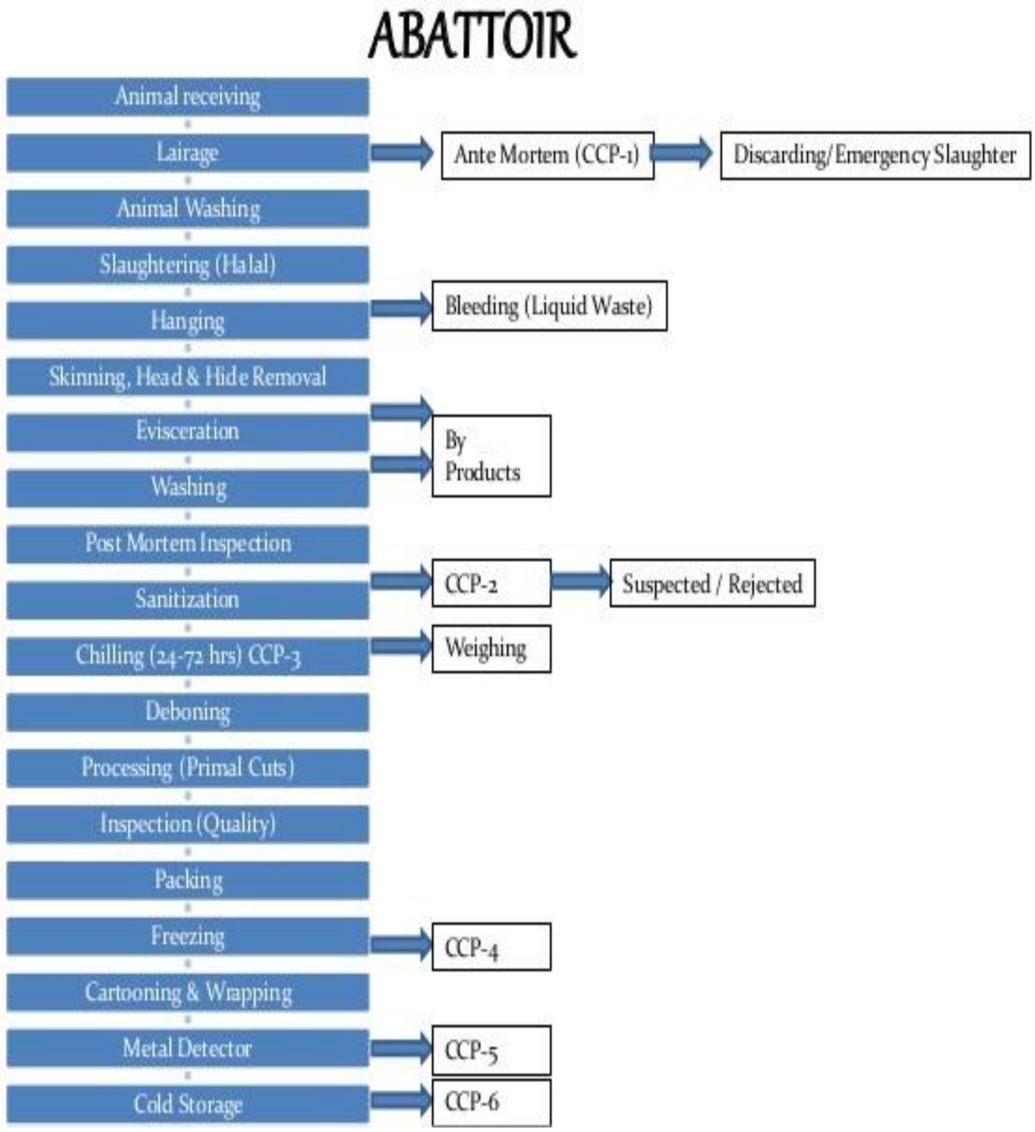
Abattoir waste

Solid waste- Condemned organs, carcasses, hides, carcass trimmings, undigested feed, bones, horns, hair, aborted foetuses, fat.

Liquid waste- Blood, bile, urine, dissolved detergents, chemicals and water.

PROCESSHYGIENE

- Facilities must be available for the isolation and removal of any animals showing signs of illness.
- Pressurized water is preferred to wash animals prior to slaughter
- Slaughtering should be done on tables or on hanging rail to reduce contamination
- The slaughtering knife should be cleaned and sterilized between each carcass at 82°C.
- The head should be removed and after skinning washed separately from the carcass
- Prevent contamination of the carcass with dirty hooks, knives and protective clothes.
- Do not puncture the viscera (alimentary tract), uterus, urinary bladder and gall bladder during separation cuts
- Regularly wash hands/aprons and sterilize knives, especially after any time contamination occurs
- Carcasses should be washed after spliting using cold potable water and if possible decontaminated by use of chlorinated water containing (50-100ppm)



DRAINAGE

- Separate for water and blood
- Separate provision for water & water containing manure
- Blood drainage- trap screen- 4 mm
- Each floor drain- deep trap
- Direction- opposite from edible product flow

LIGHT

- Adequate natural light & provision of artificial light
- Artificial light- should not disturb color/ shadow
- Window- north facing to get solar light

  Intensity
- Inspection point- 550 lux (50 ft candle)

- Work rooms- 220 lux (20 ft)
- Chilling room- 110 lux (10 ft)
- Measured at height of 0.9 M from the floor except inspection point- 1.5 M

EQUIPMENT
• Stunning Pen & boxes
• Hide or Skin Pullers
• Head & Leg Cutters
• Splitting Saw
• Carcass Washing Equipment

RENDERING PLANT Deals with extraction of fat from carcass parts, condemned carcass /diseased one by applying high temperature processing. The whole section of rendering plant may be divided into edible fat section and inedible fat section
1ETP

CLEANING PROGRAM
1. **Standard Gross Clean**
Purpose:
To remove thick deposits and pieces of product, packaging, waste, etc., which have accumulated during production. If such debris is not removed the effectiveness of the cleaning and sanitising
stages will be greatly reduced, or, in some cases, made worthless.
Method:
a) Handpick pieces of product, etc. from working surfaces, from conveyers outside and inside,from crevices in machinery etc.
b) Dismantle machinery if necessary.
c) Remove any large items stuck on shackles or splattered onto walls.
d) Using a hand scraper, scrape off any thick scum or accumulation.
e) Remove any packaging materials.
f) Sweep up any debris on the floor, shovel carefully into bags or bins and remove from the area.
Ensure that drain covers are clear and unobstructed.
g) Check the thoroughness of your gloss clean: Walk around and inspect the area and all surfaces
and crevices. There should be no particles bigger than a little fingernail remaining and no loose,
soft scum or thick layers of grease, fat, etc.
h) The floor should be clear of all pieces of product, plastic bags, cardboard boxes etc. and all rubbish bins removed from the area.
i) If necessary, arrange items of equipment so that the next cleaning stage can be carried out unhindered.

2. Standard Pre- Rinse

Purpose: To remove gross debris, e.g. layers of dirt, grease etc. from areas manual picking and loosened soil towards a drain if the particles are small. If possible, brush and shovel up any large accumulations of dirt.

g) Check the thoroughness of your pre-rinse: Walk around and inspect the area and all surfaces
and crevices. There should be no soft scum or particles larger than an apple pip remaining. All
pools of liquid water etc. should have been washed away and squeezed to drain.

Note: Do not use the pressure rinse or cold hose on any non-water proofed equipment, especially
if electrical.

8. Satisfy yourself that all is ready for the next cleaning stage.

3. Standard Foam Clean

Purpose:

To apply detergent foam to the surfaces so that the foam can stay on long enough to soften and loosen scum and grease so that they may be completely removed on rinsing. The advantages of applying detergent as foam are that it clings, it penetrates cracks and crevices, it is fast and easy to apply and it is easy to check for thoroughness.

Method:

1. After the standard Gross Clean and Pre-Rinse (if applicable) apply the foam using the foam lance.
2. Connect the injector to the high pressure rinse outlet and connect the hose to the foam lance.
3. Place a drum of the chosen foam detergent on the floor beneath the foam lance.
4. If the suction tube of the foam lance is full of detergent, which is either not known or is knownnto be different from the detergent now to be used, clean out the tube by first sucking up water before dropping the tube into the new chemical.
5. Fit the foam lance to the gun and adjust the chemical injector for the concentration required.Check with Supervisor/Manager if in doubt.
6. Point the lance to the ground and squeeze the trigger until foam appears out of the lance.
7. Apply foam as evenly as possible to all the surfaces generally, top to bottom; equipment first then walls, if included, then floors, if included.
8. On the vertical surfaces, especially smooth, metal surfaces take care not to apply too thick a foam layer as this will drag down the foam with its weight. (For stubborn scales on vertical surfaces apply the foam lightly two or three times at 5 minute intervals).
9. On the horizontal and intricate surfaces the foam will naturally cling more easily and may be] applied more heavily in a one-coat application.
10. Check the thoroughness of your foaming: Walk around your area, look at the back, the underside, the hidden parts of the equipment. Re-foam where necessary.
11. Allow the foam to act for between 15 and 20 minutes. Rinsing off earlier than 15 minutes will not allow the foam to completely do its job. If the foam is left on too long in warm conditions it may dry out. Very stubborn soils may be helped by physical agitation of the foam using brushes.

12. When finished with the foam lance, remove the suction tube from the chemical drum, suck up water for a few seconds then squeeze the gun trigger until the foam stops coming out of the lance.
13. Remove chemical injector and foam lance and prepare for rinsing.

brushing will be too time-consuming, too difficult or ineffective; or as follow-up to a manual gross clean.
Method:
a) After the standard gross clean (if applicable in that area) rinse the surfaces, using either warm,
pressurised water or cold, low-pressure water (form cold hose), according to the schedule.
b) Rinse the dirtiest area first.
c) Try not to blast particles all over the place.
d) Generally work from top downwards, from equipment to walls to floors.
e) Remember the inside and underside surfaces.
f) Direct rinsed debris toward the floor drain without blasting it. Use a squeegee to brush the may dry out. Very stubborn soils may be helped by physical agitation of the foam using brushes. 12. When finished with the foam lance, remove the suction tube from the chemical drum, suck up water for a few seconds then squeeze the gun trigger until the foam stops coming out of the lance. 13. Remove chemical injector and foam lance and prepare for rinsing.

GHP for refrigeration
• Move the carcasses into the cooler as soon as possible to speed up surface drying and hinder bacterial growth.
• Keep the carcasses on rails and without touching floors/walls and other carcasses to prevent cross contamination.
• Do not overload the cooler.
• Adjust the cooling regime optimally in terms of air temperature, speed and relative humidity, to achieve rapid refrigeration to a deep muscle temperature of 6–7 °C with no condensation or excessive weight losses.
• Do not open the cooler doors either unnecessarily or frequently to avoid temperature fluctuations.

Liquid and solid waste disposal
The easiest disposal method is to divert effluents into existing pools, rivers or lakes. However, this method cannot be recommended in view of the consequent contamination of water sources for humans, and domestic and wild animals.
For the safe disposal of liquid and solid waste, the following action should be taken:
1. Separation of blood
2. Screening of solids
3. Trapping of grease

3. The blood from slaughtered animals will coagulate into a solid mass, which may block up both open and closed drains. It is therefore recommended that the blood is collected and used for human consumption, stockfeed production or fertilizers, if the religious and cultural traditions allow the use of blood.

Meat plant sanitation encompasses different aspects:

Definitions:

Sterilization: It refers to any process, **chemical or physical** which destroys **all living microbes**. It is an absolute term, sterility means absence of all forms of life

Disinfectant: A disinfectant is an agent, **usually a chemical one** which destroys bacteria (but not necessarily bacterial spores which are much more resistant)

Antiseptic: An antiseptic is a substance, which **prevents or arrests the growth** of organisms either by inhibiting their activity or destroying them.

Germicide: It is a disinfectant usually applied in relation to disease producing bacteria. Thus, a germicide is a substance, which will **destroy vegetative bacterial cells** but not necessarily bacterial spores.

Sanitizer: A sanitizer is a **chemical agent,** which reduces the number of bacterial contaminants on surfaces in contact with food to acceptable bacteriological standards.

Recently most sanitizers on sale are made up of combined detergent sanitizing compounds.

Detergents: A detergent is a **cleansing substance**, which can remove ‘soil’ or dirt from surfaces. It may be natural (water) or artificial. Water alone is not a very good wetting agent because of its high interfacial, which inhibits close contact with other surfaces.

Sequestering agents: Sequestering agents are **used to bind calcium and magnesium and prevent the formation of insoluble calcium and magnesium salts** by the interaction of hard water or dirt with the detergent. The amount of sequestering agents needed depends on the hardness of the water, the composition of the detergent and the composition of the soil. Different chemicals have the ability to be effective as sequestering agents.

Common in use are **EDTA (ethylenediamintetraacetate), NTA (nitrilotriacetic acid) and different phosphates**. Sodium tripolyphosphate, sodium tetra phosphate and sodium hexametaphosphate are phosphates mainly used in detergent formulations. In addition to removing the minerals causing water hardness, they have varying functions in emulsification, protein peptization and dispersion

Mode of action of detergents:

A detergent **weakens the tensions** at the water surfaces, thereby lowering the amount of mechanical energy required for cleaning, breaks down particulate and greasy dirt (type encountered in abattoirs) and holds it in suspension with the aid of agitation and internal electrical forces. Detergent action is measured by the ability of a substance to removeparticles of dirt, which are first of all loosened and then enclosed in a protective films (emulsification), which are easily washed away.

Artificial detergents

1. Simple inorganic chemical compounds (e.g. washing soda (sodium carbonate), phosphate, sulphates and sulphonates of sodium and potassium)
2. Complex organic substances such as soap, or mixtures of synthetic cleansing materials and lather producing chemicals (modern washing powders and liquids)

Four main groups of **detergent or surface-active agents** (main function is to lower the surface tension):

1. **Anionic detergents**: Produce electrically positive ions in solution. E.g. soap and most modern synthetic detergents (hydrocarbon surfactants)
2. **Cationic detergents:** Produce electrically negative ions in solution. Although, possessing good wetting, foaming and emulsifying properties, they are weak detergents. E.g. Quaternary ammonium compounds.
3. **Non-ionic detergents:** Produce electrically neutral ions in solution. Some cause undue foaming but this is less serious with synthetic detergents. E.g. Polyether, alcohol.
4. **Ampholytic or amphoteric detergents**: Acts as anionic or cationic detergents depending on the pH of the solution. They are not used extensively in the food industry.

Properties of an efficient detergent or cleansing agent:

1. Power to remove soil from the surfaces
2. Capacity to effect maximum suspension and deflocculation of soil.
3. Ability to prevent scale formation in hard water systems.
4. Means to ensure optimum rinsing
5. Potential to maintain relatively stable degrees of acidity or alkalinity.
6. Ability to combine satisfactorily with sanitizing agents when used in combination.
7. Non-toxic
8. No tendency to taint meat and offal
9. Economy of use
10. Non-corrosive to surfaces, especially metal.
11. Biodegradable (no pollution problem)

Modern synthetic detergents, which can produce abundant lather, consist chiefly of sulphonated fatty alcohols and adjuvant such as carbonates, silicates and phosphates.

Choice of detergent

The type of metal and building materials (including paint) used in the area to be cleaned severely limits the choice of detergent. **Aluminum and galvanized iron,** which are frequently used, corrode rapidly in strongly alkaline or acid detergents. Although detergents can be formulated so that they do not corrode these metals, their effectiveness as cleaning agents is usually reduced. Where it is suitable to clean by hand only mild detergents should be used. Automatic cleaning techniques enable the use of stronger detergents.

Depending on the actual conditions (i.e. detergent supply, economics, etc.) it is recommended to use a single cleaning chemical. In most instances the choice of a suitable detergent must be a compromise between the efficiency of the detergent, the need to protect metals, building materials and personnel, the cleaning methods, the frequency of cleaning and economics. The cost of the detergent is not a guide to its efficiency. Only actual tests with a particular detergent during a specific cleaning operation will give some indication of its efficiency.

Sanitizing agents:

The sanitizing agents commonly used in the food industry belong to four main groups:

i) Halogen based formulations
ii) Quaternary ammonium compounds (Lysol)
iii) Amphoteric compounds
iv) Acids& alkalis

The sanitizing ability of a compound depends on

i) The concentration of the agent used
ii) Its temperature and time/duration of application
iii) The acidity or alkalinity of the solution
iv) Presence of organic matter (it retards the ability of sanitizer)

A disinfectant or sanitizer must be preceded by thorough cleansing. Cleansing must be a continuous process in the meat plant if optimum hygiene is to be activated.

Halogens:

Chlorine is the principal halogen used for sanitizing or disinfecting, treatment of water, treatment of sewage and equipment.Chlorine may be applied commercially as a gas or as hypochlorite solution (prepared from food grade salt or sodium chloride). In modern installations solutions are preferred over the gas. Fresh solutions possess maximum bactericidal activity. If required, brine solution may be acidified to a pH value of 6-7 to enhance the bactericidal action.

When **chlorine** is introduced into **water**, it produces **hypochlorous acid** and **hypochlorite ion**, the ratio of these depends on the pH of the solution. More hypochlorous acid is produced with a reduction in pH. Hypochlorous acid is more effective as a sanitizing agent than the hypochlorite ion. The strengths required for plant sanitation is **130-200 ppm**. For carcass washing, a maximum of **30-100 ppm** can be used. A solution of sodium hypochlorite containing **250 ppm** of chlorine can be used effectively to disinfect clean equipment.

The disinfectant properties of the hypochlorite depend on a number of factors:

- The concentration of available chlorine, which should be 200–300 mg/l.
- pH of the solution (to reduce the breakdown during storage, the hypo chlorite solution should be maintained at pH 9–11).
- Temperature: The hypo chlorite is more efficient if the concentration and temperature are raised and/or the pH is lowered (8.3).
- Contact time: To decrease the corrosive effect the temperature should not exceed 50–60°C and the contact time should not exceed 30–60 min.
- Absence of organic material. Organic material consumes available chlorine and reduces the disinfecting capacity.

Quaternary ammonium compounds:
They are most effective at pH levels near or just above neutral. They are inactivated by organic matter but not to the same extent as hypochlorite. They are able to form a fairly stable film, which exercises a continuing bacteriostatic action and are non-corrosive to metals. However, their efficiency is reduced with hard water and are more effective against gram-positive than gram-negative bacteria. They are more active in the presence of small amounts of organic matter than any other class of disinfectants, but are inactivated by soaps, anionic detergents and inorganic polyphosphates. They are more expensive than the halogens.

Amphoteric compounds:
They are either long chain substituted amino acids or betaines. They are surface active, compatible with all other detergents and sanitizers. They are unaffected by hard water.

Tego Compounds:
These are Ampholytic derivatives of amino ethyl (dodecyl-di) glycine and are surface active and bacteriostatic. Many substances including organic matter and hard water can reduce tego's activity and these are non-toxic and expensive.

Acids and alkalis: These are rarely used in routine cleaning operations although some may be used in specific disinfection program.

Alkali: Sodium hydroxide (caustic soda) is a common ingredient in detergents for the food industry. It is a powerful detergent and is used to suspend protein and to convert fats to soap, and it is cheap. However it corrodes aluminium and galvanized iron, strips paints, and presents a hazard to personnel using it. It is available in most geographical areas.

Sodium carbonate (soda ash) is not as efficient a cleaning agent as sodium hydroxide, but it is a cheap source of alkalinity and is used as detergent filler. It is corrosive to aluminium and galvanized iron, and forms a scale of calcium carbonate and other insoluble salts in hard water.

Sodium metasilicate is an effective detergent for many purposes. It is an excellent emulsifying and suspending agent and has reasonable wetting and rinsing properties. It possesses anticorrosive properties but tends to deposit on stainless steel surfaces. It may also deposit with soil as a grey-white coating if used in water above 70°C.

Acids are used to remove mineral deposits but they have reduced cleaning effects and are corrosive to different materials (especially galvanized iron and aluminium). The inorganic acids are principally more corrosive than the organic acids. Acidic detergents are mixtures of one or more acids and surface-active agents, and may be inhibitors too. These detergents have a reasonable cleaning effect. If acids alone remove the mineral deposits, the cleaning effect will be minimal and it may be necessary to remove fat and protein with an alkaline detergent before removing the deposits with acid.

The inorganic acids used are phosphoric acid, nitric acid, sulphuric acid and hydrochloric acid. The organic acids used are gluconic acid, tartaric acid, citric acid, acetic acid and sulphamic acid.

Emergency disinfection

For emergency disinfection, i.e. when disinfection is required during processing because of animal disease (anthrax, foot and mouth disease etc.) the following disinfectants are recommended:

a. Sodium hydroxide (caustic soda or lye) in a hot solution of approximately **2%** for foot and mouth disease and **5%** for anthrax.
b. (Avoid any splashes in eyes or on skin (caustic). If an accident happens wash clean water).
c. Sodium hypo chlorite solution: 0.5% available chlorine
d. Hot water (90°C or more) or steam
e. Chloride of lime for lairage, stables and transport vehicles: approximately **5%** solution.

Meat plant cleaning procedure:

1. Physical removal of all gross fat, skin and meat scraps (physical pick up, dry scraping, if possible). Particularly in the slaughter hall, this is a round-the-clock operation and must be associated with tidy working methods.
2. Apply cleaning compounds at proper concentration and temperature for their optimum activity.
3. Rinse with water followed by hot water.
4. Sanitize.

Cleaning efficiency depends on:

1.Temperature (application)
2.Force of agitation
3.Time
4.Chemical concentration

Many different combinations of cleansing equipments exist in different meat plants, the aim being to perform the task efficiently, quickly and with minimum of labor. An ample supply of good, hygienic, preferably soft water (**TDS 30-100 ppm**) with an adequate number of hose points are essential. An ample supply of hot water is essential at a temperature of 60ºC, preferably boosted with steam to about 82ºC. The usual method of applying hot water in meat plants is through high-pressure jet cleaners at a pressure of about 14 kgf /cm^2. For greater efficiency, hot water shall usually be in combination with a fixed amount of detergent. Thorough cleaning using high pressure hot water is usually carried out at specified times when operations are suspended or at the end of days kill. Cleaning operations must be frequent to prevent build up of bacteria.

Cleaning of slaughter equipments:

Trolleys, hooks, gambrels, knives, scabbards, cleavers, meat containers etc. may be sterilized in cabinets in sterilization room where they are immersed in batches in tanks containing hot detergent solution, hot rinses, anti rusting and oiling solutions. An efficient mechanized system of cleansing is often provided by a conveyor system on which frames holding the trolleys and gambrels in clusters are suspended and successively dipped into tanks containing hot alkaline detergent solution, hot water rinse and light mineral oil.

Methods of Cleaning:

- **Manual Cleaning**
- **Foam and Gel Cleaning**
- **Spray Cleaning**
- **Fogging**
- **Machine Washing**
- **Automated Cleaning system**

Foam and gel cleaning:

These compounds contain chemicals appropriate to the soil and surface being cleansed. The foam or gel adheres to the surface, allowing time for the chemical to break down the soil, which is then rinsed away with hot water under pressure. For proteinaceous material acidic compounds and for fatty material alkaline compounds are used.

Method of application:

Foam and gel cleaning solutions are applied through a lance (metal pipe) from a unit operated by compressed air or by an electrically operated compressor. In this way 45.4 liter of foam solution expands to 727-900 liters, sufficient to cover 55.7-92.9m^2 of surface in 15-20 min. Foam is applied in cold form and allowed to act for 15-20 min. during which time it constantly collapses, bringing a film of fresh chemical into contact with the soil. Gel does not collapse and can be applied in a very hot form which is especially useful for thin, tenacious protein or fat films where longer contact times and /or heat may be advantageous.

Cleaning schedule and the type of compound to be applied is governed by

i) Nature of the cleaning operation
ii) The degree of contamination
iii) The type of soil
iv) The form of equipment

Advantage of foam cleaning:

1. It is labor saving as large surface areas can be covered in a relatively short time. It can penetrate inaccessible areas, often eliminating the need for dismantling of equipment.
2. It is economical since the foam clings to surfaces and does not run to waste
3. It is biodegradable and does not give rise to effluent problem.

4. Foam does not splash and is comparatively safe to use. For strong alkalis and acids call for care, and protective clothing and goggles. Acids must be used with caution because of possible corrosion

Automated Cleaning system:

Three main types of automated cleaning systems have been developed:

1. **Cleaning-in-place system (CIP):**

   The CIP was first developed for the dairy industry. It is a closed system in which the cleaning solution compounds are circulated by a pump through a series of pipes to the component to be cleaned. It is basically designed for cleaning internal surfaces only. Food plants that process **fluids** extensively use this system, whereas its use in meat industry is limited.

2. **Central cleaning system (CCS):**

   CCS has a central pumping source supplying cleaning solutions under pressure to remote locations in the meat plant. In one CCS the cleaning materials may be mixed centrally and delivered to various points through one manifold, the plant water supply being used for rinsing.

   In the other CCS the detergent is transported through a separate manifold to each remote station where it is mixed with the high pressure water system as required and used through a cleaning gun. The unit should be capable of attaining pressure of 35-49 kgf/cm^2 and a flow of 136-181 liters/ min.

3. **The self- contained cleaning system (SCCS):**

SCCS has the pumping source and chemical spray systems contained in one unit and may not have facilities for foam production. Some units produce hot water while others employ steam-mixing valve or utilize the separate hot water system of the plant. Some forms of this automated cleaning equipment are potable and can be removed from one location to another. It is a flexible system in that if a pump fails a unit from another area can be used, whereas in CCS the entire operation comes to a stand still if this occurs.

BIS standards for Abattoir Disinfection:

The following agents can be used in water for routine cleaning and sanitizing: Lime water (1:3), bleaching powder (1:3), Calcium chloride (7 % chlorine). If some infection or disease is detected where the contaminating agents can be easily destroyed bleaching powder in the ratio of 1:20 shall be used. In those cases where the etiological agents are not easily destroyed (e.g.: Anthrax, BQ, Pox, Swine Fever) the carcass, blood and water shall be burnt or buried (with slaked lime) and the area in contact shall be treated with 30 % calcium chloride.

PERSONAL HYGIENE

Objectives:

To ensure that those who come directly or indirectly into contact with food are not likely to contaminate food by:

- maintaining an appropriate degree of personal cleanliness;

- behaving and operating in an appropriate manner.

Rationale:

People who do not maintain an appropriate degree of personal cleanliness, who have certain illnesses or conditions or who behave inappropriately, can contaminate food and transmit illness to consumers

HEALTH STATUS

People known, or suspected, to be suffering from, or to be a carrier of a disease or illness likely to be transmitted through food, should not be allowed to enter any food handling area if there is a likelihood of their contaminating food. Any person so affected should immediately report illness or symptoms of illness to the management.

Medical examination of a food handler should be carried out if clinically or epidemiologically indicated.

ILLNESS AND INJURIES

Conditions which should be reported to management so that any need for medical examination and/or possible exclusion from food handling can be considered, include:

- Jaundice
- Diarrhea
- Vomiting
- fever
- sore throat with fever
- visibly infected skin lesions (boils, cuts, etc.)
- discharges from the ear, eye or nose

PERSONAL CLEANLINESS -Food handlers should maintain a high degree of personal cleanliness and, where appropriate, wear suitable protective clothing, head covering, and footwear. Cuts and wounds, where personnel are permitted to continue working, should be covered by suitable waterproof dressings.

Personnel should always wash their hands when personal cleanliness may affect food safety, for example:

- at the start of food handling activities;
- immediately after using the toilet; and
- after handling raw food or any contaminated material, where this could result in contamination of other food items; they should avoid handling ready-to-eat food, where appropriate.

PERSONAL BEHAVIOUR People engaged in food handling activities should refrain from behaviour which could result in contamination of food, for example:

- smoking;
- spitting;
- chewing or eating;

- sneezing or coughing over unprotected food.

Personal effects such as jewellery, watches, pins or other items should not be worn or brought into food handling areas if they pose a threat to the safety and suitability of food.

VISITORS

Visitors to food manufacturing, processing or handling areas should, where appropriate, wear protective clothing and adhere to the other personal hygiene provisions in this section.

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ANTE-MORTEM INSPECTION OF FOOD ANIMALS

According to Thornton (1971), Ante-mortem inspection is the examination of food animals in holding pens (lairage) within 24 hours prior to slaughter with the object of providing wholesome meat to the consumers by deciding their fitness for slaughter by an authorized veterinarian.

- ❖ AMI should be conducted 12-24 hrs before slaughter in lairage pens.
- ❖ May be repeated before slaughter if the delay in slaughter is more than 2-3 days after unloading.
- ❖ AMI should be ensured that animals are not subjected to any kind of cruelty.
- ❖ All animals meant for slaughter should be rested at least for 24 hours and should not be fed for at least 12 hours before slaughter but they should be provided with plenty of water.
- ❖ Ante-mortem inspection of meat animals is necessary to produce wholesome meat and thus safeguard the health of meat consumers.

OBJECTIVES (PURPOSE)

- ❖ To make postmortem examination more efficient, accurate and less laborious. Ante-mortem examination represents 50% of meat inspection
- ❖ Detection of animals suffering from scheduled infectious diseases which are communicable to man.
- ❖ To detect certain diseases which are toxic or contagious and whose identification is either difficult or impossible during post-mortem, e.g. tetanus, rabies, listeriosis, sturdy in sheep etc.
- ❖ To implement disease control programme with more precision by tracing back the source of diseases.

FACILITIES REQUIRED FOR CONDUCTING ANTE MORTEM INSPECTION

- ❖ Properly designed holding pens and detention pens
- ❖ Proper light
- ❖ Enough space
- ❖ Trevis for clinical examination of individual animals
- ❖ Assisting staff
- ❖ A well designed "ante mortem inspection code" giving a detailed account of inspection procedures, judgment principles, disposal methods and documentation of findings.

ANTE-MORTEM INSPECTION PROCEDURE

1. Identification of the animal

- Identification can be done either by Ear Notching, hair clipping, branding or by application of Tags (usual practice in abattoir).
- Collection of the details regarding the place of purchase or origin.
- The main objectives are to: Identify the animal because of its varying source and to maintain the identity of the carcass during PMI, and in case of outbreaks/ Notifiable disease it becomes easy to trace back to the location from which the animal were procured.

2. Examination of the animal

The animals are placed in properly lighted holding pens where they can be moved about freely so that the inspector can view them at rest and motion.

Ante mortem inspection should be carried out in two stages:

Stage I - General examination:

- Meat animals should be observed in the lairage pens during rest as well as in motion.
- Observed for cleanliness
- The general behavior, reflexes, fatigue, excitement, gait, posture, evidence of cruelty, level of nutrition, symptoms of diseases or any other abnormalities should be closely observed.
- Both sides of the animals are observed so that the condition of face, skin and presence of wounds can be detected.
- While animals are in motion, it is observed for lameness.
- In case of female animals, they are observed for mastitis, metritis and retained placenta.
- In case of calves they are mainly observed whether they have matured enough to produce quality beef.

Stage II - Detailed examination:

❖ Suspected or diseased animals should he segregated for detailed examination.

Their temperature, pulse rate and respiration rate should be recorded. Animals showing elevated temperature and systemic disturbance should be detained for further inspection and treatment in the isolation pen.

Conditions to Be Recorded

❖ **Temperature:** Normal temperature is sign of health. An increase in temperature may indicate a systemic disturbance. If the temperature is high, it usually indicates the possibility of a septicaemia.

❖ **Pulse Rate:** Pulse rate reflects the condition of the heart. In a healthy animal the pulse rate is fairly constant. It is often faster in young animals.

❖ **Respiration:** Healthy animals take regular and easy breaths and the process of respiration is effortless and noiseless. To measure the respiratory rate, observation should be made at the flank region of the animal (Type of respiration - abdominal and occasionally costoabdominal). If the breathing pattern differs from normal, the animal should be segregated as a 'suspect'.

❖ **Appearance / General condition:**

- Bodily condition: Emaciation, cachexia, weakness, injuries.
- Healthy animals generally have bright eyes. Unhealthy animal may show reeling, dull and anxious expression in the eyes and aggressiveness
- Alert and responsive ears in healthy animals.
- Glossy skin - Skin gives a good indication on the general appearance and level of hydration of the animal. In healthy animals, it’s glossy with smooth hair coat.
- Moist muzzle in healthy animals; otherwise dry or scaly.

❖ **Behaviour:**

- "Response to stimuli" is judged.

- Normal animal reacts and responds very promptly, with the movement of its head, tail, eye, ears etc.
- Animals with abnormalities tend to separate out from the flock and are not responsive to stimulus.
- Changes in behaviour may either be expressed as depression or anxiety.

Abnormalities in behaviour may be manifested by one/more of the following signs:

- Walking in circles .
- Pushing its head against wall .
- Charging at various objects and acting aggressively (Rabies, Heavy metal poisoning)

❖ **Posture:**

This condition is usually observed when the animal is in standing and lying positions.

- Abnormality in posture is observed as tucked up abdomen.
- The animal may stand with an extended head and stretched out feet.

❖ The animal may be lying and have its head turned along its side. It is called as "Downer Syndrome". These animals should be handled with caution in order to prevent further suffering.

❖ **Gait**:

- This condition is generally observed when the animal moves/walks.
- This helps to detect conditions like pain in the legs, chest and abdomen.
- Changes in gait are generally associated with pain in the legs, chest or abdomen or are an indication of nervous disease.

❖ **Structure (Conformation):** It requires close observation of the animal from both front and rear side i.e. the abdominal contour.

- Abnormalities in structure is manifested as:
- Swelling (abscesses) seen commonly in pigs
- Enlarged joints
- Hernia
- Umbilical swelling
- Enlarged jaw (Lumpy jaw - Actimomyces)
- Bloated abdomen
- Mastitis, etc.

❖ **Discharges and Protrusions:**

- Secretions and excretions from natural orifices (nose, mouth, bloody diarrhoea etc).
- Protrusions like prolapsed uterus and prolapsed rectum are indicative of bodily abnormalities.

❖ **Abnormal colour:** Such as

- Black areas in swine
- Red areas on light colored skin (inflammation),

- Dark blue areas on the skin and udder (Gangrene).

❖ **Abnormal odor:** It is very difficult to detect on routine AMI.
  - The foul odor may be due to discharges or due to some abnormality in the animal digestive physiology.
  - Acetone odor – ketosis
  - Purulent exudates with medicinal odor - abscess
  - Putrefactive exudates- putrid rhinitis, putrid bronchitis

❖ **Physical exhaustion:**
  - The condition usually observed in animals travelled for long distances under unfavourable conditions.
  - The most consistent clinical finding is dehydration reflected as loss of skin elasticity, sunken eyeballs and dry mucous membranes.

The other conditions usually associated with this are fatigue, weakness, stiffness and pain.

Principles of Ante-mortem inspection

To differentiate normal, healthy animals from the diseased or abnormal animals.

The abnormal or diseased animals can be grouped under three categories:

1. "Category - A" : UNFIT - Totally unfit or transitory or allowed for treatment - unfit for slaughter.
2. "Category - B" : Animals with localised conditions (e.g.,light fracture, a tumour or a lesion).
3. "Category - C" : Conditions not advanced to a point of rendering the animal unfit but which might influence the disposition of carcass on post-mortem inspection.

Diseases and Abnormalities Encountered in Ante Mortem Inspection

i). Diseases of Cattle:-

1. **Viral diseases** -Rabies, Foot and Mouth Disease, Rinderpest.
2. **Bacterial diseases**-Tuberculosis, Actinobacillosis, Actinomycosis, Anthrax, Black Quarter, Tetanus, Haemorrhagic Septicaemia, Mastitis, septic metritis.
3. **Parasitic diseases**- Ringworm mange, emaciation and tumours

ii) Diseases of calves :-

1. Immaturaity
2. Calf diptheria
3. Ringworm
4. White scour

iii) Diseases of sheep :-

1. Emaciation
2. Sheep scour
3. Caseous lymphadenitis
4. Pneumonia
5. Enteritis
6. Gid
7. Tetanus

iv) Diseases of swine:-

1. Salmonellosis

2. Swine fever - Reddening of skin especially abdomen,
3. nervous symptoms such as circling, in coordination, muscle tremors & convulsions
4. Rabies
5. Actinomycosis of udder
6. Hernia
7. Tumour
8. Tuberculosis of lumbar vertebrae.

Significance of Ante-mortem Inspection

Tag 'U' :Listeriosis, Forage poisoning, Rabies, Pseudo-rabies, Scrapie.

Tag 'S' : Vesicular exanthema, Vesicular stomatitis, Salt poisoning, ergotism, hyperkeratosis, Fluorine poisoning.

Tag 'P' : Parturient paresis, Rail road sickness, Shipping fever, Selenium poisoning, Infectious bovine rhinotracheitis.

Tag 'CU' : Swine erysipelas, Actinomycosis, Actinobacillosis, Haemorrhagic enteritis.

Tag 'D' : Anthrax, Tetanus, Black quarter.

Inference

Terminology used to identify the results of Ante-Mortem Inspection:

U - Unfit for slaughter

D - Destroy and dispose

CU - Conditionally unfit

P - Postpone slaughter

S - Suspect

Examplesofspecific diseases and their judgements

S.No	Condition	Significance
1.	Listeriosis	U
2.	Polioencephalomyelitis	U
3.	Anthrax	UandD
4.	White scour in calves	CU
5.	Pseudorabies	U
6.	Tetanus	U and D
7.	Blackleg	U and D
8.	Swine erysipelas	U
9.	Vesicularstomatitis	U
10.	Vesicularexanthema	U
11.	Parturientperesis	U
12.	Grasstetany	F
13.	Hypomagnesemia	D
14.	Transportsickness	D
15.	Epitheliomaofeye	CU
16.	Actinomycosis	CU
17.	Actinobacillosis	CU
18.	Shipping fever	U

19.	Infectious bovine rhinotrachitis	D
20.	Selenium poisoning	U
21.	Fluorine poisoning	S
22.	Hyperkeratosis	S
23.	Salt poisoning	U
24.	Cutaneous streptotrichosis	S
25.	Ergotism	U
26.	Brisket disease	U
27.	Nasal granuloma of cattle	S
28.	Chlorinated hydrocarbon poisoning	U
29.	Organophosphate poisoning	U

AM of Poultry

Live poultry should be subjected to ante-mortem inspection in the holding pens by a qualified veterinarian on the day of slaughter. Enough space and water should be provided in the holding pens. Adequate light is an essential requirement during inspection. The birds are carefully examined and those in good health and alert condition are declared fit for slaughter

The diseases commonly encountered in ante-mortem inspection of poultry are as follows:

	Diseases	Symptoms
1	Ornithosis	Inflammationoforbit.Liquid,yellowishand bloodtingeddroppingsandeversionof cloaca.
2	Respiratory Disease Complex	Absence ofmusculardiaphragm.
3	RanikhetDisease	Respiratorydistress,coughing,depression, gaspingimpaired appetiteand prostration.
4	ChronicRespiratory Disease (CRD)	-do-
5	InfectiousBronchitis	Dropineggproduction,nasaldischarge, gasping, andcoughing.
6	InfectiousLaryngeotracheitis	Gasping,coughing,depression,sneezing, expulsionofbloodymucousfromtrachea, cyanosis onhead.
7	Infectious Coryza	Inflammationandaccumulationofmucous andserous exudates inrespiratorytract, oedemaofhead,distress,gaspingand coughing.
8	Neural Lymphomatosis	Progressive paralysis ofleg,wingandneck.
9	Infectious Enterohepatitis (Black head)	Sulphurcoloureddiarrhoea,drowsiness, weakness,droppingofwingsandruffled feathers.
10	Fowl Typhoid	Darknessofcombandwattles,greenishyellowdia rrhoea, loss ofappetite and high fever.

11	Fowl Pox	Lesionsoncomb,wattlesandface in cutaneous form,lesions in mouth in diphtheriticform,coryzalikesignsincoryza, formandoccasionally poxlesionsonleg,feet and otherparts ofbody.
12	Fowl Cholera	Emaciation,lameness,hot,edematousand painful wattles, difficult breathing and diarrhoea.
13	Botulism(Limberneck)	Paralysis,droppedwing,headhangslimpand loosefeathers.

POST –MORTEM EXAMINATION OF FOOD ANIMALS INCLUDING POULTRY

1. Introduction

The postmortem inspection (PM) constitutes a detailed and complete examination of the carcass in order to provide a wholesome and disease free meat to the consumer and forms a part of responsibilities of the meat inspector (Veterinarian) in a slaughterhouse. The other subsidiary aspect of post mortem inspection is checking the efficacy of slaughter, carcass dressing and diagnosis of disease condition for disease control purpose. Many abnormalities affecting animals, which are not evident on ante-mortem examination, may be detected at postmortem. Inspection procedures are so organized that they not only provide a thorough post-mortem examination but also to eliminate contamination of meat during each step of intricate slaughtering operations.

The PM inspection must be carried carefully; in hygienic manner and avoiding unnecessary cuts which may impair the market value of the carcass. The nature of the disease lesions as acute, sub acute and active spreading must be given adequate attention. The ante-mortem report, the necessary laboratory tests, general condition of the carcass should be considered in overall perspective before making a final Judgment. Responsibility of the consumer must be uppermost in inspector’s mind at the same time there must be no unnecessary wasting of meat.

2. Facilities for PM inspection

Proper infrastructure and mechanical facilities for smooth movement of carcass and their parts for PM inspection include adequate light, protective clothing, sharp knives, hand washing units with hot water liquid soaps, towels, sterilizers for knives, saws, cleavers etc. These are essential at each inspection point (head, viscera, and carcass) and also at detained area where carcasses are detained for further examination. There should be a perfect co-ordination between the workers on slaughter line and inspector along with synchronization of conveyor lines for identification of carcass and their individual viscera, organs. The clips, tie-on labels stickers, marking inks, knife marks on superficial muscles are generally used in carcass identification. The preprinted gelatin strips are struck on the carcass fat. This is edible, waterproof, non-smearing, non-dissolving and abrasions resistant. Any system of identification must be clearly legible, easily applied, cheap, non-toxic and non-corrosive. Generally beef carcass should be labeled on rib cage, veal on the leg, and lamb on front of shank and pork on the hock, away from ham. The offal and the carcass must be labeled with the corresponding numbers for easy identification.

3.PM inspection procedure

The essential conditions for PM inspection include

1. The carcass should not be quartered unless the inspection is complete
2. The carcasses that are dressed without flaying (e.g. pig) must be thoroughly washed before making an incision

3. No serous membrane must be removed or stripped obliterated before the inspection is complete
4. No organs or parts should be removed without the permission of the inspector

The visual examination of the carcass is done for

- The state of nutrition of the carcass (e.g. poor, emaciated, moderate etc.)
- Evidence of bruising, pigmentation or discolouration, local or general edema
- Bleeding efficiency, body cavities, condition of pleura, peritoneum visceral organs, their size, colour and shape

Palpation: for satisfactory examination of an organ the fingers should be pressed firmly into the organ / part of the carcass. This enables the inspector to locate smallest abscess or cyst.

Incision: whenever necessary the lymph nodes or carcass / organs may be incised to observe lesions or cysts but care must be taken not to distort the carcass or organs. Incision should only be deep enough to satisfy the inspector and to prevent any risk of contamination.

Cattle

Inspection of carcass and organs proceeds in following order: Head - An inspection of the gums, lips and tongue for foot and mouth disease, necrotic and other forms of stomatitis, actinomycosis and actinobacillosis follow an examination of the outer surface and eyes. The tongue being palpated from dorsum to tip for the latter disease. Incision for the internal and external massetor muscle for cysticercosis should be made parallel with the lower jaw. Routine incision of the retropharyngeal, submaxillary and parotid lymph nodes should be made for tuberculosis, abscess and actinobacillosis. The tonsils of cattle and pigs often harbor tubercle bacilli and shall not be used in meat product.

Lymph nodes

Detailed examination of the lymph nodes (bronchial, mediastinal, mesenteric) is necessary for the detection of tuberculosis

Lungs

Visual examination, palpation should be carried out for the pleurisy, pneumonia, tuberculosis, fasciolasis, hydatid cyst etc. The bronchial and mediastinal lymph nodes should be exposed by a long deep incision from the base to apex of each lung, if necessary.

Heart

The pericardium should be examined for evidence of pericarditis, hemorrhages etc. The heart ventricles are incised surface observed for petichial hemorrhages on epicardium or endocardium and for cysticerci, hydatid cysts and occasionally *linguatulae* in the myocardium. A flabby myocardium in cow is often associated with septic condition.

Liver

A visual examination with palpation should be made for fatty change, actinobacillosis, abscess, telangiectasis and parasitic infections as hydatid cysts, *cysticercus bovis*, and fascioliasis. The larval stages of *oesophagustomum radiatum* may be occasionally being found in the cattle liver. Portal lymph nodes should be examined and gallbladder be palpated and incised if necessary taking care not contaminating the carcass.

Oesophagus, stomach and intestine

Observe and if necessary palpate these organs. The serous surface may show evidence of or actinobacillosis while anterior aspect of the reticulum may show evidence of penetration by foreign body. Mesenteric lymph nodes should be incised for tuberculosis.

Kidney

EEC law requires kidney, adrenal lymph nodes inspection and enucleation.

Spleen

The surface and substance should be examined for tuberculosis, anthrax, haematoma or the presence of infarct with viewing palpation and if necessary incision.

Uterus

The uterus should be viewed, palpated and if necessary incised. In brucellosis reactors the uterus should not be incised or handled.

Udder

Udder is sold for human consumption and should be carefully examined for abscesses or septic mastitis. Multiple incisions may be made. Supra mammary lymph nodes should be incised for tuberculosis or abscesses. In brucellosis cases the udder is removed intact without incision.

Testes

Testes for human consumption should be viewed and palpated.

Carcass

The cut surfaces of bone and muscle, carcass exterior and interior, pleura, peritoneum and diaphragm should be observed for efficiency of bleeding, colour, cleanliness, odours and evidence of bruising and any other abnormalities. If necessary palpation and incision of parts is indicated e.g. triceps brachii muscle for *Cysticercus bovis.* The superficial inguinal, external and internal iliac, prepectoral and renal lymph nodes should be observed and incised. If tuberculous lesions have been detected in the viscera, the main carcass lymph nodes must be examined. The thoracic and abdominal cavities should be inspected for inflammation, abscesses, actinobacillosis or tuberculosis. The diaphragm should be lifted for tuberculosis lesions may be hidden between the diaphragm and thoracic wall. Where *Cysticercus cellulosae i*s prevalent the external masseter muscle and the root of the tongue are incised and the organs palpated.

Calves

The routine postmortem examination is usually the same except with special attention to particular sites. A visual examination of the mouth and tongue should be made for foot and mouth disease, calf diphtheria; abomasum for peptic ulcers; small intestine for white score or dysentery; liver, portal and mediastinal lymph nodes for congenital tuberculosis. The lungs, kidneys and spinal cord should be examined for melanotic deposits and the umbilicus and joints for septic omphalophlebitis. The appearance and consistency of renal fat should be carefully noted.

Sheep and Goats

Sheep and goat carcass should be examined on the same line that of cattle however more attention is paid for evidence of parasitic infestation.

Pigs

The skin lesions are an important diagnostic feature of swine erysipelas, swine fever and utricaria. The skin should be examined for shotty eruption; the tail for necrosis, the feet for abscess formation and the udder for mastitis or actinomycosis. The viscera requires detailed inspection as cattle with a particular attention to pneumonia and the secondary complication that develop in virus pneumonia, mainly pleurisy, pericarditis and to lesser extent peritonitis. The submaxillary, bronchial and mesenteric lymph nodes should be always examined for tuberculosis. The liver need not be incised except when it appears cirrhotic but portal lymph nodes should be incised as routine procedure. The kidney surface should be examined.

Equines

Post mortem inspection of equines follows the same general pattern as for cattle and other livestock.

Poultry

Inspection is done first for whole carcass after de-feathering and washing and second after evisceration. The overall procedure remains the same. The inspector in charge should regulate the line speed correctly.

The normal appearance of the organs is mentioned below to appreciate the changes resulting form diseases.

- The eyes: bright, free from opacity and discharge
- Skin: varying form off-white to brown
    1. Comb and wattles: Uniform in colour
    2. Bones and joints: uniform and without enlargement
    3. Abdomen: free from distension with fluid or blood
    4. Crop, proventriculus, gizzard: firm, solid

  Viscera: the gut should firm and dry to touch without leaving deposits on hand
  Liver: uniform, smooth and sharp borders
- Spleen: reddish brown without enlargement
  Kidney: firm, deep red
  Lungs: pink, slightly moist, light in weight

If the un-eviscerated carcass is presented for inspection, a cut must be made on the right side below the last rib to expose the liver. When the abnormalities are found in the liver, carcass should be retained for full examination.

While considering total condemnation of the carcass it must be remembered that carcass should be condemned totally if:

- The disease is accompanied by emaciation or dehydration

Advanced pathological conditions are seen

The condition has so spread that affected organ or portion cannot be separated easily

Decision at post-mortem examination

The final Judgment of the veterinary inspector is a result of total evidence produced by observation, palpation, incision, smell, any ante-mortem signs and any laboratory tests if conducted. The main decisions and the symbols are are as follows:

- Approved for human consumption, **(A)**
- Totally condemned for human consumption, **(T)**
- Partially condemned for human consumption, **(P)**
- Conditionally approved for human consumption refers to treatment of meat under the supervision of meat inspector to make safe for human consumption (Freibank system). **(K)**
- Inferior meat, (such as from the carcasses of incomplete bleeding, off colour, off odour or taste, slight edema etc.) certainly hygienic is sold as inferior meat especially in those countries where there is a scarcity of protein. **(I)**
- Approved for human consumption, with distribution restricted to limited areas, includes the meat from animals in an area under quarantine because of an outbreak of contagious animal disease. In this case there must be no risk to public health and the meat must be restricted in sale to the affected area to avoid the possible spread of disease. **(L)**

P.M. judgment in some important diseases

S. No	**Disease**	**Judgment**
1	**Tuberculosis**	Tissues affected by TB or contaminated by the products of a tuberculous process is unfit for food. An organ is unfit for food when it contains a lesion of TB or when a lesion of TB present in a lymph node draining it. The following principles are considered as guidance in judging the TB affected carcass. Carcass is condemned when Lesions of TB are generalized Animal had fever on ante mortem inspection and

		found to be associated with acute TB lesions on post mortem examination. There is associated cachexia (general physical wasting and malnutrition usually associated with chronic disease) TB lesions are found in muscles, bones, joints and respective lymph nodes. An organ or part of the any carcass affected with localized TB lesions is condemned and carcass is passed for food. Carcass passed for cooking When the TB lesions are not so severe or numerous as described in A., The unaffected portion is passed for cooking. When the carcass is tuberculin reactor but free of TB lesions. Carcass is passed without restriction for food, when the animal was not a positive tuberculous reactor and is free of lesions on postmortem inspection. TB is considered localised if there is no evidence of recent passage of numerous bacilli into the blood stream. If the animal is well nourished and in good condition, in such cases there if there is no proof or reason to suspect the flesh as unwholesome and the disease may be regarded as localised-all the affected organ need to be condemned. If only the lymph node is affected, the whole organ may be destroyed even apparently normal. Generalized TB Where there is evidence of recent passage of numerous bacilli in to the circulation e.g. Milliary TB in organs. Acute generalisation justifies condemnation of whole carcass and organs. Examination of carcass lymph nodes – typical lesions in the bronchial, mediastinal and mesenteric lymph nodes is of great importance in detecting TB during post mortem inspection. Confirmatory tests – Direct microscopic examination, culture of lesion material (pus and sputum) – Acid Fast on Ziel Neelsen's Stain.
2	**Brucellosis**	Brucella remains for a very short time in the muscles of a slaughtered infected animal, as are easily destroyed by the lactic acid formed after slaughter. Distinction must be

		made between the reactor and infected animals. It is likely that, provided care is taken in handling and dressing of reactor animals with the condemnation of genital organs, udder and associated lymph nodes, carcass form such animal is safe for humane food. But where Br. melitensis is encountered total condemnation or heat treatment may be appropriate, depending on local incidence and economics. Confirmation: Cultural examination of blood or lesion material (placenta, fetus, milk, and semen).
3	**Anthrax**	The entire carcass including blood, hide, fat and all intestinal organs must be destroyed along with the contaminated animals and meat. Zoonotic importance The type of anthrax in man usually depends on the mode of infection. It can be a cutaneous, pulmonary or intestinal disease. Cutaneous form is a common and results form the handling of infected material. Incubation period is 1-5 days. It commences with a small pimple which rapidly develops into a large vesicle with a black necrotic center; the so-called malignant pustule. If organism enters blood stream septicemia results. Pulmonary form is more serious and results in about 100% mortality. Death is mostly due to asphyxia. Confirmation: Gram staining, Ascoli's precipitation, culture of blood or tissue, direct inoculation in pigs (s/c) – death in 48 hrs is + ve test. Mcfadyean's reaction 1% methylene blue stain → capsule – purple, bacilli blue in colour.' Procedure when anthrax is detected The anthrax order forbids cutting of the skin of a diseased or suspected carcass. The detection of anthrax in the dressed carcass is much more serious than in the uneviscerated dead animals. As the skinning and opening of the carcass favours the formation of spores, with increased danger of humane infection any further dressing is forbidden. If the dressing of suspected carcass is completed the carcass in contamination should be condemned. The knives, saws and other instruments should be boiled in water for half an hour and thoroughly cleansed in a hot 5% solution of NaOH or 10% formaldehyde or

		ethylene oxide in plastic bags. The clothing must be disinfected with 10% formaldehyde. Hg (Cl) $_{2,}$ 1: 1000 is more effective as a disinfectant when HCl (0.5%) is added. The abattoir premises, pen, trucks, cars and other areas are disinfected with 10% NaOH or 5% formaldehyde. Hands / arms of the workers are washed with liquid soap followed by hot water and immersed for about one minute in an organic iodine solution or 1 ppm $HgCl_2$, followed by potable water rinse. Clothes should be sterilized by boiling. Method of disinfecting of hides and skins in anthrax: - I. Seymourjones method –in this hides are immersed in 1: 5000 solution of $HgCl_2$ with 1 % formic acid for 24 hours. The hide should be detained for a period of 2 weeks before being released. II. Shatterforb method –hides are soaked for 48 hours in 2% hydrochloric acid and 10 % NaCl solution.
4	**Swine erysipelas**	Total condemnation of carcass in septicemic form while in urticarial form affected tissues are rejected and carcass is passed for food.
5	**Glanders**	On account of the infectivity of the disease to man, either by handling or by consumption of meat affected carcass must be totally condemned. Positive cases should not be admitted to the abattoir. EEC regulation (64/ 433 / EEC) states that head of the solipeds must be split along the mid line and the nasal septum, mucosa of upper respiratory tract must be care fully examined
6	**Leptospirosis**	Carcass showing lesions of leptospirosis is unfit for food. However in recovered cases carcass is passed after rejecting kidneys, liver etc.
7	**Pox disease**	Total condemnation of the carcass in acute stage (poor bleeding) while in recovered cases passed for food.
8	**Rabies**	Total condemnation of the carcass from the animals that shown symptoms of the disease. The danger from such an animal is to those handling such meat rather than to

		those who consume it. WHO expert committee on rabies states that the flesh may be consumed if the animal was bitten eight days before slaughter / within 48 hrs. Of slaughter. The bite area and surrounding tissue must be condemned. The danger from such animal is to those handling such meat rather than to those who consume it.
9	**Tetanus**	The poor keeping quality and change of colour of the musculature necessitate total condemnation. However it is extremely unlikely to be acquired by man consuming the flesh of tetanus affected animal.
10	**Foot and Mouth disease**	Total condemnation of the diseased carcass or subject to heat treatment. Virus is susceptible to acid and destroyed within 48 hours of slaughter due to lactic acid production. Virus retains infectivity for 100 days in frozen meat. Virus also survives in meat and other organs like heart and lymph nodes. The commercial refrigeration of imported beef plays an important role in maintaining the viability of the virus in infected carcasses and organs. The viruses are the intracellular and present in minute quantities within the tissues. They do not alter the organoleptic qualities and do not cause any overt changes in appearance of the meat.
11	**ORF (contagious pustular dermatitis)**	Mildly affected cases may be passed for food rejecting the head, however total condemnation in severe cases.
12	**Cysticercus bovis**	Total condemnation of the carcass in cases of generalized condition due to its zoonotic importance. In localized cases affected tissue is removed and carcass is passed for food. A double cycle in blast freezing tunnel at maximum temperature of -30 ^{o} C would kill the cyst in boxed beef (30 kg. Carton) after 48 hr. cycle followed by 72 hrs cold storage at – 23.3 ^{o}C. Cooking temperature of above 60 ^{o} C and salting for 21 days kills the cyst. In offal stored at 6.5 ^{o} C for not less than 3 weeks or -10 ^{o} C for 2 weeks destroys the cyst.
13	***Cysticercus cellulosae***	Total condemnation of carcass in generalized cases.

14	**Hydatidosis**	Carcass is passed for food after removal of affected organs.
15	**Trichinosis**	Total condemnation and proper disposal of the carcass is required due to its zoonotic importance.
16	**Sarcocyst**	There is no proof that consumption of infested meat leads to human sarcosporidiosis and it is probable that man is an unsuitable intermediate host. The condemnation of the specific muscle or tissue or carcass in generalized cases is only on the aesthetic grounds. Life cycle of the sarcosporida requires two omnivore animals such as dogs, foxes and cats ingest muscle cyst from herbivorous animals and pass sporocyst in their faces. Herbivorous animals ingest the sporocyst, which develop in the intestine and pass to the muscle. It is possible that blood sucking and also non-blood sucking flies may act as transport hosts. Over 35 % of sheep dogs and up to 75 % of hunting dogs have found to be infected. The cystic stage in the muscles are regarded as non pathogenic but the earlier stages which develop in the endothelium of blood vessels may be very pathogenic, causing abortions in cattle and sheep and may even prove fatal particularly in calves.

Conditions detected at post mortem inspection and their judgment

S. No	Organ / condition	Judgment
	Skin	
1.	Primary – Ringworm, acne, papillomata, screwworm flies, warble flies, insect, tick bites, transit erythema.	Localised condemnation
2.	Secondary – A sequel of disease, swine erysipelas, diamond skin disease, toxaemic condition, sheep pox, contagious pustular rheumatitis, Utricaria (allergic reaction), cutaneous neoplasm (lymphoma, melanoma), photosensitization, subcutaneous emphysema (black quarters)	Generalized condemnation in case of systemic diseases.
	Alimentary Tract	
	Head tongue	
	Actinomycosis	
	F & MD	
	Blue tongue	
	Atrophic rhinitis	
	Fowl pox	
	Cysticercus bovis	
	Sarcosporidiasis	
	Multiceps multiceps	
	Black mottling (flourosis)	
	Lymphosarcoma	
	Oesophagus	
	Chocking due to foreign bodies	
	Sarcocyst tanella	
	Cysticercus bovis	
	Warble fly larvae	
	Stomach & intestine	
1.	Peptic ulcers	
2.	Petechial haemorrhages	
3.	Gut oedema - E.coli	
4.	Acute enteritis – salmonella, fever, intoxication	
5.	Haemorrhagic entertitis – calf dysentery, salmonella	
6.	Foreign bodies	

7.	Parasites – worm load, amphistomes	
	Liver (usually secondary in nature)	
	Telengiectasis (plum pudding liver) numerous bluish black patches especially in old cows	
	Fatty liver	
.	Bacterial necrosis – yellowish, spherical areas of	
	Coagulation necrosis	
.	Liver abscess	
.	White liver disease (cobalt/ vitaminB12 deficiency)	
.	Black liver – feeding of certain leaves in sheep	
.	Sawdust liver – (minute yellowish necrotic foci)	
	Due to Vitamin E deficiency	
	Cirrhosis of liver (Proliferation of Fibrous connective tissue in liver)	
	Fasciola hepatica (cattle / sheep)	
	Migration of *Ascaris sum* larvae in pig	
	Toxic mold Penicillium rubrum, Aspergillus flavous	
	Parasitic Affection of liver	
	Fasciolosis	
	Hydatid Cyst	
	Cysticercus tenuicollis	
	Ascaris suum (Milk spots on liver)	
	Peritoneum	
	Acute Peritonitis in calves due to perforation of peptic ulcers (straw colored inflammatory exudates with fibrin deposits in peritoneal cavity & feeble adhesions)	
	Chronic cases	
	Yellowish cheese like thick covering of the abdominal organs & foul smell.	
	Tuberculos peritonitis (Pearly disease)	
	Respiratory System	
	Nasal Cavities	
	Nasal Catarrh (rhinitis) e.g. due to corynebacterium pyogenes causes a purulent nasal discharge in calf pneumonia	

	Bronchi	
	Chronic bronchitis due to parasitic nematodes (dictyocaulus viviparus)	
	Bacterial Bronchitis (corynebacterium pyogenes)	
	Affection of upper respiratory tract is not generally important in meat inspection.	
	Lungs and Pleura	
	Interstitial emphysema (old cattle) characterized by a marked dilution of the pulmonary interstitial tissue in which mobile air vesicles of varying size are arranged in strings.	
	Pneumonia	
	Usually bacterial or viral origin but may be due to accidental inhalation of liquid. Bronchopneumonia is characterized by small pneumonia foci around the bronchioles and intermixed with areas of normal lung tissue Lobar pneumonia usually caused by bacteria and involves whole or greater part of a lung lobe, hepatized or consolidated tissue and sharply demarcated from normal tissue. Both lobar and bronchial pneumonia may be associated with defined necrotic foci of varying size. Even gangrene may develop if putrefactive organs are present. It is a common sequel to septic pneumonia of the bovine lung. Pneumonia is seen in specific diseases like T.B, swine, fever, shipping fever, glanders, CBPP, enzootic pneumonia	Simple lobar or bronchial pneumonia is nearly an indication for total condemnation but incases of gangrenous or specific disease like military T.B there is a definite reason for total condemnation.
	Heart	
	Cysticercus bovis / Cysticercus cellulosae	
	Vegetative endocarditis – chronic swine erysipelas	
	Ulcerative Valvular endocarditis – Corynebacterium pyogenes, sarcocystis	
	Spleen	
	Acute enlargement (Anthrax, anaplasmosis, salmoneollosis, swine erysipelas, septicemia)	
	Abscess – (tuberculosis)	
	Urinary System	
	Nephrosis is the term used to denote any kidney condition especially those associated with degenerative changes.	
	Nephritis appears in several different forms.	

	Glomerulonephritis	
	In acute form shows an enlarged kidney, pale yellow in colour due to fatty changes with haemorrhages on the surface	
	Interstitial Nephritis	
	Results from the ingestion of various toxins (bacterial/chemical/plants). Kidneys are pale & enlarged.	
	Pylo-nephritis (Inflammation of kidney Pelvis)	
	Arises as an ascending infection e.g., from cystitis. Mostly bacterial origin (streptococci, Corynebacterium renale). Pylo-nephritis can be unilateral / bilateral. Manifested by catarrah and dilation of renal pelvis, which filled with a slimy fluid, which may contain fibrinous clots or pus and strong ammoniacal odour.	
	White spotted kidney – salmonella or brucella in young calves	
	Pulpy Kidney - Enterotoxaemia in goats	
	Kidney worm - Stephanurus dentatus	
	Uraemia	
	Obstruction of urethra due to many reasons results in strong urine odour from carcass, which render it unfit for consumption.	
	Reproductive System – Testicles	
	Orchitis – Brucella abortus in bulls (Necrosis of testicular substance)	
	Suppurative necrosis – T.B, Corynebacterium	
	Uterus	
	Metritis, Placentitis (brucella), T.B, Vibrio, Campylobacter, toxoplasma, leptospira, mycoplasma	
	Udder	
	Mastitis – Streptococci, E. coli, Staphylococci, Corynebacterium, Pseudomonas, Klebsiella, Mycoplasma, Aspergillus, Candida. In cow commonest form of mastitis is caused by Streptococci agalactiae.	
	Carcass affected with acute septic or gangrenous mastitis with systemic invasion should be condemned. “Seedy belly”, a form of melanosis in the udder of cows shows black spots in the mammary tissue.	
	Muscular System	
	Xanthosis – chocolate colored pigmentation	
	Splashing – Muscular pin point haemorrhages	

	Myositis – Black leg	
	White muscle – selenium poisoning	
	Pale / Strip muscles – Tetanus	
	Nodular necrosis (Rockle's granules) mostly in older cows, characterized by gray yellow nodules in muscles of neck, shoulder & back.	
	Parasites	
	Sarcosporidia / C.bovis / C. cellulosae / trichinella spiralis / multiceps – characterized by departure condition, greenish foci, calcified	
	DFD / PSE	
	Bruise	
	Injection damage	
	Implants of growth promoters	
	Wet carcass Syndrome – lambs in South Africa	
	Characterized by translucent jelly like encapsulated fluid over bullock's, brisket.	
	Brain	
	Encephalitis – Listeria, T.B	
	Cyst. cellulosae – BSE, Multiceps multiceps	
	Viral encephalomyelitis, hog cholera, African Swine fever, rabies, etc.,	
	Pathology	
	Inflammation	
	The type of inflammation and its stage is important. Consideration in meat inspection, acute inflammation in general being more serious than chronic form.	
	Acute – Anthrax, Black leg, braxy	
	Chronic – Chronic TB, Actinobacillus	
	Hypothermia	
	Due to excessive stress, poisons there is generalized congestion in internal organs and muscles.	Total condemnation approval after heat treatment
	Hypothermia	

	Due to severe trauma, pain, dehydration, electrocution, prolapse of uterus, internal rupture. There is congestion of lung, liver, and oedema, in intestine, peritoneal cavity.	Total condemnation, detailed examination is necessary, may be approval after heat treatment
	Imperfect bleeding	
	Left ventricle usually contains blood, subcutaneous blood vessels appear infected, flesh is dark, and organs are congested, flabby & watery.	Poor keeping quality, If imperfect bleeding is due to febrile condition, carcass must be totally condemned.
	Necrosis	
▪	Liquefaction	Total / partial condemnation
▪	Coagulative	
▪	Zankers	
▪	Caseatious	
	Gangrene	
	It is the invasion of necrotic tissue by putrefactive bacteria, mostly seen in ears, tail, lungs or udder.	Partial / total condemnation
	The tissue is soft, dark or greenish color with fowl smelling fluid.	
	Pigmentation	
	Melanosis –	
	Xanthosis – Deposition of lipofuschin pigment in skeletal muscles & organ imparting dark brownish or yellowish color – Localised condemnation	
	Yellow Fat disease	

	Seen in pigs fed on fish diets. The pigmentation is due to the deposition of a waxy substance called ceroid, which is similar to lipofuschin, being resistant to fat solvents, acid – fast, negative to iron stains and sustainable with fat stains. Carcasses must be distinguished from jaundice	Approval in absence of jaundice or fish smell.
	Haemorrhages	
	Petechial – small pin point	
	Ecchymoses – larger than petechiae	
	Hematoma – haemorrhages in the form of large clots	
	Anaemia Pale colour of the meat, watery blood	Totally condemned if associated with poor condition or haematoma, conditional approval in less severe cases.
	Odema	Detain the carcass for 12 hrs. Anasarca if accompanied by emaciation needs total condemnation
	Hydrothorax (Thoracic cavity)	
	Ascites (Abdominal cavity) ------- Less serious	
	Anasarca (subcutaneous / connective tissue) -----------More serious	
	Toxaemia	
	Importance depends on the type of organ & their metabolic activity generalized effects affecting all major organ.	Total Condemnation.
	Septicaemia / Viraemia	
	Serious condition of Toxaeamia, fever and the presence of large number of pathogenic bacteria in blood stream. Characterized by high fever, petechial haemorrhages in myocardium, liver, kidney, enlarged edematous lymph nodes, cloudy swelling of liver, kidney, blood stained serous exudates in thoracic cavity abdominal cavity. Meat become alkaline, soft, dark in color, sweetish repugnant odour.	Total condemnation for two reasons Poor keeping quality and pathogenic organism or toxin.

	Icterus / jaundice - is a symptom of presence of bilirubin (orange / yellow) and biliverdin (green) bile pigments in the circulating blood. It can be a obstructive jaundice or hemolytic jaundice **Lesions** Yellow coloration of fatty tissues, serous, connective tissue, nerve trunks, endothelial linings of arteries such as iliac, brachial & femoral, cortex of kidney, sclerotic membrane of eyes, liver, tendons and cartilaginous extremities of long bones. Carcass may render repugnant odour. Test – Boiling test or Remington Fowrie test - far presence of abnormal odour & bile pigments. Distinguish jaundice with carotene in cattle or xanthophyll in sheep (sheep are unable to oxidize xanthophylls in its food)	Carcass may be detained for 24 hrs save this color fades due to action of the muscular enzymes. In absence of these changes the slightly icteric carcass may be released.
	Abscesses	
	Injection, wound, jowl abscess or organ specific lesions.	Local condemnation
	Flesh of fetuses and stillborn animals	
	Flesh watery may be affected with specified organising like brucella / leptospirosis / listeriosis / mycosis/ toxoplasmosis	Flesh of stillborn / unborn/immature carcass is condemned if it is edematous / poor physical conditions
	Poorness	
	Due to inadequate nutritional intake trance is a less development of fatty tissue muscular tissue is firm/darker than normal, tough due to more connective tissue.	Condemnation depending on depress of poorness

	Emaciation- It is a pathological condition, which develops during the course of an illness, which, either lowers the food intake or increases the metabolic rate of the animal. May be due to disease like JD, TB, muscular tissue atrophies, and parasite or simply due to prolonged starvation. **Characterized by** Reduced size of organ, loss of body fat, jelly like consistency. Water protein ratio is less than 4:1.	On depress of cause ofemaciation in absence of disease the carcass may be passed to following heat the oddment.

The diseases commonly encountered in P.M. inspection of poultry

	Diseases	**Lesions**
(i)	Erythroblastosis	Diffuse condition of liver, kidney and spleen, paleness of affected tissues, petechial hemorrhages in small intestines.
(ii)	Granuloblastosis	Gross pathological changes are similar to erythroblastosis, paleness of bone marrow.
(iii)	Visceral Lymphomatosis	Almost all organs including skin are affected, spleen is enlarged three times of its normal size and has grayish brown colour.
(iv)	Neural Lymphomatosis	Soft swelling of peripheral nerves, crop contains fetid material.
(v)	Osteopetrosis (thick leg disease)	Irregular thickening of long bones.
(vi)	Ranikhet Disease	Inflammation of trachea, bronchi and cloudiness and yellowish thickening of air sacs, spleen pale and small.
(vii)	Infectious Laryngotracheitis	Cyanosis of head, blood stained exudate in trachea and larynx.
(viii)	Infectious-Coryza	Catarrhal inflammation of nasal passage and sinuses, conjunctivitis, subcutaneous oedema on head, face and wattles.
(ix)	Chronic Respiratory Disease	Hypertrophy of mucous membranes, hemorrhage and exudation in the paranasal sinuses and trachea, bronchial thickening and pneumonia.
(x)	Ornithosis and psittacosis	Thickening and inflammation of air sacs, mesentery and serous surface of intestine, enlargement of liver, pericarditis.
(xi)	Salmonellosis	Inflammation of duodenum, hyperemia of liver,

		kidney, gall bladder and myocardium, straw coloured fluid in pericardial sac, caseous plug in caecum and dehydration of muscle.
(xii)	Fowl Typhoid	Anaemia of-muscles, comb, wattle, skin and kidney, hypertrophy of liver and enlarged and congested spleen.
(xiii)	Pullorum Disease	Haemorrhagic enteritis, cyanosis of musculature and necrosis, hyperemia, anaemia and inflammatory changes in the organs.
(xiv)	Listeriosis	Necrosis of myocardium, pericarditis, generalised oedema, necrotic foci in the liver and spleen inflammation.
(xv)	Tuberculosis	Irregular, greyish-yellow or grayish-white nodules of different size are found in liver, spleen, intestines and bone marrow.
(xvi)	Fowl Pox	Lesions on comb, wattles and face in cutaneous form, lesions in mouth in diphtheritic form; and occasionally pox lesions on leg, feet and other parts of body.
(xvii)	Coccidiosis	Epithelial cells of intestinal tract are invaded and cause hemorrhages, petechial hemorrhages on caecal wall.
(xviii)	Aspergillosis	Miliary to large size nodules in lungs, localised hepatization in the lungs.
(xix)	Favus (white comb disease)	Minute mould like grayish-white growth on the comb, wattles around the eyes.
(xx)	Thrush	Crop mucosa is thickened with whitish, circular, raised ulcers covered with diphtheritic membrane.

Poultry Postmortem Inspection The principal purpose of post mortem inspection is to supplement ante mortem inspection in identifying diseases of public or animal health significance and monitoring animal welfare to remove meat that is unfit for human consumption.

Its purpose is to detect:

- Diseases of public health significance
- Diseases of animal health significance
- Residues or contaminants in excess of the levels allowed by legislation
- Non-compliance with microbiological criteria

- Other factors which might require the meat to be declared unfit for human consumption or restrictions to be placed on its use
- Visible lesions that are relevant to animal welfare
- Evidence of animal welfare problems such as beating or long standing untreated injuries

Additional examinations and tests

Where it is thought necessary, additional examinations are to take place such as palpation and incision of the carcass and offal and laboratory tests to:

(I) To reach a definitive diagnosis; or

(ii) To detect the presence of:

1. An animal disease,
2. Residues or contaminants in excess of the levels allowed,
3. Non-compliance with microbiological criteria,
4. Other factors that might require the meat to be declared unfit for human consumption or restrictions to be placed on its use,
5. Animal's welfare being compromised.

The body cavity of every dressed bird is opened through the transverse incision and visceral organs are drawn out. Now, the carcass is inspected externally for signs of disease, bone abnormalities, wounds, muscular atrophy, tumours etc. followed by body cavity. Liver is examined for consistency, texture, lesions and colour changes. Spleen is also palpated for texture and abnormalities. A cut is given on the hock to see synovial fluid for sinusitis.

Modernized Poultry Inspection Program (MPIP)

MPIP is a HACCP and science-based inspection system that focuses on the slaughter process within the gate to plate food safety continuum. MPIP represents the latest Canadian advance in the evolution of poultry inspection methodology.

Table 2: Postmortem significance of poultry disease

Disease	Unfit for food/condemned	Partially condemned/passed for food
Avian leucosis complex	All affected carcasses	-
Erysipelas	All affected carcasses	-
Tuberculosis	All affected carcasses	-
Ranikhet disease	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.

Infectious	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.
Infectious coryza	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.
Chronic respiratory disease	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.
Fowl typhoid	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.
Pullorum disease	Carcasses with systemic involvement	If lesions are localized, only affected parts are condemned.
Listeriosis	Carcasses with acute septicaemia	If lesions are localized, only affected parts are condemned.
Salmonellosis	Carcasses with active septicaemic lesions	In chronic cases affected parts are condemned. Rest are passed for food
Fowl pox	Carcasses with progressive lesions and systemic changes	Recovered birds may be passed for food after removal of scabs
Fungal disease	-	Only affected parts are condemned
Fowl cholera	All carcasses are condemned	-

All the condemned carcasses and parts thereof should be destroyed by chemical denaturing with crude carbolic acid any phenolic disinfectant or completely by incineration.

EXAMINATION OF LYMPH NODES AND THEIR IMPORTANCE IN MEAT INSPECTION

Lymph is the medium by which oxygen and nutritive matter are transferredfrom the blood to thebody tissues and waste products removed. Lymphnodes consist of a reticular framework of elastic and smooth muscle fibers enclosing lymphatic tissue, which contain lymphocytes. Besides acting as filters, lymph nodes manufacture lymphocytes. The size of lymph nodes varies from that of a pinhead to that of a walnut. The posterior mediastinal lymph nodes are generally round or oval and somewhat compressed; in the ruminants they are large and few in clusters while in horse nodes are small and occur in large numbers forming a grape like clusters.

The size of lymph node is relatively greater in the young growing animals than in the adult. The colour of lymph nodes shows considerable variation, and may be white, greyish, blue or black. **The mesenteric lymph nodes of the ox** are invariably **black** but in the pig the lymph nodes are lobulated and almost **white** with the exception of the head and neck, which are reddish. The largest efferent lymph-collecting vessel is the thoracic duct, which commences as a thin walled dilatation about 19 mm in width and known as the *receptaculum chyli*.

The consistency of lymph nodes varies in different parts of the body, the nodes of the abdomen being generally softer than those of the thorax.

The response of a lymph node to an irritant is normally rapid, involving enlargement, congestion and possibly tissue breakdown; thus the size, colour and consistency of lymph nodes form a valuable guide in the estimation of disease progress in the animal body.

Haemal lymph nodes

These are deep red or almost black in colour, oval and up to the size of pea, but differ from lymph nodes in their anatomical structure and in the absence of afferent and efferent lymphatic. They contain numerous white blood corpuscles with red blood corpuscles in various stages of disintegration, hence the red colouration of the nodes. **Haemal lymph nodes** are **numerous** in the **ox and sheep** but **not** found in the **horse and pig**. In cattle they occur especially along the course of the aorta and in the subcutaneous fat, while in sheep and lambs they are commonest beneath the peritoneum, the sub lumbar region, being large and more numerous in animals suffering from anaemia and cachectic condition.

Lymph nodes of cattle

S. No.	Name	Location Head and neck	Number and size
1	Submaxillary	One on each side on the inner side of mandible just inside the angle of jaw, embedded in fat	2, 1-2 cm long
2	Parotid	One on each side on the edge of masseter muscle, behind articulation of jaw covered by parotid salivary gland	2, 7.5cm long 2.5cm wide
3	Retropharyngeal	Internal retropharyngeal—between the hyoid bones; Lateral retropharyngeal-beneath each wing of the atlas and located at the neck of dressed carcass	2-4, 2 cm long
4	Middle cervical	In the middle of the neck on each side of the trachea in the jugular furrow. Often absent in cattle	1-7 1 cm long
		Chest and forequarter	
1	Prepectorals/ Lower cervical	Embedded in fat along the anterior border of the first rib at the entrance of the thorax	2-4 on each side
2	Costocervical	Inner side or just anterior side of the 1^{st} rib and close to its junction with the 1^{st} dorsal vertebra, adjacent to the oesophagus and trachea	1
3	Prescapular	This a large node situated about 10 cm. in front of the shoulder joint	1,elongated 7.5-10 cm long
4	Intercostal / Dorsocostal	In the intercostal spaces at the junction of the ribs with their vertebrae, and deep seated, being covered by the intercostal muscle	6-8
5	Subdorsal	Lies in fat between the aorta and the dorsal vertebrae	4-6. 12-25 mm in length
6	Suprasternal/ Sternocostals	Lie between the costal cartilages and covered by muscle	
7	Bronchial (Inspectors node)	Right, left and middle bronchial, found on the either side of the right and left bronchiole, irregular in shape. Right is partially hidden by the right lung and absent in 25% cases. The middle bronchial is situated in the middle line above the bifurcation of the trachea, but absent in 50% cases	3, 4 cm X 2.5cm

8	Anterior mediastinal	Numerous lying in the mediastinal space anterior to the heart, and related anatomically to the oesophagus, trachea and anterior aorta.	Numerous
9	Posterior mediastinal	Situated in the fat along the dorsal wall of the oesophagus.	8-12 in nos, up to 20 cm long
10	Axillary/ Brachial	Covered by scapula and situated in the muscle external to and about midway along the second rib.	2, 2.5cm long
11	Xiphoid/Ventral mediastinal	Found in the loose fat at the junction of the sternum and diaphragm at the level of the 6^{th} rib	2,absent in 50%
		Abdomen and hind quarter	
1	Lumbar	Situated in the fat covering the lumbar muscles and related anatomically to the aorta and posterior venacava.	4 nos
2	Portal /hepatic	These form a group around the portal vein, hepatic artery and bile duct, and covered by the pancreas	10-15 nos.
3	Renal	Found in the suet at the entrance to the kidney	Vary in size and number
4	Mesenteric	Comprise a large no of elongated nodes which lie between the peritoneal folds of the mesentery and receive lymph from intestine	10-50 and 6 to 121cm in length
5	Splenic	Absent in cattle	
6	Gastric	Numerous and difficult to group, forms a chain along the right and left longitudinal grooves of the rumen. Usually found in caul fat or attached to the stomach	Numerous
7	Iliac	Situated near the terminal branches of the aorta and embedded in fat	2
	Internal iliac	Junction of the sacrum and last lumbar vertebra	Several, 12-50mm in length
	External iliac	Situated laterally to the internal iliac, embedded in fat in the pelvis	1-2, 13-25 mm in length
8	Superficial inguinal (Male)	Lie in the mass of the fat about the neck of the scrotum and behind spermatic cord	2 nos

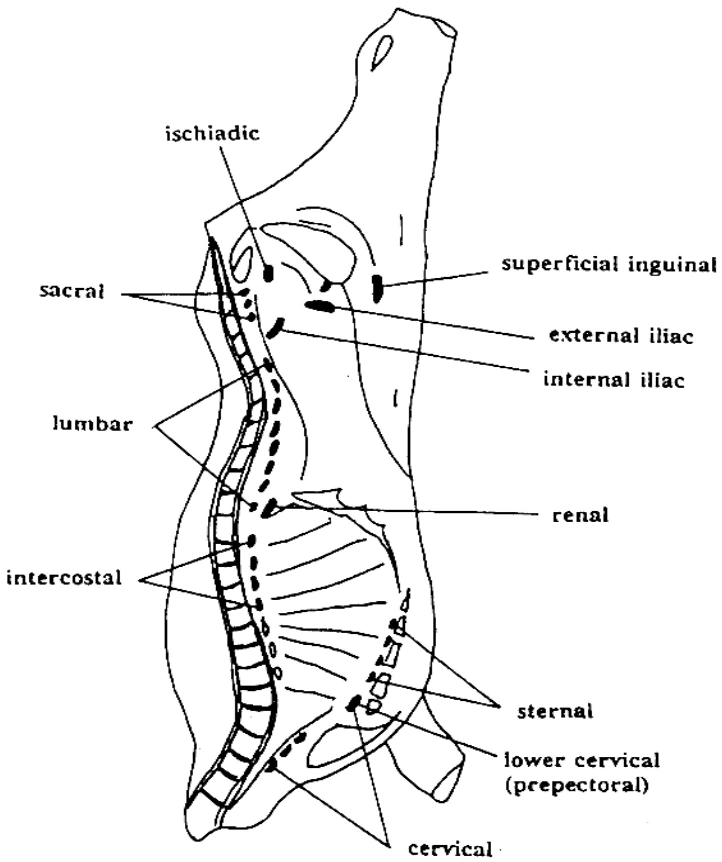
9	Supramammary	Situated above and behind the udder	2 on each side, 7.5cm
10	Deep inguinal	In the inguinal canal and frequently absent	
11	Ischiatic	Lie on the outer aspect of sacrosciatic ligament between anus and outer portion of ischium near coccygeal vertebrae	2 nos
12	Sacral	Not constantly present	
13	Precrural/ Pre-femoral	Embedded in fat and can be exposed by a incision at the edge of the tensor fascia latae muscles immediately above patella	1 no.
14	Popliteal	Deeply seated in the round of the beef in popliteal cavity posterior to the knee or stifle joints. This is usually exposed between joints embedded in fat.	2 nos.

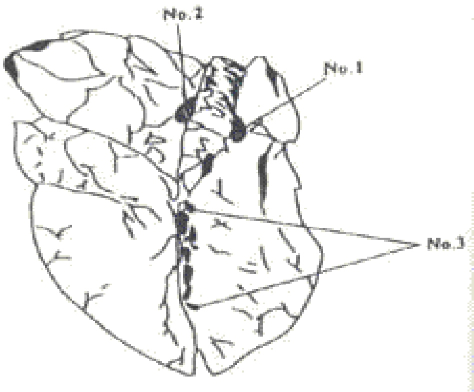
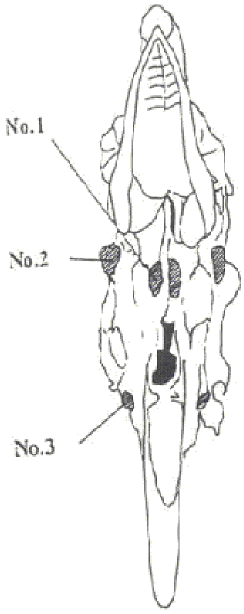
Lymph nodes of pig

S. No	**Name**	**Location**	**Number**
		Head and neck	
1	Submaxillary	Situated anterior to the sub-maxillary gland near to the angle of the jaw, and partially covered by the lower part of the parotid salivary gland	2 on each side
2	Anterior or upper cervical	Corresponding to pharyngeal node of cattle, situated a short distance behind and above the preceding nodes	2 nos.
3	Parotid	Several on each side, red in colour, one of the largest being situated just posterior to the masseter muscle of the lower jaw and partly covered by the parotid salivary gland. In order to expose, a deep incision must be made just in front and below ear	Several
4	Prescapular	Lie close to the parotid salivary gland, being partly covered by it’s posterior border	Several in chain
		Other nodes	
5	Precrural	Embedded deep in fat, about 2.5 cm. in front of the stifle joint	2 nos, 5X2.5cm

6	Popliteal	Absent in 50 % cases, but when present superficially placed on the posterior aspect of the limb about hands breadth above the tuber calcis	2 nos,
7	Gastric	Situated on the lesser curvature of the stomach	
8	Bronchial	In additional to the right and left bronchial, this group include one on the bifurcation of the trachea and another at the apical bronchus of the right lung.	4 nos.
9	Portal	Several along the portal vein, the largest about 2.5 cm.	Several

Major lymph nodes of bovines

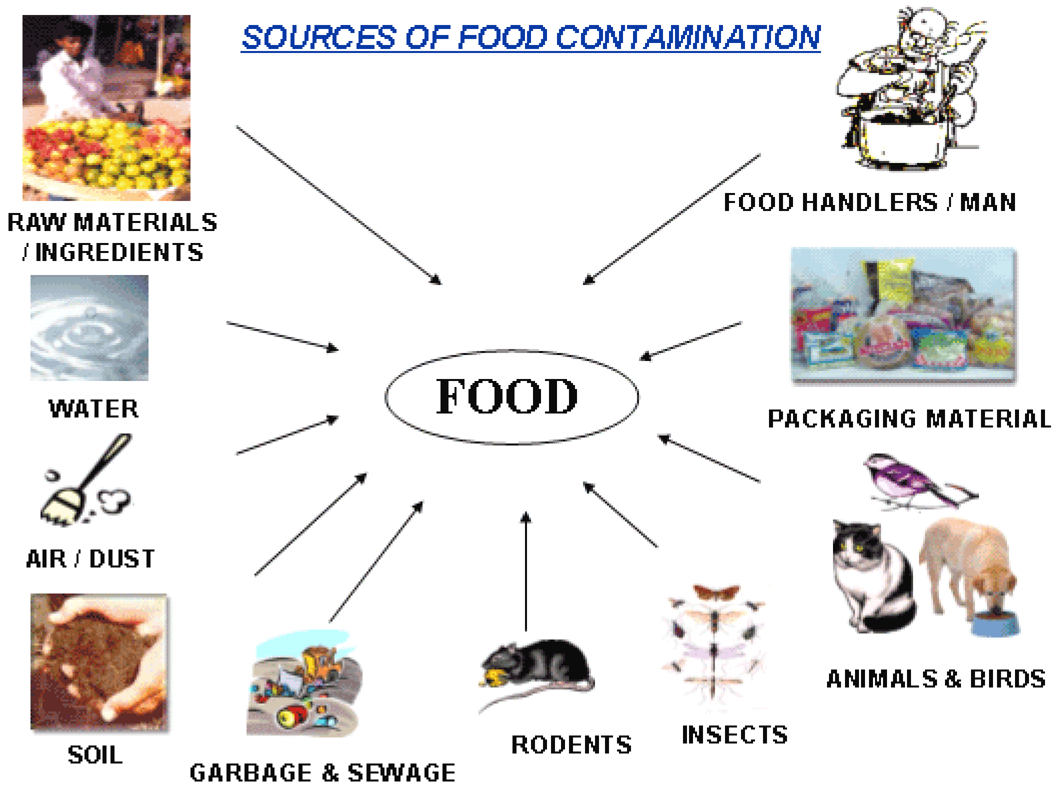


1. Right Bronchial 2. Left Bronchial 3. Mediastinal	1. Retropharyngeal 2. Parotid 3. Sub maxillary
	

Sources of contamination of meat

What are contaminants?

Contaminants- Any substance not intentionally added to food, which is present in such food as a result of the production, manufacture, processing, treatment, packaging, transport or holding of such food or as a result of environmental contamination.



MAJOR SOURCES OF CONTAMINATIONS:

Contaminant can include

- BIOLOGICAL(MICRO ORGANISM)
- CHEMICAL (Heavy metals and environmental contaminants, Residues of pesticides and other agricultural chemicals)
- PHYSICAL(PIN,HAIR etc)
- Naturally-occurring toxins

Biological Contamination

- Biological contamination is when **bacteria or toxins** contaminate food and is a common cause of food poisoning and food spoilage.
- Food poisoning can happen when harmful bacteria, also called pathogens, spread to food, and are consumed. Bacteria are small microorganisms that split and multiply very quickly.

- Some bacteria such as salmonella, staphylococcus and listeria are extremely toxic by themselves and sometimes it's not the bacteria that are toxic to humans, but the process of the bacteria multiplying and **producing waste**. However, **not all bacteria are harmful to humans**; many are quite beneficial, such as those found in yoghurt. Primary sources and routes of microorganism to fresh meats:
- 1. The stick knife
- 2. Animal hide.
- 3. Gastrointestinal tract.
- 4. Hands of handlers.
- 5. Containers
- 6.Handling and storage environment
- 7. Lymph nodes.

Chemical Contamination

- Chemical contamination occurs when food comes into contact with chemicals and can lead to chemical food poisoning.
- Chemical contaminants can be present in foods mainly as a result of the use of agrochemicals, such as residues of pesticides and veterinary drugs, contamination from environmental sources (water, air or soil pollution), cross-contamination or formation during food processing, migration from food packaging materials, presence or contamination by natural toxins or use of unapproved food additives and adulterants.
- Kitchen cleaning agents:
- Food containers made from non-safe plastics:.
- Pest control products:
- Chemicals used in equipment maintenance:

Pesticide Residues

Insecticides, fungicides or herbicides - over 1,400 known pesticides. Some of them should no longer be used but may still be present in the environment

Veterinary Drug Residues

The major classes of veterinary drugs include antibiotics, anthelmintics, coccidiostats, nonsteroidal anti-inflammatory drugs, sedatives, corticosteroids, beta-agonists and anabolic hormones.

Environmental Contaminants

Environmental contaminants can be man-made or naturally occurring substances present in air, water or soil. They can enter the food chain and even bioaccumulate.

Examples of environmental contaminants that enter the food chain include heavy metals, polychlorinated biphenyls (PCBs), "dioxins" (polychlorinated dibenzodioxins and dibenzofurans), persistent chlorinated pesticides (e.g., DDT, aldrin, dieldrin, heptachlor, mirex, chlordane) brominated flame retardants (mainly polybrominated diphenyl ethers), polyfluorinated compounds, polycyclic aromatic hydrocarbons (PAHs), perchlorate, pharmaceutical and personal care products or haloacetic acids and other water disinfection byproducts.

Food Processing Contaminants

Certain toxic or undesirable compounds can be formed in foods during their processing, such as during heating, baking, roasting, grilling, canning, hydrolysis or fermentation.

Precursors of these contaminants can occur naturally in the food matrix, such as in the case of acrylamide being formed during the Maillard reaction between the amino acid asparagine and a reducing sugar

Nitrosamines, Carcinogenic and genotoxic chlorpropanols, such as 3-monochloropropane-1,2 diol (3-MCPD) are formed during processing.

Migrants from Packaging Materials

Direct contact of foods with packaging materials can result in chemical contamination caused by migration of certain substances into foods. Examples of migrants of health concern may include bisphenol A or phthalates from plastic materials, 4-methylbenzophenone and 2-isopropylthioxanthone from inks, mineral oil from recycled fibers or semicarbazide from a foaming agent in the plastic gaskets that are used to seal metal lids to glass packaging**.**

Toxins

Toxins are naturally occurring substances that are produced by various organisms, with mycotoxins and marine biotoxins typically representing the major concerns in foods. Other examples of toxins in foods may include bacterial toxins (e.g., staphylococcal toxins) or certain plant toxins, such as pyrrolizidine alkaloids that can be found in honey, milk or eggs. While the bacterial/fungal contamination can be eliminated with heat treatment, the toxins can remain in the food product as contaminants.

Unapproved Food Additives and Adulterants

Food adulteration can happen accidentally when unapproved additives are introduced to the food, or the wrong additive is introduced through formulation error. This results in mislabeled food.

The addition of melamine to whey and other protein concentrates to increase their apparent protein content analyzed as total nitrogen. Other examples include the use of toxic Sudan dyes in adulterated chili powders or adulteration of virgin olive oil with hazelnut oil.

Physical Contamination

- **Hair:** Always wear hair neatly tied back and use a hair net if possible.
- **Glass or metal:** This can occur when kitchen items are not maintained. Cracked or broken crockery and utensils should be thrown away, as well as any food that might have come into contact with it.
- **Pests:** Pests - such as mice, rats and cockroaches - leave droppings that can contaminate food. Also, pests themselves - such as flies and insects - can also make their way into food.
- **Jewellery:** Always keep jewellery to a minimum when preparing and handling food.

- **Dirt:** Because dirt is so small, it's easy not to notice that it's contaminating your food. It usually gets into the food from unwashed food and vegetables.

  - **Fingernails:** Always keep nails short and clean to prevent contamination.

Methods of carcass decontamination

CONTAMINATION

The healthy inner flesh of meats have been reported to contain few or no organisms,. Although they have been found in lymph nodes, bone marrow, & even flesh. Staphylococci, streptococci, Clostridium, & Salmonella have been isolated from the lymph nodes of red meat animals. So normal slaughtering practices would remove the lymph nodes from edible parts. Approved "humane" methods of slaughter- mechanical, chemical, electrical- have little effect of contamination. But each method is followed by sticking & bleeding, which can introduce contamination.

The important contamination comes from external sources during bleeding, handling & processing. Sources of contamination may be Knives, cloths, air, & clothing of workers, during handling of meat, boxes, containers, air, & other contaminated meat, during bleeding, skinning & cutting main source of micro-organisms are the exterior of the animal & intestinal tract.

In the market, knives, saws, cleavers, slicers, grinders, chopping blocks, scales & containers are the sources of contamination. In Home, refrigerators previously used containers used to store meat. Growth of organisms on surface, touching the meats & on the meat themselves increases their numbers.

Important isolated micro-organisms from meat product micro-organisms isolated fresh & refrigerated meat bacteria:- Pseudomonas, Aeromonas, Micrococcus, & Alcaligenes MOLDS:- Cladosporium, Geotrichum, & Mucor YEASTS:- Candida, Torulopsis, & Rhodotorula PRECESSED & CURED MEATS BACTERIA:- Lactobacillus & other lactic acid bacteria, Bacillus. Micrococcus, & Staphylococcus MOLDS:- Pencillium, Aspergillus, Rhizpus YEASTS:- Candida, Torula, Torulopsis

Animals carry bacteria on their skins or hides and in their intestines. These bacteria such as Salmonella, E. coli or Clostridium perfringens are pathogenic and can cause illness. The transfer of bacteria from the hide or the intestine can occur during the slaughtering process. When cattle are slaughtered, their hide is removed. If great care is not taken, any faecal matter on the hide can come into contact with the flesh.

During evisceration, i.e. the removal of the offal, the stomach or intestines may burst, releasing bacteria which can come into contact with flesh. This flesh is the raw meat which is prepared for human consumption. All raw meat delivered to retail premises may be heavily contaminated with bacteria.

Where chickens or turkeys are eviscerated by the same machine or knife this can result in the transfer of bacteria such as Campylobacter and Salmonella from one infected bird to several others. Proper disinfection of equipment is therefore vital.

Antimicrobial intervention The application of an antibacterial chemical or temperature treatment to carcass surfaces in order to reduce the potential presence of enteric pathogens that may contaminate tissue surfaces during /processing. Hot water carcass treatment

Treatment of meat surfaces with hot water, typically 80–96 °C, to raise carcass surface temperatures to levels capable of destroying enteric foodborne pathogens.

Washing

(a) Water Washing:

Since bacteria are not necessarily removed by trimming of visible contamination on the affected area of the carcasses, alternative methods such as carcass washing with water or sanitizing agents are considered. Normally, carcasses are washed with water on the slaughter floor, under pressure, after weighing and before chilling to minimize physical or microbial contaminants. Several studies suggested showering or spraying the carcass or meat cuts with water as a physical method for microbial decontamination.

(b) Hot Water Washing (HWW):

Hot water washing of carcasses before chilling is an alternative to ambient water washes, and is used by about 90% of beef processing facilities (AMI, 2000). Researchers in this area found that hot water temperature/time combinations were found effective in removing bacteria without permanently impairing the appearance of the treated meat, especially color.

Steam pasteurization A process in which beef carcasses are introduced to a pressurized, closed chamber and sprayed with steam in order to raise the surface to a temperature sufficient to destroy enteric foodborne pathogens. This process is an online treatment where a controlled treatment of saturated steam is applied to surface of carcasses, instantaneously raising the surface temperature of the product to 190/F (88/C) for <u>10 sec</u>. Immediately following treatment, the carcasses or trimmings are chilled using cold water as it enters the cooler. Two steam pasteurization systems are commercially available on the market, SPS 400 and SPS 60, with the former allowing treatment of up to 400 carcasses/h while the latter can accommodate 60 carcasses/h

Steam vacuum A hand-held device consisting of a vacuum wand equipped with a steam nozzle that can deliver a stream of steam to the carcass surface while vacuuming the area.

Organic acid carcass treatment Over the years several organic acids have been used in an attempt to reduce the microbial flora of animal carcasses and subprimals. These include lactic, acetic, citric, propionic, ascorbic, formic, and peracetic acid. The treatment of meat surfaces with dilute organic acids (lactic, acetic, and citric) in n effort to dramatically reduce the surface pH and destroy enteric foodborne pathogens. Similar to knife trimming and water washing, organic acid sprays also are approved by FSIS . Acetic, lactic, and citric acid solutions at 1.5-2.5% are approved by USDA-FSIS as acceptable interventions for reducing carcass contamination. <u>Acetic and lactic acids are the most commonly used sanitizers, and both are generally recognized as safe compounds.</u>

Ozonation

Ozone (O_3) is a soluble and unstable blue gas commercially produced by passing electric charges or ionizing radiation through air or oxygen. It is a powerful compound used to inactivate bacteria through its oxidizing properties, and was the first oxidizing agent to be used for disinfection of water. Ozonation was found to be effective for the inactivation of bacteria (e.g., E. coli O157:H7, S. typhimurium, Pseudomonas fluorescens) on several muscle foods and fish.

Hydrogen Peroxide

Hydrogen peroxide (H_2O_2), is a strong oxidizing agent that can rapidly kill microorganisms if used in adequate concentrations. Although it inactivates microorganisms, H_2O_2 is not permitted for use as a food additive in many countries because it has a bleaching effect and can oxidize food constituents. Scientists studied the antimicrobial activity of H_2O_2(5%) and ozone (0.5%) as meat decontaminant, and found that these sanitizers reduce bacterial counts by more than 2.5 log10 CFU/cm2

Irradiation Pasteurization

Irradiation of food generally uses γ rays or electron beams. Antimicrobial activity of ionizing radiation is due to direct damage of DNA and the effect of generated free radicals. A 1-kGy dose of electron beam radiation reduced inoculated E. coli O157:H7 on beef carcass surface cuts, by at least four orders of magnitude without affecting sensory. Application of irradiation at adequate dosages seems also to be effective, but costs for the infrastructure and the acceptance by the consumers should also be considered.Other pasteurization technologies for reducing microorganisms on muscle foods include pulsed light or PureBright ®TM, infra-red irradiation and **ultraviolet light**. **The ultraviolet light** has poor penetration, so can be used only for sterilizing surfaces of carcasses and meat products.

Sodium Chloride (NaCl)

Sodium chloride (NaCl), is generally recognized as safe (GRAS) antimicrobial and a nonintentional food additive. Numerous studies examining NaCl as an antimicrobial agent indicated its effectiveness in reducing or inhibiting the growth of a number microorganisms. Generally, foodborne pathogenic bacteria are inhibited at 13% (w/v) NaCl (equivalent to water activity, aw, of 0.92 or less) with the exception of S. aureus. In fermented meat products, NaCl restricts the growth of spoilage microorganisms and allows growth of lactic acid bacteria. Growth of lactic acid bacteria in fermented meats inhibits proliferation of foodborne pathogens such as staphylococci. Bacteria have different levels of tolerance to salt depending on their inherent properties and intrinsic and extrinsic growth factors such as nutrients, water activity, pH, oxygen availability, and temperature. Overall, Salmonella spp. can survive in salty environment with NaCl concentration as high as 3.25.3% , C. perfringens 8.0%, L. monocytogenes 8.0-12%, C. botulinum 11-12%, and S. aureus, 1820%, respectively (Stein, 2000). The preservation mode of NaCl as an antimicrobial agent can be attributed to a number of factors including (i) dehydration, (ii) direct effect of chloride ion, (iii) removal of oxygen from the medium, (iv) sensitization of microorganisms to carbon dioxide (CO_2), and (v) interference of NaCl with the rapid action of proteolytic enzymes

Cetylpyridinium Chloride

Cetylpyridinium chloride (CPC) is a water soluble, colorless and odorless compound with a 250M6 - 4 neutral pH that has been found effective as an antimicrobial agent. This compound has been used for many years as an oral hygiene product because it reduces bacterial attachment and plaque formation on the tooth surface. Recent research studies indicated that CPC is effective against E. coli O157:H7, S. typhimurium, C. jejuni and L. monocytogenes inoculated on meat.

Acidified Sodium Chlorite (Sanova®Process)

Chlorite stabilized in acids (e.g., acidified sodium chlorite, ASC, or $NaClO_2$) has been shown to provide an excellent method of carcass decontamination resulting from a combination of antimicrobial effect due to the acid content (low pH) of the spray, and the anti-microbial properties of chlorine. ASC is formed from the reaction of sodium chlorite with citric acid, and it is a powerful oxidant. ASC solutions, including the commercial antimicrobial compounds of Sanova®, are approved by the Food and Drug Administration as a direct food additive to be used for decontamination of poultry and red meat carcasses (CFR, 1998). It altering cecal Salmonella colonization in broiler chickens, and chicken skin decontamination. ASC was found effective against S. typhimurium and E. coli O157:H7 on beef

Chlorine Washes

Chlorinated water solutions have been used as carcass rinses during the chilling process of carcasses to reduce or prevent proliferation of any remaining bacteria on the surface of carcasses. Chlorinated water at 200 ppm was found to reduce total aerobic bacteria (APC), and greater reductions were obtained after 24 h of treatment. Calcium hypochlorite ($CaCl2O2$) and chlorine dioxide ($ClO2$) also reduced contamination on beef forequarters. Unfortunately, the rapid inactivation of chlorine in an organic system such as meat is a disadvantage for this treatment**.**

Trisodium Phosphate (TSP)

TSP application as an antimicrobial treatment for beef carcasses before chilling (24 to 48 h before fabrication). The ability of TSP to inactivate pathogenic and nonpathogenic bacteria, and to minimize microorganisms from adhering to carcasses or meat trimmings has been extensively studied. A 1.6-1.8 log10 reduction of Salmonella per carcass was reported when post-chilled chicken were dipped in 10% TSP at 50/C for 15 sec. TSP reduced the population of artificially inoculated pathogens on lean and adipose tissue with greater reductions observed in gram negative (e.g., E. coli and Salmonella spp.) than gram positive (e.g., L. monocytogenes) microorganisms.

Lactates

Sodium and potassium lactates that normally are used as humectants or flavor enhancers in meat and poultry products, have been reported as an effective agents in controlling growth of aerobes and anaerobes in meats, and in antibotulinal and antilisterial activities. Scientists found that sodium lactate delayed Clostridium botulinum toxin production in cook-in-bag turkey products; sodium, potassium, and calcium lactates exerted antilisterial activity, and the effect of calcium salt was greater than sodium or potassium salt.

Packing Innovations

These technologies are applied to packed meat only, and are more applicable to meat trimmings than to whole carcasses. These technologies include oxygen (O2) adsorption method, modified atmosphere packaging (MAP), and vacuum technology. Oxygen adsorption involves including an O2 scavenger (e.g., potassium permanganate) in a small pouch within the meat package where the O2 binds with scavenger instead of the meat. This extends the shelf life of meat products as the unavailability of O2 prevents the meat color from turning into brown, and renders proliferation conditions difficult for aerobic microorganisms. When the MAP technology is used, the red meat in a barrier package is flushed with an O2 (80%) and carbon dioxide (CO2; 20%) mixture. Oxygen in the package allows the meat to remain red in color while CO2 delays microbial growth. Thus, by modifying the atmosphere in the package, the shelf life of red meat cuts can be extended for an additional 7-12 days. Finally, vacuum technology uses a high barrier packaging material and removes the atmosphere within the package. This extends the shelf life of the product to 21 days or more; however, the meat does not bloom or turn red inside the vacuum package. Additional packaging innovations under progress include development of biosensors that detect chemicals, indicators of decomposition, and packaging that can detect changes in temperature.

Physical decontamination treatments

Cold/warm water washing

Carcass rinse with water before evisceration or after hide removal has been suggested to decrease the carcass surface's microbial load by reducing the bacteria's ability to attach on meat surface Furthermore, washing at the end of the slaughtering process and before chilling, is used as a decontamination method .Under commercial conditions, cold and warm water spraying using wash cabinets reduced aerobic bacteria, coliforms and E. coli on beef carcasses by 0.5-1.0 orders of magnitude . Carcasses are washed with cold (10-15ºC) or warm (15-40 ºC) potable water to remove bone dust and blood clots. As, in many cases, washing simply redistributed bacteria from one area to another area therefore, water spray direction, temperature and pressure are critical factors that should be taken under consideration. Use of cold or warm water is less effective than hot water as it only removes bacteria, while hot water application leads to their injury or death. Therefore, washing with cold or warm water is not considered to be a decontamination step as its effects are related solely to improving carcass appearance and not food safety .

Hot water washing

Water at 75 to 85ºC may be applied for 10 to 15 s to the carcass under pressure (9.7-13 Pa) as a spray or using a deluge system which delivers sheets of water at 85ºC for 10 s onto the carcass reduce bacterial contamination of beef carcass tissue

Combination of hot water at the temperature of 70ºC and with low pressure (1.36 atm) and warm water application (30ºC) with high pressure show the best results. Hot water is easier and more economical to generate than steam. However, a lot of water is wasted and heat may discolour cut surfaces of the carcasses

Chemical decontamination treatments

Organic acid application Organic acids, such as lactic or acetic acid, are usually applied using a spray cabinet. Organic acids are widely used in USA, but have not been permitted under EU Regulations until now. Lactic acid solution 1% application on beef carcasses at the temperature of 55 oC reduced microbial population to levels similar to that described for decontamination techniques. Applications of 3% acetic acid solutions were reported to have similar results. Other researchers claim that there is no clear evidence that organic acid application has a significant lethal effect on its own. Acid kills some cells and damages many others on meat surface, while the carcasses are **discoloured** and operators may experience **skin/eyes irritation when acetic acid** is used.

Chlorine dioxide at concentrations of up to 20 ppm was found to be no more effective than water in reducing bacteria of fecal origin on beef carcass tissue.

200 ppm sodium hypochlorite

Phosphate buffer, ethanol, sodium chloride, sodium hydroxide, and potassium hydroxide. Phos- phate buffer, ethanol, and sodium chloride produced reductions by less than 1 log cycle, whereas sodium and potassium hydroxides effectively reduced the populations of the inoculated bacteria by as much as 4 log cycles.

Acidified sodium chlorite (ASC) solutions have also been evaluated for their ability to reduce the presence of E. coli O157:H7 and S. typhimurium on beef carcass surfaces different types of meat surfaces

composition of the surface microflora

Concerns about decontamination

- Toxicity of compounds?
- May lead to poor slaughterhygiene?
- Discolouration of meat
- Excessive water consumption (Hot Water Decontamination).
- Consumer acceptance
- May not be installed in small slaughterhouses (Practical/economical reasons)
- Small slaughterhouses – significant suppliers of fresh meat in DK
- Unknown effect on human ilnesses

Food-borne diseases remain responsible for high levels of morbidity and mortality in the general population but particularly for at-risk-groups such as infants, young children, pregnant women, elderly or immunocompromised people (http://www.who.int). There is increasing interest in effective decontamination treatments because healthy food- producing animals can harbor food-borne pathogens and complete prevention of contamination during slaughter can hardly be warranted.

SPECIATION OF MEAT

Fraudulent substitution is a malpractice in meat industry, in which inferior or cheaper quality meat is mixed into superior quality meat. These practices are not new but existed since long back as the first case of fraudulent substitution was recorded in thirteenth century A.D. at Florence in Italy . A common fraud in the meat industry in which uninspected meat is substituted for meat that has undergone inspection and been branded as satisfactory. The other frauds are the substitution of meat of another species, i.e., horse for beef especially in Britain and Ireland, beef in kangaroo meat in Australia, cat for chicken or rabbit, goat meat for mutton, mutton for venison, dog meat and cat meat for chevon in other countries including India These practices are more common in comminuted meat products. For detection of meat species in adulterated meat have several techniques started from simple physical tests to recent sophisticated molecular techniques.

Identification methods

- Physical method
- Anatomical techniques
- Histological techniques
- Chemical techniques
- Electrophoretic techniques
- Immunological techniques
- Chromatographic techniques
- PCR based techniques

I. Physical method This is based on general appearance, colour, texture, odour and tenderness of different species of meat. Besides the general characteristics of body fat, its colour, marbling, firmness of fat can be identified. Dentition formula, vertebral formula and articulation pattern, rib number and degree of curvature, characteristics of long bones will help to identify the species if the carcass is intact.

Kind of meat	Colour of meat	Colour of fat	Consistency of fat
Beef	Cherry or bright red	Yellow or pale tallow	Fairly firm
Cara beef	Brick red	Pure white	Firm
Camel meat	Red	Pure yellow	Fairly firm
Mutton / lamb	Light red	Very white	Very firm and flakier
Chevon	Brick red	Very white	Very firm and flakier
Venison	Light red	Greyish white	Fairly firm
Pork	Whitish grey to red	White or yellow	Very soft
Horse meat	Dark red and sticky	Yellowish white	Soft and smears
Dog meat	Red, soft and smeary	White to whitish grey	Oily and greasy

II. Anatomical techniques

- Dental formula
- Bone percentage

- Rib number and degree of curvature
- Characteristics of long bone
    - Female –udder and gracilismuscle,larger pelvic cavity
    - Male –bulbocavernosus muscle

Differentiate between sheep and goat carcass

Features	Sheep	Goat
Back and withers	Round and well fleshed	Sharp, little fleshed back n withers
Thorax	Barrel shaped	Flattened laterally
Tail	Fairly broad	Thin
Radius	1.25 times length of metacarpus	Twice as long as metacarpus radius
Scapula	Short an broad, superior spine, bent back and thickened	Possess distinct neck. Spine straight and narrow, lateral border thin and sharp
Sacrum	Lateral borders thickened in the form of rolls	Sharp
Flesh	Pale red and fine in texture	Dark red and coarse with goatyodour. Sticky subcutaneous tissue, which may have adherent goat hairs.

Differentiation of beef and carabeef-physical and chemical method

Characteristics	Beef	carabeef
Colour	Dark red with slight brownish tinge	Brick red
Meat consistency	Firm and cut surface are shiny	Firm
Marbling	Present	Absent or poorly present
Fat colour	yellowish to white	Pure white
Consistency	Firm	Slightly firm
Location of fat	Intramuscular	Mostly inter muscular
Carotene content	present	absent

III.Histological techniques

- Muscle fibre diameter
- Density and pattern of muscle fibres

- Diameter and number of muscle fibres determined by a fibre optic microscope.
- Buffalo >ox
- Size of Muscle fibre decrease –pig,buffalo,sheep,goat and poultry

Disadvantages of physical, anatomical and histological methods

- Samples may not be of full size to give interpretation.
- Reliable only in unprocessed raw meats.
- No use in ground meats
- Affected by age, sex, plane of nutrition etc.
- Difficult to interpret in adulterated samples

IV. Chemical method

Horse fat contain 1-2% linoleic acid. Linoleic acid content in other animals' fat is not more than 0.1%. Thus adulteration of lard or beef and mutton fat with horse fat can be identified by estimation of the linoleic acid concentration.

Iodine Value: Estimation of iodine value is a valuable test for the detection of horse fat.

Iodine value is the amount of iodine absorbed by the unsaturated fatty acid present in the fat. Good lard has an iodine value of 66. The iodine value of the fat from various food animals are: Horse - 71-86 Ox (cattle) - 38-46 Sheep - 35-46 Pig - 50-70

Refractive index: Refractive index is an another valuable test for the detection of fat of different animal species. Fat is liquefied by heat and converted into oil for estimation of refractive index. All liquids including oils possess a specific refractive index. Horse - 53.5 Ox - less than 40 Pig - not above 51.9

Melting Point: The melting point of fat varies with the species of food animals and the kind of feed fed to the animal. External fat Kidney fat Beef tallow

Myoglobin content: The myoglobin content of different species are:

- Rabbit - 0.02
- Broiler chicken - 0.02
- Sheep - 0.25
- Pig - 0.06
- Cattle - 0.50
- Buffalo - 0.60
- **Horse-0.77**

Other chemical tests for differentiation of different meat species are:

- Estimation of muscle nitrogen fraction
- Glycogen content
- Muscle enzymes
- Composition of fat
- Carotene content
- Fatty acid composition

Disadvantages of chemical methods:

Chemical methods are of little use in mixed and processed meats due to chemical changes of the constituents

V. Electrophoretic method

In electrophoresis, the migration of the protein moiety according to their molecular weight under the influence of electric field principle is applied. The band patterns produced in this technique is them visualized for result interpretation.

- Agar gel electrophoresis
- Starch gel electrophoresis
- Polyacrylamide gel electrophoresis
- Sodium dodecyl sulfate polyacrylamide gel electrophoresis
- Iso-electro focussing

Agar gel electrophoresis (AGID)

Electrophoretic separation of proteins on the basis of mobility of proteins in a supporting gel of agarose at a constant pH and electrical field is termed as Agar gel electrophoresis.

Polyacrylamide gel electrophoresis (PAGE)

Polyacrylamide gels are prepared using acrylamide and bis which provides cross linking between the polymerized long chains of acrylamide. The separation of proteins on the basis of their mobility in a polyacrylamide gel is comparatively better since the resolution of different proteins are optimum. Polyacrylamide gel electrophoresis also requires a constant pH and electrical field to provide high resolution of protein**.**

Iso-electric focussing (IEF)

It is an electrophoretic technique which utilizes the charge at the surface of the protein to derive it through a gradient gel. The pH gradient is setup by polyacrylamide compounds, the ampholytes. The proteins applied on to the gel reach a point where the surface charges becomes neutral at their iso electric point

Disadvantages:

- Electrophoretic techniques presuppose that the protein composition of meat is similar within the species and has differences between them.
- Electrophoretic profile is influenced by several factors
- Results are inadequate and ambiguous in processed and adulterated meat sample

VI. Immunological method

These techniques are based on the principle of antigen-antibody reaction

- Tube precipitin test

- Haem agglutination test (HAT)
- Complement fixation test (CFT)
- Agar gelimmuno diffusion test (AGID)
- Dry disc immuno diffusion test
- Enzyme linked immuno sorbent assay (ELISA)

Agar Gel Immuno Diffusion Test (AGID)

Based on a simple double diffusion method (Ouchterlony, 1948). Species specific antiserum (antibody) and unidentified meat extract (antigen) are allowed to diffuse towards one another in an agar gel slab. If the antigen and antibody are homologous a precipitin band is formed along the line where the two meet.

Enzyme Linked Immuno Sorbent Assay (ELISA)

The technique involves the application of species specific antibodies to the proteins (antigen) coated on the plastic surface of micro-titre plate. The recognition of antigen results in the formation of antigen-antibody complex, which are detected by either enzyme linked immunoglobulin or protein A (antibody detector) producing visible colour reaction with added substrate. The colour intensity is measured objectively at a specific wave length in a micro ELISA reader as absorbance value.

Counter Immuno Electrophoresis (CIE)

The principle of this technique is very similar to agar gel immunodiffusion. The diffusion of antigen and antibody is facilitated by the application of a low voltage current. Further, the endosmosis created by the agar gel changes the charge of γ-globulins and moves them towards cathode. The meat proteins moving towards anode provide immuno precipitate with homologous γ-globulin at the point of equivalence.

Disadvantages:

Cumbersome process of isolating species specific proteins hampers the effectiveness of ELISA and other immunological tests.

VII. Molecular Techniques

Cation exchange chromatography: A simple sample preparation procedure consisting of an extraction step with Milli-Q water as extraction solvent for hemoglobin's from meat samples is followed. The filtration is done with a cellulose acetate filter. Then cation exchange chromatographic separation is done and diode array detection obtained. Then different peak patterns for extracted hemoglobin's of different species may be obtained. Other heme-group containing proteins like myoglobin or cytochrome C which may also be detected with diode array detection at 416 nm. These may be chromatographically separated from the hemoglobins. By the use of these characteristic peak patterns, the species of the meat can be specified.

DNA based molecular techniques: DNA is molecule of choice for species specifications due to its stability during heating and processing. DNA molecular based species specification is possible in the foods obtained from identification rendered meat products, genetically modified foods etc

DNA hybridization technique: It is a qualitative or semi-quantitative technique of meat species speciation in which species-specific DNA sequence is detected. For this purpose probes prepared from DNA or cloned DNA are hybridized with target DNA and detected by colour development or audiography.

Polymerase Chain Reaction (PCR): PCR is a rapid method because in this technique we can obtain multiple copies of specific piece of DNA sequence *in vitro* and it has high degree of selectivity and sensitivity. PCR amplifies a target DNA sequence in an exponential phase, being capable of detecting even a single copy sequence from a single cell sample. It is a qualitative test for meat species specification. There are two main techniques for amplification of genetic marker, i.e., mono-locus-specific primers for amplification of a concrete DNA fragment and multi-locus amplification of non-targeted DNA. By this technique closely related meat species can be identified with the discrimination between male and female raw meat.

Animal Welfare and Public Health Issues

What is animal welfare?

Animal welfare science is the science of animal suffering and animal satisfaction.
One way of evaluating different causes of suffering is to measure the animal's stress responses. A *stress response* is a physiological reaction in an animal to threatening or harmful situations. *Distress* is the emotional state that is created by the threatening or harmful situations.
There are three *reasons for being concerned about animal welfare*:
• respect for animals and a sense of fair play;
• poor welfare can lead to poor product quality;
• risk of loss of market share for products which acquire a poor welfare image.

Five freedom all the animal deserve from the Animal welfare point of view are

- freedom from thirst, hunger and malnutrition;
- the provision of appropriate comfort and shelter;
- the prevention or rapid diagnosis and treatment of injury, disease or infestation with parasites;
- freedom from distress;
- the ability to display normal patterns of behaviour.

Some examples of genetic issues which affect meat production traits and animal welfare are:

- dystocia, conformation and body size in particular cattle breeds;
- exercise stress disorders and muscularity in double-muscled cattle;
- osteochondrosis and growth rate in pigs;
- stress-induced deaths and muscularity in particular pig breeds;
- PSE meat and muscularity in particular breeds or strains of pig and
- turkey;
- leg disorders, lameness, conformation and growth rate in poultry;
- green muscle disease and muscularity in turkeys and chickens;
- ascites and genetic selection for breast meat yield and growth rate in chickens.

Poor animal welfare has effect on the in the fresh meat or carcass as given below
• abnormal meat colour;
• pale soft exudative (PSE) meat in pork and turkey;
• dark firm dry (DFD) meat in pork, beef and lamb;
• poor shelf life;
• dry meat;

• heat shortening in poultry;
• bruising;
• torn skin;
• broken bones.
In some situations poor welfare may also aggravate problems with:
• gaping in meat;
• boar taint.

In many countries due to poor animal welfare practices the consumers are showing concerns, there are protests by the organizations like PETA, RED CROSS, SPCA. The consumers are turning in to vegetarian food habits. The meat trade is also affected. Some EU countries have banned import of animal products form developing countries following poor animal welfare.

Animal welfare is also linked with the public health due to spread of diseases, occupational hazards to animal workers and veterinarians, environmental pollution due to improper disposal of waste.
Environmental hazards, stagnation of drug development, climate change, human population growth, emerging infectious diseases (EIDs), world hunger, and violence are among our most urgent public health concerns. The health and feeding care of animals , following the withdrawal periods is important from the meat quality point of view and transmission of diseases to humans.
Animal welfare laws in India

- The Constitution of India
  - The DPSP – Part IV – Art. 48 & 48A
  - The Fundamental duties – Part IVA – Art. 51A (g)
  - "It shall be the duty of every citizen of India … to protect and improve the natural environment including forests, lakes, rivers and wildlife, and to have compassion for living creatures."
- The Indian Penal Code, 1860
  - S. 47 – Definition of "animal"
  - S.289- Negligent conduct with respect to an animal
  - S.428- Mischief by killing or maiming animal of the value of ten rupees.
  - S. 429 – Mischief by killing or maiming cattle etc of any valueor any animal of the value of fifty rupees-up to five years.
- The Criminal Procedure Code, 1973
- The Prevention of Cruelty to Animals Act, 1960
- The Wildlife Protection Act, 1972
- The Police Acts
- The Municipal Corporation Acts.

OCCUPATIONAL HEALTH HAZARDS IN MEAT SECTOR

OCCUPATIONAL HAZARDS

Occupational hazards can arise in four main areas of the work environment. These are physical, chemical, biological and ergonomic factors.

1. Physical factors

Problems can arise from noise, vibration, various types of radiation, heat, cold and extremes of pressure**.**

2. Chemical factors

This may be in the form of dusts, fumes, fibres, liquids, mists, gases or vapours. Most chemicals are absorbed in to the body by being inhaled in to the lungs. They may also be absorbed through the skin or taken in by mouth.

3. Biological factors

These will range from insects through to yeasts, fungi, bacteria and viruses. Zoonoses are diseases and infections which are transmitted between vertebrate animals and humans.

4. Ergonomic factors

This will cover areas such as posture, movement, work pressure, repetitive actions, psychological stress and illumination and visibility.

The nature of the materials used; products and by products involved; possible points of release or emission of hazardous agents; posture and movements of the employee; hours worked and the frequency and duration of rest periods; the type of protective equipment provided.

Back disorders

Back disorders are the main cause of injury-related time off work and persistent disability in the meat industry. Some situations are particularly conducive to back injuries eg pelters backing off, changing legs on the mutton chain. In some workplaces' there may be a specific design problem eg a situation where there has to be heavy, awkward lifting that could be overcome by modification to the workplace.

Prevention

Lifting education programmes to minimise the risk of injury .Providing advice on worksite modification, work practice alteration

Burns

Burns are a common source of injury to meat workers. Hot water and steam are used extensively in cleaning and sterilisation. The water in the sterilising units is at 82.Z°C which is sufficient to cause burns. One of the more frequent sources of burns is the use of hot water hoses. These are sometimes misused in an attempt to clean waterproof boots. If trousers are tucked in to boots there is an opportunity for hot water to get in to the boots causing extensive burning. Besides hygiene considerations (meat and fat getting in to boots), this is another good reason not to tuck trousers in to boots. The physical properties of steam mean that it is more hazardous than boiling water and it can easily cause substantial burns. Electrical burns can often be deceiving with significant tissue damage underneath skin that may only exhibit slight burning or damage.

Prevention

By using a 'cold pack' or by immersing burns immediately under cold water (running water from a cold water tap is quite adequate) for at least five minutes, the amount of blistering and inflammation will often be greatly reduced. Burns readily become infected and need cleaning and dressing.

Hypothermia

Export standards require boning room temperatures to be 10°C. Temperatures as low as -30°C may be experienced in blast freezing conditions while chillers operate between 3°C and 5°C. Cold storage requires a minimum o f -20°C. Problems related to these low temperatures are rare but there is a potential for chilblains, frostbite and hypothermia. Chilblains are the result of a disordered version of the normal reaction for blood vessels to constrict when exposed to cold. Frostbite is the freezing of tissue as a result of exposure to extreme cold. Cheeks, nose and ears are the commonest areas to be affected

Prevention

The most effective treatment iS early recognition, removal from cold and rewarming The clothing specified by export standards for boning room workers, People are usually in freezers for only brief periods of time and management will often rotate these people among other jobs.

Hyperthermia

These are conditions where the person's temperature is raised significantly above normal due to a combination of exertion, high environmental temperatures and failure off the body's cooling mechanisms. This can occur with a hot, humid environment; wearing waterproof or heavy clothing or with inadequate fluid intake.

Prevention

The first aid treatment is to reduce temperature (by sponging or bathing in cool water) and correcting the dehydration (by drinking or using intravenous fluids).

Cuts

Cuts are one of the commonest injuries in the meat industry. Most cuts occur from overcutting leading to injury to the non-knife hand, or, where the cutting is towards the person, in to the abdomen or leg. The 'run-through' laceration can occur when the knife hand slips down the handle onto the blade of the knife. Lacerations may also occur when passing a knife from hand to hand or attempting to grab a dropped knife.

Prevention

Knives should be designed with a hygienic non-slip handle and a guard to stop the hand slipping on to the blade. Adequate training will teach methods that are both safe and efficient. Emphasis needs to be given to knife-sharpening skills as inability to sharpen a knife properly iS a common cause of cuts. If a knife is not sharpened adequately, workers tend to slash or saw, instead of cutting cleanly and easily. Steel mesh gloves and aprons will have a significant effect on the incidence of lacerations.

Skin disorders

The commonest skin disorders are cuts and abrasions. Many of the other skin diseases are caused by chemicals. The commonest form of inflammation of the skin is an irritant dermatitis caused by agents such as solvents and hand cleansers which produce problems by direct action at the site of contact. Some chemicals may cause an allergic dermatitis. Such chemicals are rarely used in the meat industry.

Prevention

Occupational skin disease will be reduced by preventing dangerous substances coming in to contact with the skin. High standards of cleanliness (to remove potentially hazardous substances) should be encouraged by the ready availability of good washroom facilities and protective clothing. High standards of housekeeping should be maintained to limit exposure. Barrier creams can be of use as a form of protection.

Overuse Injury

Overuse injuries can be a significant problem in the meat industry. The term covers a group of preventable muscle and tendon disorders, characterised by recurrent or persistent pain syndromes of the back, neck, shoulders or limbs. The problems can be caused by repetitive actions, awkward posture or non-specific factors.

Prevention

Initiate changes to the ways of working. Avoidance of the activity that caused the problem. Gentle muscle strengthening and stretching exercises.

Noise

Within the abattoirs one of the noisiest areas is usually around the backbone splitting saw.

Prevention

Hearing protection to be most effective requires.

Ammonia

Anhydrous ammonia is wider used as refrigerant in industrial facilities such as meat, poultry and fish processing unit.

Precaution to be taken against leakage of ammonia

- Vacate the affected area
- Cover your nose and mouth with wet hand kerchief
- Breath normally trough the wet hand kerchief
- Try and reach the open ground away from the place of leakage
- Move against the direction of the wind

- If required to reach the place of leakage wear oxygen mask
- Start sprinkling water over the place from where the ammonia is leakage
- Close all the valves
- Open all the doors and window and start the exhaust fans
- Report the matter to the fire bridge immediately
- Inform hospital for ambulance
- Keep the first aid equipment ready for any eventually.

Contact with the eyes

Even if only a small amount of ammonia enters the eyes, irrigate the eyes with an abundance of water for a minimum of 15 minutes. Continually and thoroughly flush the entire eye surface and the inner lining of the eyelids. Eyes affected by ammonia close involuntarily, so the eyelids must be held open so that water can flush the entire eye surface, as well as the inner lining of the eyelid.If there is no physician available, continue irrigation for an additional 15 minutes.

Do not wear contact lenses when handling anhydrous ammonia. If ammonia gets in the eyes, the ammonia will get trapped under the lenses causing even more damage. They may also prevent immediate flushing of the eye surface.

Serious eye injury should be treated by an ophthalmologist, but in an emergency, wash with large quantities of water for 15 minutes or more as quickly as possible. In fact, the only real hope for preventing permanent eye injury lies in quick and generous washing.

One suggestion for those likely to be exposed is to carry a small, eight-ounce squeezable squirt bottle filled with water, which can be used to get excess ammonia out of the eyes until a larger water supply can be reached. This small amount of water is not sufficient to remove all the ammonia. It is essential that the eyes be irrigated for a minimum of 15 minutes as soon as possible.

Another emergency method is to duck the head in water and rapidly blink AND move or rotate the eyes about.

Contact with the skin

It is essential that any ammonia spilled on the worker be removed immediately and that the worker be moved to an uncontaminated area quickly. Clothes that have been saturated by liquid ammonia may freeze to the skin. In any case, the victim, still clothed, should get immediately under a shower, if available, or jump into a stock tank, pond, or into any other source of water. Time is important! Remove clothes only after they are thawed and they can be freely removed from frozen areas. If the clothing is removed incorrectly, whole sections of skin can be torn off. No salves, creams, ointments, or jellies should be applied to the skin during a 24-hour period following the injury since this will prevent natural elimination of the ammonia from the skin. After the 24 hour period, the medical treatment is the same for thermal burns. A physician should view any second- or third-degree freeze burns of the skin.

Taken internally

Take the exposed workers at once to a clean, uncontaminated area. Watch workers exposed to low concentrations for a short period of time. They will usually require no treatment and can be released.

For severe exposure to higher concentrations:

- Administer oxygen by an individual who is trained and authorized to do so by a physician. This will help relieve pain and symptoms of lack of oxygen.
- Begin artificial respiration immediately if the patient is not breathing.
- Keep victim warm (but not hot) and rested until transported to the hospital.

Electric shock

Use of common sense can help reduce electrical injury. People who work with electricity should always check that the power is off before working on electrical systems. Avoid use of any electrical device near water. Be careful of standing in water or when working with electricity.

Do not touch a person who is still in contact with electric wire! This may cause your death as well as his. The person should be removed from electric contact as quickly as possible. This may be accomplished by cutting off the current that is going to the patient or by disconnecting the patient from the wire contact by use of a dry stick or rope that is thrown around him. An axe may be available to cut the wire that is causing the contact with the patient. When using an axe, be sure that your hands are dry and that the wood handle of axe is dry.

- ✓ Artificial respiration should be instituted as soon as possible.
- ✓ The patient should be kept quiet and warm and supplied with oxygen if this is available.
- ✓ The burn area, which is often present at the site of contact, must be treated in the same manner as any burn.
- ✓ Place the patient on his back with his feet higher than his head.
- ✓ If there is any active bleeding contributing toward the shock, it should be stopped.
- ✓ The patient should kept warm. Supply him with adequate belts or other covering.
- ✓ If there is severe pain that can be relieved by the first–aider, this should be done immediately.
- ✓ Pain is one of the strongest contributors toward the development of shock. If a fracture is present, it should be splinted.
- ✓ If it can be determined that there is no injury or wound to the abdomen, the patient may be given warm fluids to drink.

Removal from contact

If the person is still in contact with the apparatus that has given him shock. The resquer should if possible stand on a dry wooden chair while removing the victim. Otherwise pull him free by using a dry coat, dry rope coconut malting of stick, preferably standing on rubber mat or any other dry mat handy.

Prelimary step

Extinguish any sparks if the patients clothes smouldering, ascertain if he is breathing, and send for a doctor. If apparently not breathing, proceed as follow

To recover patient if there are any burns, avoid if possible, so placing the patient as to bring pressure on them. It is far preference to operate as in face downward. If badly downward. If badly burnt in front turn to the second method shown later

First motion: expiration kneel over the patient, rest the hands flat in the small of his back, let your thumbs nearly touch, spread your fingers on each side over his lower ribs. Now lean firmly but gently forward over patient exerting a steady pressure downward

Second motion: inspiration rock yourself gently backward and do not remove your hands merely keep them position for expiration pressure. Continue these two movements.

The double movement continue through about fifteen times per minute. The object is to keep expending and contracting the patient lung so as to initiate slow breathing. If the operator himself also breathes slowly letting the air out as he press forward and drawing it in as he rock backward. He will naturally arrive at the proper rate and understand the reason of the movements.

Do not cease operation until natural breathing is re-established it may take half an hour or even longer to produce any effect.

Alternative method

Should be expedient to place the patient on his back, first loosen the clothes around the chest and stomach. Then place a rolled-up coat or the improvised pillow, beneath the shoulder so that the head fall backwards. The tongue should then be drawn.

First motion: the operator must kneel in the position clasp the patient just below the elbows, and draw his arm over his head until horizontal, retaining them therefore about two seconds.

Second motion: next bring the patient arm down on each side of his chest and pressing inwards upon it leaning upon his arms so as to compress his chest.

Retain thus for two seconds and keep repeating the two motion at the same time. The lung inflating effect is assisted if the arms be swung outwards as they are lifted.If more than one person can be present, the patient tongue should also be drawn out each outward or lung – inflating stroke and released during each inward or lung-deflating stroke.

Be careful to avoid violent operation, as injury of the internal organ may result from excessive and sudden pressure.

Upon recovery: - burnt if serious should be treated with proper dressing. Avoid exposing patient to cold

Zoonoses

Zoonoses are those diseases and infections which are naturally transmitted between vertebrate animals and man. Zoonoses affect the meat sector is given below;

Disease	Causative agent	Mode of transmission	Prevention
Brucellosis	Cattle :*Br. abortus* Goat and sheep:*Br.melitensis* Pig :*Br. suis*	Airborne infection from the inhalation of contaminated aerosolS iS probably a major source o f infection.	The key to prevention is the identification of affected stock by blood testing, followed by slaughtering using

		Contact with infected tissues, blood, urine, vaginal discharges and placentas. Ingestion o f raw milk and dairy products iS a possible source	adequate safeguards. Care taken in the handling and disposal of the placenta, discharges and foetus**.** Ensure that aerosol production of potentially contaminated matter does not occur.
Leptospirosis	*L. hardjo L. pomona L.icterohaemorrhagiae, L.grippotyphosa and L. canicola.* These will affect humans and may be carried by animals such as cattle, sheep, horses, dogs, cats, rats and other rodents.	It is transmitted through the skin, especially if it is cut or abraded, after contact with infected urine or contaminated water, moist soil or vegetation.	Animals can be immunised but it must be with vaccines containing the particular local strains. Employees need to be protected with boots and gloves where practicable. Even minor cuts and abrasions need to be protected with waterproof dressings.
Q fever	*Coxiella burnetii* -cattle, sheep and goats are common sources of infection t o man	Placenta and birth fluids, the urine and the faeces (which is probably from eating contaminated feed).	The vaccination of susceptible individuals is a safe and effective method of prevention. There needs to be a high standard of hygiene in abattoirs where there may be infective material and processes should be developed to minimise aerosol or dust production.
Anthrax	Anthrax is caused by a bacterium, *Bacillus anthracis*.	workers who handle hides, hair (especially from goats), bone and bone products and wool.	Efficient ante-mortem inspection. Dispose the affected animal

Ringworm (Dermatophytoses)	Dermatophytes which live in the dead superficial layers o f the skin. horns, hooves and feathers	The infection will be caught from contact with the fungal spores, which may be on 'animals' coats or contaminating floors or clothing. Handling skins and hides is a common cause of skin infections.	People need to avoid touching obvious skin lesions. Good personal hygiene will also decrease the opportunity for infection. Floors in dressing rooms' need to be frequently cleaned with an agent containing a fungicide and showers need to be rapidly draining to minimise wet, infective surfaces. The use of a footpowder containing an effective fungicide will be helpful in reducing foot infections

Skin infection Swine Erysipelas (bone scratch fever)	The most commonly involved bacteria are *Staphylococci* and *Streptococci* Group A Streptococci,	Contact with infective material	-Avoiding contact with infective material -Wearing waterproof gloves will keep the skin dry Washing the hands with soap and hot water will remove infective organisms -Prompt cleaning and dressing (with a waterproof bandage) of even minor cuts will minimise the risk of infection.
Erysipeloid	Erysipelothrix rhusiopathiae. The disease is transmitted from pigs, lambs, turkeys and fish.	People are usually infected via cuts or abraded skin from infected material.	
Orf or Scabby mouth or ContagiousPustular Dermatitis.	virus	Transmitted by contact with sheep and goats.	Prevention will be based on good personal hygiene Washing exposed areas with soap and water. Animal housing areas must be kept clean.
Salmonellosis	*S.typhimurium* *S.dublin, S.virchow*	People working in the runners room, the paunch room and those who slice mesenteric lymph nodes for inspection have the greatest exposure t o Salmonella.	Animals need to be handled so that there is negligible contamination of the carcass by faecal material, intestinal contents, hide or fleece. If contamination occurs, it needs to be removed. Individuals with diarrhoea need to be excluded from food handling.
Tetanus	*Clostridium tetani* the organism resides in the intestine of herbivorous animals, including human	The tetanus spores are found commonly in the soil and people working in stockyards and lairages may easily get infected material in to minor	Prior immunisation-prior to the start of employment and boosters at 10 yearly intervals. All wounds require fastidious cleaning and dressing and, in some

		wounds	instances, antibiotics.
Tuberculosis	*Mycobacterium tuberculosis* transmitted by human to human contact. *M. bovis* which iS primarily transmissable from cattle but can also infect pigs and other species	*M. bovis* from drinking infected milk, inhaling infected aerosol droplets	avoidance of high risk activities such as the unnecessary slicing of lesions resembling tuberculosis identification and testing of at - risk herds. BCG vaccination of employers Employees may require screening from time to time if they are considered at risk
Warts	Human Virus	Direct touch	They usually disappear spontaneously wit in months to years When they do not resolve they can usually be removed by diathermy, surgery, freezing or the use of wart paints
hydatid Disease	Hydatid Disease is caused by one of the dog tapeworm parasites, *Echinococcus granulosus*.	Dogs become infected by eating the liver , lungs and other tissues of sheep (and cattle, goats, pigs, kangaroos and wal lables), that contain the cysts of the tapeworm. The dogs develop the adult tapeworms in the intestines and pass out tiny eggs in their faeces. The eggs readily contaminate the environment.	Within an abattoir area, dogs need to be muzzled to ensure that they can't eat potentially infected offal. Dogs in high risk areas may also need periodic tapeworm treatment.

As regards the known diseases, there are some established measures which need to be followed. Immunization is recommended against Q fever and medical screening of both employees and livestock is essential. Employees must maintain high standards of personal hygiene, ranging from clean clothes and hands to proper and immediate treatment of cuts, wounds and skin lesions. In order for employees to maintain high standards of personal hygiene, they must be provided with adequate facilities that include basins for hand washing and shower rooms.

Another important factor to prevent zoonosis is that plants must be designed so that yards, slaughtering areas and skin sheds can be cleaned and drained to maintain hygiene. The use of high-pressure water should be minimized to reduce the generation of aerosols which could contain harmful micro-organisms.

Finally, appropriate personal protective equipment over the mouth, eyes and hands can guard against inhalation and contamination of disease-carrying micro-organisms. If these measures are strictly observed at all times, zoonotic diseases will be reduced.

The design of control measures to improve working conditions involves:

-the redesign of the work process and procedures

-substitution of safer material

-shielding or screening of the worker against the hazard

-reducing the intensity of the hazard by distancing the process

-using ventilation systems to extract or dilute airborne toxins

-reducing employee exposure by job rotation or providing rest periods

-Using protective clothing.

MEAT BORNE ILLNESSES, THEIR PREVENTION AND CONTROL

Meat borne illnesses are the infections or diseases, which are transmitted through meat either by direct consumption or handling of contaminated meat.

Meat borne illnesses include food poisoning (Salmonella, Staphylococcus aureus, Bacillus cereus, Clostridium perfringens, Campylobactor jejuni or Clostridium botulinum) or transmission of diseases like tuberculosis, brucellosis, Anthrax and parasitic infestations like cysticersus bovis, cysticersus cellulosae and trichinella etc.

The method of prevention and control depends upon the way of transmission of the disease or infection, type of the causative agent and the type of meat and meat products.

Classification of meat borne illness

1. Meat borne illness of chemical or toxicological agent

1. Intrinsic toxicity: fish toxin, animal eating toxic plants
2. Toxic chemicals from the soil: plant grown on the seliniferous soil / fluorite soil
3. Chemical preservatives: when added above limits e.g. nitrites
4. Accidental or criminal contamination: lead, zinc, cadmium, arsenic

2. Endogenous animal infection transmissible to man and animal

- Zoonosis acquired occasionally through the intestinal tract: Pasteruellosis, Leptospirosis, Erysipelothrix, Listeriosis, miscellaneous (Brucellosis, Vibrio foetus, F and MD, Tularemia, Rickeettsia burnetti.)

- Zoonosis acquired chiefly through the intestinal tract. : Salmonellosis, Shigellosis
- Zoonosis acquired commonly through the occupation: Anthrax, Bovine TB, Brucellosis, ORF, Louping ill, Ring worm, Avian psittacosis, Streptococcal meningitis.
- Meat borne helminthic zoonosis: Trichinosis, Taeniasis, Echinococcosis
- Rare meat borne zoonosis possibly acquired by infection: Toxoplasmosis, Sarcosporidiasis, and Intestinal myiasis.
- Food poisoning: Salmonellosis, Shigellosis, Staphylococcosis, Botulinum, Coliform (E. coli-0H-157)
- Miscellaneous food poisoning: *Cl. perfringens, Bacillus cereus, Streptococcus*
- Viral food poisoning: Poliomyelitis, Infectious hepatitis

Prevention: Inhibiting the introduction of a disease agent into an area. A specific population, or an individual.

Control: Efforts consists of steps taken to reduce a disease problem to a tolerable level and maintain it at that level.

Eradication: Is the elimination of a disease producing agent form a population or a geographic area.

General Steps for zoonoses Control and Prevention

- Animal identification: Identification of source of infection farm, herd, and area.
- Health maintenance; public health, surveillance.
- Consumer protection strategies: avoid food poisoning / Hazard Analysis and Critical Control Points (HACCP)
- Increasing host resistance: immunization / stress reduction / genetic selection
- Strict sanitary measures
- Cleaning and the disinfecting of the contaminated objects and materials.
- Restricts the movement of the animals, unauthorized people, animal by-products feeds etc.
- Reducing contact potential: Separating vector / host; isolation of affected host; population control.
- Establishment of the local quarantine of infected premises and the initiation of the epidemiological studies to ascertain the extent of infection.
- Reservoir neutralisation: Test slaughter / mass therapy.
- Immediate slaughter of all infected and the exposed animal and their disposal by deep burial in quick lime or by burning.
- Communication: mass media
- People education: training programs.
- After reasonable interval of time, susceptible test animals are introduced on the premises and periodically examined for the reappearance of infection.

General precautions in avoiding food poisoning

a. Food should be handled in a hygienic manner by all food handlers and infected handlers should be kept away
b. Cross contamination from raw to cooked foods can be prevented by washing hands and all equipment or surfaces in contact with raw foods.
c. The time gap between preparation and service of food should be reduced to avoid long storage in warm environment.
d. Large masses of food, which have to be reheated after should be cooled quickly to 15^0C and refrigerated immediately.
e. Food should be reheated thoroughly so that the center of the food gets heated to a temperature (80^0 C) high enough to destroy bacteria.
f. Frozen foods should be thawed carefully at temperatures between 10- 15^0 C and frozen food should not be cooked till it has thawed. Foods once thawed should not be refrozen unless it has been cooked well after thawing.
g. Cooked foods, which are to be served hot, should be stored above 63^0 C. avoid cooling and heating food repeatedly.
h. Leftover foods should be refrigerated immediately to keep it out of the danger zone.
i. Food should be prepared in quantities required and quantities for which adequate refrigerated storage space is available. This will prevent perishable or high-risk items from spoiling.
j. Suspected food should be discarded immediately without tasting it.
k. The kitchen and cooking equipment should be cleaned daily and regular pest control measures should be taken.
l. Adequate toilet and washbasin facilities with a continuous supply of water should be provided.
m. High-risk foods like meat, poultry, egg and milk should be purchased from certified dealers only.

The prevention and control of some important diseases

Tuberculosis

Control measures in animals

1. Provide clean well-ventilated shelter.
2. Appropriate stocking density
3. Segregation of calf at birth and feeding with clean pasteurized milk.
4. Intra-dermal testing of calves at the age of 3 months and segregation of reactor animals
5. Periodic examination of all the animals and segregation of clinically suggestive cases.
6. Remove clinically suggestive and reactor animals from the herd.
7. Try to identify fresh cases and observe
8. Routine check up of farm labourers and exclude the positive cases from the farm.

In human beings

Screening, confirmation, registration and isolation of the affected individuals and animals.
Treatment of individual cases. --- (DOT - direct observation therapy)
Reduced no and period of infectious cases.
Vaccination- (Bacille Calmette –Guerin, BCG vaccine): this vaccine produces cell mediated and not humoral immunity. It stimulates the body defense than inducing antibody system. Hence after BCG vaccination, a primary infection will still take place.
Follow up of the treated cases
Diagnosis and community search (smear, culture, sensitivity and full plate X-ray). A productive cough especially if sputum positive is the most important sign of TB. but loss of weight and anemia are often present can be adequately judged on clinical examination.
Pasturisation of the milk –test and slaughter tuberculin reactor animals.

Brucellosis- Control and Prevention

Isolation of infected cases, segregation of pre parturient females, serological surveillance, vaccination (19 or 45 / 20 strain for heifer, calves and Rev. 1 for goats). Vaccine from Br. abortus strain RB 51 is a new stable and rough live vaccine, presently used successfully in USA. Proper pasteurization of milk, and cooking of meat, wear protective clothing while handling abortion or dystokia cases. There is no vaccine for human beings. Other measures include test and slaughter, quarantine, hygienic precautions and education of the public.

Swine erysipelas - Control and Prevention:

- Use of protective clothing, gloves while handling animals and cleaning fishes.
- Rodent control and Immunisation
- Control movement of animals from the premises.

Glanders

Control and Prevention: isolation of the patient, test and slaughter of equines, proper disposal of the carcass. Malline test (intrapalpebral) of equines in endemic areas. There is no vaccine available hence effective quarantine and other controls are needed.

Leptospirosis

Control and Prevention: Chlorinating swimming pool water regularly. Rodent control and avoid post preparation contamination of foods.

Listeriosis

Control and Prevention: - Treatment of the individual. Testing of ensiled feed.Feeding low level of tetracycline.Sanitary measures. Do not consume raw milk. Efficacy of the vaccine is doubtful. Commercial, killed vaccine is available.

Tularemia

Control - Wear protective clothing, avoid control with ticks, cook meat thoroughly, boil drinking water. Dipping of sheep. Organism does not survive in carcasses for long time unless frozen. Organism persist in mud and water, may proliferate. Persistent treatment may be necessary because the organisms are intracellular. Vaccines are available for people at high risk of infection. Recovery confers long lasting immunity.

Pox disease

Control –Isolation of affected individual, wear protective clothing, cook meat thoroughly, and boil drinking water and hygienic practices.

Pox disease affects man (small pox or variola major and alastrim or variola minor and mammals (sheep, goat, swine, horse, rabbit and fowl).

In their virulent form small pox in man and sheep and goat pox are severe diseases with high mortality, the reminder are generally benign.

With exception of poxviruses of sheep and goat all the pox viruses listed above are immunologically benign. Small pox, sheep pox and rabbit pox are transmitted by inhalation, the others by direct contact.

Rabies

Prevention: 100 % vaccination of the dog, control of stray dog population, and protective gloves while treating rabies cases.

Wash the bite wound with 20% soap, followed by water and 0.1% Iodine, penicillin injection; observe the biting dog for 10 days for the development of the disease. The area around the wound should be infiltrated with antiserum and suturing should be avoided or delayed to allow bleeding and drainage.

Tetanus

Control: Clean the wound with antiseptic solution.Tetanus toxoid injection immediately. Take care during farm management procedures. Promote prophylactic immunization

F & MD.
Control and Prevention

1. Regular vaccination in endemic areas.
2. Isolation and treatment of affected animals.
3. Disinfection of premises
4. Disposal of cotton swabs, syringes.
5. Use of protective gloves by farms workers, vets.
6. Pasteurisation of milk, proper cooking of meat, testing of meat meant for export.

Q fever

1. Identification of infection in domestic animals
2. If suspected in sheep flock care must be taken to avoid the spread if infection by proper disposal of placenta, fetus, discharges and disinfecting area by 1 % Lysol or 5 % Hydrogen peroxide.
3. Vaccination in human
4. Pasteurization of milk.

Cysticercus bovis

A double cycle in blast freezing tunnel at maximum temperature of -30 ^{o} C would kill the cyst in boxed beef (30 kg. Carton) after 48 hr. cycle followed by 72 hrs cold storage at – 23.3 ^{o} C.Cooking temperature of above 60 ^{o} C and salting for 21 days kills the cyst. In offal stored at 6.5 ^{o} C for not less than 3 weeks or - 10 ^{o} C for 2 weeks destroys the cyst.

Cysticercus cellulosae

Cyst destruction by: -

Heat (45 - 50 ^{o} C), salt (20% concentration), freezing (- 30 ^{o} C, 30 kg carton, 48 hrs cycle)

Control (T.solium and T.saginata) :

- Identification, isolation and treatment of human case harbouring adult worms with proper disposal of faeces. Pig should not be accessed to human faeces.
- Adequate inspection of carcasses with condemnation of generalised cases and appropriate freezing / heat treatment of lesser infections.
- Education to public, sanitary measures

Hydatidosis

Control: Regular inspection of slaughtered animals and destruction of organs showing hydatid cyst. Sheep lung cysts are majority fertile and lay a major role in spread of disease. Registration of dogs and their periodical examination and deworming.

Control of stray dogs and prohibition of their access to slaughter premises.

Proper disposal of affected carcasses and offal by burial

Trichinella

Prevention

1. The parasite's lack of host specificity and ability to complete its life cycle within the body of a single host makes the control of human trichinosis theoretically simple.
2. Ban on illegal slaughters of all animals especially pigs.
3. The free roaming pig population should be controlled.
4. Construction and maintenance of community toilets and people must be educated against indiscriminate defecating and unhygienic habits.
5. Commercial rearing of the pigs must be encouraged.
6. Swine should be fed only grain or cooked garbage because uncooked garbage may contain contaminated pork scraps.
7. Pork should be cooked at the internal temperature of 70 0 C for at least 5 min.
8. A temperature of – 15 o C will destroy cyst in muscle block less than 15 cm thick. The temperature of – 18 o C for 24 hrs will render pork safe.

9. Salting (15 %) and smoking destroys the cyst.
10. Freezing at – 200 C for five days also kills the larvae in the cyst.
11. Regular trichinoscopic examination at the abattoir and disposal of affected carcasses.
12. Since low-dose irradiation (0.03 M rads) is sufficient to inactivate encysted trichinae, feasibility of use of this procedure on pork carcasses after slaughter also is being studied.

Treatment

No routine treatment for infected swine before slaughter has been developed that will clean the animals of trichina cysts. In human infections, thiobendazole and other supportive treatments are used. Similar drugs have been used experimentally in swine and have been found effective, primarily against the adult worms in the intestine and less effective against muscle larvae.

Sarcocyst

Control Disposal of condemned material from a slaughterhouse.

Cooking of beef, pork at 70^0C for 15 minutes, freezing at – 4^0C for 2 days or at – 20^0 C for 1 day.

Bone taint

Bone taint in cattle is reduced by avoiding bacterial contamination of the carcass and rapid cooling it to 1.5 ^{o} C. Experimentally tetracycline injection immediately before slaughter and antibiotic spraying of the carcass can prevent taint. The smarter butcher aids in dissipation of heat from the carcass of beef by removing fat from the kidney and pelvic cavity. He also incises stifle joint to promote air access and rapid cooling, thus preventing anaerobic bacterial growth. To avoid bone taint the temperature at the center of the round must not exceed 4.5 ^{o} C after 48 hours.

Food poisoning

1. Personal hygiene
2. Clean kitchen, work places
3. Clean raw material
4. Use potable water for cooking
5. Proper heating of foods to an internal temperature of minimum 80^0C.
6. Preserving cooked foods in refrigerator if the interval between cooking and consumption is more than one hour
7. Avoid post-cooking contamination by proper covering the foods, use of clean stirrers, spoon etc.
8. Avoid contact between raw and cooked foods.
9. Reheat cooked foods thoroughly.
10. Protect foods from insects, rodents and other animals.
11. Wash hands repeatedly.

The food processing plants must adopt Hazard Analysis Critical control points (HACCP) system to ensure safe and quality foods. HACCP system consists of an evaluation of all procedures concerned with the production, processing, storage, distribution and use of raw materials and food products.

Food-borne pathogens causing food poisoning

Most cases of food poisoning are caused by bacteria which arise from animal, human or environmental sources. Viral infections are unlikely to be from animals and may be due to direct human contamination or indirectly through the environment, for example, from shellfish contaminated by discharged human sewage. **Bacterial food poisoning may take one of two forms:infection with living organisms or intoxication with preformed toxins s**uch as with Staphylococcus aureus. The feature which chiefly distinguishes the two types clinically is the incubation period, that is, the interval between eating the food and the development of symptoms. Where pre-formed toxins are present, the conditions are somewhat analogous to chemical poisoning, and symptoms will develop very rapidly, usually within a few hours. If living organisms are ingested, time will elapse before their multiplication in the body has proceeded sufficiently to provoke the usual reactions of diarrhoea and vomiting.

Campylobacter spp. The traditional animal strains of *Campylobacter fetus fetus* and *C. fetus venerealis* rarely cause human infection and the most common types are*C. jejuni*, *C. coli* and *C. lari*. Unlike many other food-borne organisms, Campylobacter spp. are fastidious in their growth requirements. They are Gram-negative and microaerophilic, requiring a low oxygen concentration (5%). They are slender, curved and highly mobile and grow best at 42–43°C and not at all below 30°C. They are most unlikely to multiply on food at room temperature. They are not heat resistant and can be killed by cooking. As though to compensate for their inability to grow on food, they have an extremely low infective dose, with as few as 500 cells being sufficient to cause human infection.

Infection in humans The normal incubation period is 3–5 days, but some evidence exists that this can be considerably extended. A characteristic of infection is acute abdominal pain, which can be so severe as to be mistaken for appendicitis, sometimes precipitating unnecessary surgery. This pain is often accompanied by bloody diarrhoea and fever, but vomiting is rare. Most infections resolve spontaneously within one week, while others can be prolonged. Complications during the acute phase are unusual and only exceptionally does death occur. There is, however, increasing evidence that there may be longer-term sequelae to campylobacter infection including both reactive arthritis and even peripheral polyneuropathy (Guillain–Barré syndrome).

Source of human infection

The Advisory Committee on the Microbiological Safety of Food investigated infections with campylobacter and reported in 1993 that **poultry was the most common vehicle** of infection. While it is now considered that 60–80% of human infection can be linked to poultry, it should be remembered that the types of campylobacter affecting humans can also be isolated from all species of animals (including pets) and a wide range of foods including, milk, water and cooked meats. **Person-to-person spread** is also important. The low infective dose for campylobacter means that cross-contamination is a particular risk since a relatively small amount of contamination may be sufficient to establish human infection. This cross-contamination must be prevented throughout the whole food chain with efforts being made to reduce the prevalence of the bacteria on farm through good biosecurity as well as through interventions during transport of the birds, in the slaughterhouse and during storage and retail.

Salmonella spp.

The salmonellae constitute a large group of over 2200 different serotypes. They are members of *the Enterobacteriaceae*, are Gram-negative and can readily grow on a wide range of media including foods. They are temperature sensitive and readily destroyed by cooking. The different serotypes are identified by means of the **somatic (O) and flagellar (H)** antigens using the **"Kauffmann–White scheme".** Further sub-typing can be carried out by phage typing and, increasingly, molecular methods such as plasmid profiling.

Some salmonellae **usually only affect a single animal species** including, for example, *S. typhi* in humans, *S. abortus ovis* in sheep and *S. pullorum* in poultry. Most salmonella species including those associated with food poisoning can infect many species of animal, although they may not cause illness in all of these. A good example of this is S. enteritidis, now the most common salmonella in humans, which is widespread, but usually causes no illness in poultry. . While not as resistant as the spore-forming organisms such as the *Clostridium spp., Salmonella spp*. can exist for many months in the environment, especially if protected from extremes of temperature and sunlight. This means that recycling through the environment is an important route for animal (and human) infection. There is considerable ongoing effort being expended by the poultry industry to reduce salmonellae causing human illness, in particular *S. enteritidis*.

Infection in humans The incubation period in people is variable but is usually between 12 and 36 hours. The typical presenting symptom is diarrhoea, but this may be accompanied by nausea and abdominal pain, although vomiting is not usual. There may also be headache and fever. While the infection is normally self-limiting and does not require antibiotic treatment, occasionally, with more invasive salmonellae such as *S. virchow*, bacteraemia can occur. The infection is rarely fatal in people.

Source of human infection People can become infected following a failure of personal hygiene after contact with infected animals (or other infected people). Environmental contamination, especially untreated water, is also important. Most cases are thought to be the result of food-borne infection and highlight the importance of controlling hygiene in the food chain. Meat can become contaminated during the slaughter process either from intestinal contents or from faecal contamination on the hide. As with any faecally spread organism, it is essential that clean animals are presented for slaughter to help minimise the latter and that the abattoir operates hygienically to prevent both.

Escherichia coli O157:Verotoxigenic E. coli infections, particularly serogroup O157, were first reported in the United Kingdom in 1983. Since then, despite a relatively low notification rate of 0.45 cases per 100,000 population across the EU, they have become recognised as a major source of human morbidity and mortality due to the potential serious consequences of infection. E. coli are ubiquitous inhabitants of the intestinal tracts of animals and man. A variety of serogroups cause infections, usually in a single animal species, for example, bowel oedema in pigs.

In humans, a number of different disease syndromes are recognised, including:

- Enteropathogenic E. coli (EPEC)
- Enteroinvasive E. coli (EIEC)
- Enterotoxigenic E. coli (ETEC)
- Enterohaemorrhagic E. coli (EHEC)

Some of these **EHEC** have the ability to produce one or more toxins (verotoxins), which can be detected using tissue culture (Vero cells), and are often referred to as **verotoxin-producing E. coli (VTEC)**. Together with other virulence factors, they have the ability to cause human illness but apparently no animal illness. While several serogroups such as O111 and O26 have been involved in the United Kingdom, the majority of infections are caused by **serogroup O157**. In other countries, other serogroups have been reported as more important than O157. As with campylobacter, a small infective dose – perhaps as few as 10 cells – is required to cause human illness, but unlike campylobacter, they can multiply on food.

Infection in humans:Incubation is normally 1–10 days and a considerable spectrum of symptoms can be seen. These range from asymptomatic infection, abdominal pain and diarrhoea, bloody diarrhoea and haemorrhagic colitis to haemolytic uraemia resulting in renal failure. Serious systemic disease is a particular feature in the old and the young.Conversely, many people appear to carry and shed the bacteria yet show no signs of infection.

Source of human infection:Three main routes of transmission: consumption of contaminated food or water, direct or indirect contact with animals and person-to-person spread. Beefburgers were recognised as a significant vehicle of infection following a large outbreak across several states in the United States in 1993 in which 732 people were affected and 195 hospitalised with four deaths. However, a wide range of foods have been implicated, including apple juice, vegetables, potatoes, bean sprouts and water. Contamination of the food with animal faeces is thought to cause most of the problem. Cattle are considered to be the primary reservoir of VTEC with an estimated 0–17% of cattle carrying VTEC O157. As with all organisms found in animal faeces, control is based on minimising carcase contamination. This begins on the farm by reducing the number of animal carriers, but at present, not much is known about the natural transmission of E. coli O157 to devise a control strategy. The most important measure is to ensure that only clean animals, with the minimum of faecal contamination, are slaughtered. This was a major recommendation of the Pennington group which reported after a large outbreak in Central Scotland (at Wishaw) in 1996 in which 496 people were affected and 20 elderly people died. Products from a single butcher's premises, including cooked meat, were implicated as the cause of the outbreak. Slaughter of clean animals must be accompanied by the hygienic operation of the whole slaughterhouse andeffective control of the food chain. Ultimately, thorough cooking will kill the organisms. The low infective dose means that contamination unlikely to cause problems when Salmonella spp. or Campylobacter spp. are present may be enough to cause infection if E. coli O157 prevails.

Yersinia enterocolitica Yersinia enterocolitica was identified as a human pathogen in the late 1930s. It is a Gram-negative non-sporeforming bacterium which grows over a wide range of temperatures (0–40°C) and optimally at 29°C. The range of growth temperatures allows multiplication at refrigeration temperatures. It is widespread in the intestinal tract of animals and is readily recovered from the environment, including water and soil. It can be divided into biotypes, serotypes and phage types. Different types are associated with different parts of the world. In Europe, serotype O3 is most commonly recorded in humans and is also associated with pigs.

Infection in humans The incubation period is about 3–7 days, and infection causes acute diarrhoea, abdominal pain, fever and vomiting. It is more common in children, although cases in adults may be followed by longer-term problems including skin rashes and arthritis. Infection is usually self-limiting, but in people with some other underlying pathology, septicaemia with a high mortality may follow. About 500 cases a year are reported in the United Kingdom.

Source of human infection This organism is often associated with pig meat products, either fresh or cured. The ability to grow at refrigeration temperature means that contaminated food, even if properly chilled, can cause infection. Such foods can also cross-contaminate other foods in the kitchen.

Listeria monocytogenes Of the several Listeria species, only L. monocytogenes is thought to cause human infection. It is widely distributed in animals, birds, humans and the environment but has only been recognised as a food-borne pathogen since the 1980s. Like Yersinia spp., L. monocytogenes can grow at a wide range of temperatures (0–42°C) and especially well at 30–37°C. Listeria spp. can also grow slowly at refrigeration temperatures. A wide range of serotypes exist, but most human infection is associated with types 4 and 1/2.

Infection in humans Most human cases are sporadic, and the extended incubation period (several weeks) can make identification of the source very difficult. Unlike the case with most food-borne pathogens, the illness produced is systemic rather than intestinal. Most infected people are symptomless, and as many as 5% of the population are faecally excreting at any one time. Illness is most likely to be seen in people with reduced immunity, when the symptoms range from a ‘flu-like’ illness through fever and septicaemia. There is a particular risk to pregnant women (who have a naturally reduced immune capability) and the unborn child, when the mortality can be as high as 30%. Although there was a significant rise in reported cases of listeriosis in the late 1980s, this was not maintained in subsequent years. Listeriosis is now relatively uncommon, with about 100 cases reported each year in the United Kingdom.

Source of human infection This is often not identified because of the length of the incubation period. The organisms are ubiquitous and widespread in animals and the environment. Listeriosis can cause animal disease, including abortion in ewes and meningitis in younger sheep. A wide range of foods have been implicated including cheese, paté and poultry meat. Vulnerable groups, especially pregnant women and the immunocompromised, are advised not to eat risk foods such as patés and soft cheeses.

Clostridium perfringens The mode of action of food poisoning with Clostridium perfringens is through **the consumption of large numbers of bacteria which rapidly form enterotoxin in the small intestine**. It is a Gram-positive spore former, requiring anaerobic conditions for growth. Five types (A–E) of Cl. perfringens are known to exist, but only type A has been implicated in food poisoning. The clostridia form part of the normal intestinal flora of animals and are widespread in the environment.

Infection in humans The rapid production of enterotoxin in the intestine results in a short incubation period of 10–12 hours. The toxin causes diarrhoea and abdominal pain but not usually vomiting. Illness normally lasts 24 hours and is self-limiting. Subsequent complications and death are rare.

Source of human infection The clostridial spores survive normal cooking, and the heating process may in fact stimulate them to germinate. As food cools, very rapid multiplication can take place, since optimal growth occurs at 43–47°C. The heating process of cooking also drives off oxygen, creating the anaerobic conditions necessary for growth. These conditions are most likely to be found when large volumes of food are cooked and cooled, particularly large joints of meat or stews and casseroles. Thorough reheating (above 75°C) will kill any vegetative cells present and prevent food poisoning. Most cases occur as outbreaks, which may be large and are often associated with hospital or commercial catering. About 50 outbreaks are reported in the United Kingdom each year.

Staphylococcus aureus Unlike most bacterial food poisoning, illness caused by Staph. aureus is **due to the consumption of pre-formed toxin and not the bacteria,** which may be absent. The toxin is heat stable and may survive for 1½ hours at boiling temperature, even though the staphylococci themselves are destroyed. Five major enterotoxins are known to be produced (A–E) of which type A is the most common. The Gram-positive cocci can grow over a wide range of temperatures (10–45°C) with the optimum at 35–40°C.

Infection in humans The presence of pre-formed toxin results in a short incubation period of 2–6 hours. The symptoms are primarily nausea and vomiting, with additionally diarrhoea. The illness may also be so acute as to cause fainting or collapse. Most patients recover within 12–48 hours.

Source of human infection Staph.aureus infection usually follows contamination from a human source. The organism can cause wound or skin infections and is present in the nose of up to 40% of healthy people. A failure of basic hygiene, including covering skin lesions, results in contamination of food and subsequentmultiplication and toxin production. The foods usually implicated are cooked meats, poultry and dairy produce. This has been an uncommon cause of food poisoning in the United Kingdom since the 1950s, with around 10 outbreaks reported each year.

Clostridium botulinum Botulism is one of the most feared causes of food poisoning because, although it is exceptionally rare, it is a severe disease with a high mortality. Clostridium botulinum is a Gram-positive spore-forming obligate anaerobe producing one or more of seven toxins (A–G). Toxins A, B and E have been associated with human disease. Types A and B are more commonly linked with meat and vegetables, while type E is associated with fish. The toxin is not thermostable and can be destroyed by cooking at 80°C for 30 minutes.

Infection in humans The clostridia produce potent neurotoxins and the symptoms reflect muscular paralysis. Symptoms can appear within 2 hours but may take as long as 5 days. There can be an initial short period of diarrhoea with vomiting and subsequent constipation. Blurring or double vision is often the first systemic sign, accompanied by dry mouth and difficulty in swallowing. The patient is usually mentally alert and there is no loss of sensation. Paralysis can extend to the limbs and eventually result in respiratory failure. The effects of the toxin may persist for several months.

Source of human infection Cl. botulinum is ubiquitous and can be found on a wide range of foods. Toxin production only takes place when growth occurs. This can happen during preservation processes, such as smoking or fermentation of fish and meat, which result in a suitable anaerobic environment. Improperly bottled or canned foods can also allow growth to take place. Botulism is very rare in the United Kingdom. The first reported outbreak occurred in 1922 in Loch Maree in Scotland when eight people became ill after eating potted duck paste and all died. The largest outbreak occurred in 1989 when 27 cases (one death) were caused by hazelnut yoghurt. In total, only 10 outbreaks have been described in the United Kingdom.

Vibrio parahaemolyticus*Vibrio parahaemolyticus* is a Gram-negative rod usually found in seawater where the temperature is above 10°C. It can multiply rapidly at ambient temperatures but is easily killed by cooking.Infection in humans The incubation period is 12–24 hours, after which profuse diarrhoea and abdominal pain develop. Less commonly, fever and vomiting may be present. The symptoms may persist for 7 days.

Source of human infection The organism is found in seawaters, especially if the temperature is greater than 10°C. Fish and shellfish from affected waters may cause illness if inadequately cooked or if subsequently re-contaminated. It is rarely acquired in the United Kingdom, although one incident involving locally caught crab was reported on the south coast of England.

Bacillus cereus Bacillus cereus has been recognised as an uncommon cause of food poisoning since the 1970s. It is a Gram positive, spore-forming, motile bacillus producing **two different toxins**. A **heat-sensitive enterotoxin** causes a diarrhoeal illness, while the **heat-stable 'emetic' toxin** causes vomiting. The spores are heat resistant and can survive normal cooking temperatures. The bacteria are widespread in the environment, including soil and water, and consequently contaminate many foods, particularly dry foods such as cereals. Any food such as meat, dairy products and vegetables can be vehicles of infection.

Infection in humans The two toxins produce different disease syndromes. The more common illness is **vomiting** caused by the 'emetic' toxin after an incubation period of 1–5 hours. This is particularly associated with rice and pasta. The diarrhoeal illness caused by the enterotoxin has an incubation period of 8–16 hours. Both forms of infection usually last no longer than 24 hours and complications are rare.

Source of human infection Many foods are contaminated with a few spores and these can survive the initial cooking. If food is subsequently kept at ambient temperature, the spores germinate and the vegetative cells multiply rapidly, producing toxin in the food. The 'emetic' toxin is not destroyed by further heating, but the enterotoxin can be inactivated by thorough reheating. This latter toxin can, however, also be produced in the intestine of the patient following consumption of the vegetative cells. The disease is **controlled** by keeping cooked foods either hot or at refrigeration temperatures.

Zoonoses -Occupationally Acquired By Abattoir Workers

Zoonoses are described as those diseases and infections which are naturally transmitted from animals to humans. They represent about 70% of the number of emerging infectious diseases in recent time . There are over 300 zoonotic diseases of diverse etiologies which cause high morbidity and mortality. Zoonotic diseases occur in both sexes, in all age groups, in all seasons, in all climatic zones and in rural and urban settings .Increasing demand for meat and meat product by human population has made human contact with animals unprecedented, coupled with movement of animals across international frontiers to supplement the local supply and increasing the risk of zoonotic diseases especially from endemic zones Transmission of zoonotic infections occurs through various routes However, direct contact seems to be the most common mode of entry of infectious agent in the employees working in slaughter houses The employees of meat industry are at particular risks of acquiring many zoonotic infections, due to the close contact that exists between them and animals/tissue of animals during slaughtering or processing.

Erysipelothricosis-This is an infectious bacterial disease caused by *Erysipelothrix rhusiopathiae*.The disease is sometimes called Fish finger, Pork, finger or Whale finger . Human infections are usually acquired by occupational exposure in meat or chicken slaughterhouses or fish plants. Localized skin lesion in man is termed **"Erysipeloid**" which is characterized by a self- limiting, painful, red swelling of the fingers with or without lymphadenopathy. It is worldwide indistribution, and the major reservoir of*E.rhusiopathiae* is domestic pig. Infection is recorded in pig, sheep, cattle, horse, fish, birds and reindeer .Disease is most common in abattoir workers, fish handlers, butchers, and meat inspectors, and poultry processing workers .Infection is acquired through abraded skin and bite wounds of fish and crustacean.

Brucellosis -This is one of the major **anthropozoonoses of public health significance** and is caused by *Brucella abortus, B. suis* and *B.melitensis* It has been reported in buffalo, cattle, camel, horse, pig, sheep, goat, deer, and poultry . Human brucellosis commonly referred to as **'undulant fever' or 'Malta fever'** is important zoonoses that often coincide with livestock infection. Serological surveys in many countries have revealed antibodies to B. abortus in abattoir workers and veterinarians. Man gets infections by direct contact with infected discharges and materials of animals and by inhalation. All slaughter house workers who are engaged in the actual handing of livestock, dressing of carcasses, disposal of condemned organs are at greater risk of acquiring brucellosis. Clinical signs of brucellosis in man include anorexia, fever, headache, backache, weakness and sweating .Death may occur due to endocarditis .

Listeriosis The etiologic agent, *Listeria monocytogenes* is a rod-shaped bacterium, first recovered by Murray is 1926 from rabbit and guinea pig . Infection occurs in buffalo, cattle, goat, sheep, house, bird, rabbit and fish . Veterinarians and abattoir workers can acquire a primary cutaneous listeriosis from direct exposure to diseased animals or infected discharges/ tissues . Cutaneous listeriosis is manifested with an initial reddish rash which later develops into vesicular or pustular lesions, about 1-2 mm in diameter with either dark or light centre . Sometimes, it may lead to a generalized form of disease. The disease appears to be more common in temperate zones. The incidence of cutaneous listeriosis has been reported to be higher in U.K. About 1,700 cases of sporadic listeriosis with 550 deaths occur annually in USA .The mortality is higher in immune compromised individuals. The pregnant women should not be allowed to work in meat processing plants .

Glanders- It is a bacterial zoonosis caused by *Burkholderia mallei,* a non-motile,Gram negative, non-capsulated organism . The first case of human glanders was reported in 1723 by Dr Sainbel of England. In the past, it has caused serious mortality both in equines as well as human beings. Glanders in man is contracted in most cases, as an occupational exposure among persons working in close association with diseased horses or wound infection during autopsy or handling of meat of diseased animal . The mode of transmission of disease in abattoir worker is though the abrasions in the skin. The nodule develops at the seat of inoculation. Later, the lymphatic vessels and lymph notes becomes inflamed and may extend infection to the nasal mucous membrane or lungs. Inhalation of organisms may lead to fatal pneumonia. Biohazard cabinet must be used to work with*B. mallei*

Anthrax-This is an occupational disease caused by *Bacillus anthracics*, a Gram positive, aerobic, sporulated organism .The disease is recorded in all food animals. The spores under favorable condition may remain viable for 40-50 years in contaminated soil and 150-250 years in the bones of dead hosts. Global incidence of anthrax in man is estimated to be 20,000 to 100,000 cases per annum. The cutaneous form also known as malignant pustule accounts for 95-99% of human cases throughout the world . It is endemic in many countries including Asia and Africa. Abattoir workers usually contracts infection by inoculation through a skin wound. Cutaneous anthrax appears to occur more commonly on the hands and arms of meat handlers . It is characterized by the small pimple that rapidly develops into a large vesicle with a black necrotic center- the so called “Malignant Pustule”. Cutaneous form may become generalized in 10% of cases when fatal septicemia ensues. In untreated cases, the outcome is fatal . Pulmonary form is responsible for febrile respiratory tract disease, mild onset, than sudden onset of second stage with dyspnea, sweating, cyanosis, and death with 24 hours (100% case fatality rate). Additionally, the intestinal form results in febrile gastrointestinal disease. Transmission is usually by direct contact with infected animal. Also, by ingestion of undercooked meat or, occasionally, by inhalation of spores in wool from infected ruminants. Spores are found in bone meal, soil, water, and on vegetation. Spores form readily when vegetative form is exposed to air, are long-lived, and are very resistant to environmental extremes and disinfectants .

Leptospirosis It is a global bacterial zoonosis known by several synonyms such as Weil’s disease, Mud fever, Canicola fever, Rice-field worker’s disease; and is caused by pathogenic spirochetes in the genus *Leptospira.* The disease affects both humans and animals, occurring widely in developing countries and is reemerging in the United States . Leptospirosis is encountered in many species of food animals such as cattle, buffalo, camel, horse, goat, sheep, deer and pig . Poultry are said to be resistant to *Leptospira* infection. Transmission occurs by directed contact with diseased animals and their tissues or indirectly with contaminated environment especially water contaminated with urine of infected food animals . In Jamaica, Brown and co-workers studied the environmental risk factors associated with leptospirosis in butchers. Infection usually takes place through the penetration of the abraded or traumatized skin with the urine of the sick animals. Rodents serve as natural reservoir of leptospirosis .Organisms can survive for weeks in the soil and water contaminated by urine of carriers, particularly if the pH is alkaline. After an incubation period of 7-14 days, the patient exhibits signs of a flu-like syndrome characterized by sudden high fever, headache, vomiting, nausea, chills and conjunctivitis. If immediate treatment is not given, patient may develop hepatitis, jaundice, nephritis, pneumonia, anemia, kidney failure, internal bleeding and meningitis. Death may occur due to renal failure .

Tularemia- Disease is also known as Deerfly fever, Rabbit fever and is caused by*Francisella tularensis,* a Gram negative, non-sporulated, aerobic organism .The disease is reported in rabbit, deer, horse, pig and calf . It is an occupational disease of rabbit butchers. The skinning of infected rabbits and hares is the common means of human infections. In USA, about 2000 cases of human tularemia are reported each year .The first sign in man is usually a papule at the site of initial infection, often a finger which ulcerates. The organisms are carried by lymph notes which enlarge, become painful and may suppurate. The disease is accompanied by fever, headache, muscular pain and lasts for 2-4 weeks. Mortality rate in ulceroglandular form is 5%. Transmission occurs by direct inoculation of organisms into the skin, mucous membrane or conjunctiva with diseased animals or their excretions/ tissues. Bite of infected arthropods can also transmit the disease .

Tetanus- It is a bacterial disease, caused by *Clostridium tetani* , a Gram positive, spore forming, anaerobic organism. Natural infection is recorded in horse, sheep, cattle and pig . The pathogen enters the body through the infection of the wound, injury or laceration with soil or dust contaminated with spores of *C. tetani*. Incubation period is 4-10 days. The tightening of jaw muscles is the earliest sign of tetanus. The disease in man is called "Lock Jaw". The fatality rate is high in patients who delay medical attention. The disease is prevalent in all parts of the world as the organism is a common inhabitant of the soil. Endospores ofC.tetani can be killed by autoclaving at 1210C for 15 minutes. Immediate attention to wound/injury is imperative .

Melioidosis- It is caused by the bacterium, *Burkholderia pseudomallei*, a Gram negative, motile, aerobe .Man acquires the infection by direct contact of injured skin with contaminated soil or water. Inhalation of infectious dust through respiratory tract can also produce disease. Disease occurs in camel, goat, horse, pig, sheep and kangaroo. Organism can survive for many months in soil and water. The patient develops vesicles and pustules on hand and feet. Pulmonary, extrapulmonary and septicemic forms are seen. Disease is highly endemic both in man and animals in Australia.

Staphylococcal infection- It is primarily caused by*Staphylococcus aureus*, a Gram positive, non-sporulated and non-motile bacterium .The disease is recorded in cattle, buffalo, camel, horse, goat, pig, rabbit and poultry. Abattoir workers usually contracts infection while dealing with carcass of food animals. Lesions may appear as boil, pustule, furuncle and abscess on hands and arms of the workers. In India,*S. aureus* was isolated from the skin lesions of a meat inspector and a butcher

Streptococcal meningitis-The disease is caused by*Streptococcus suis*type-2. The infection is reported in meat factory workers, abattoir employee's butchers and veterinarians. The bacteria enter the skin through cut, wound or abrasion. The patient may show febrile condition with severe headache, numbness of fingers, rigors, erythema besides meningitis, septicemia, arthritis, lymphangitis. Patients respond slowly with antibiotic treatment over a period of several days. The disease is widespread in pigs in U.K. The infection has also been encountered in goat, horse and sheep.

Tuberculosis- *Mycobacterium bovis* is the cause of bovine tuberculosis which is a major zoonosis. Global distribution of *M. bovis* infection in animals and humans varies widely. In some of the developing countries, *M. bovis* is responsible for 5-10 % of all human tuberculosis .Disease is reported in cattle, buffalo, sheep, goats, horse, pig and deer.

Transmission of *M. bovis* from animal to man takes place due to direct exposure to infected animal or carcass at abattoir (occupational exposure). Tuberculus lesions occur in the skin, tendon and localized lymph mode among the persons who handle infected carcass in the slaughterhouse. The cutaneous tuberculosis was diagnosed and treated in an abattoir worker in India

Necrobacillosis- It is a bacterial disease of cattle, goat, horse, pig and man. Transmission of disease in human takes place by direct contact of the injured or abraded skin or wound to infected tissues of animals. Disease is caused by*Fusobacterium necrophorum* which is a Gram-negative, non-sporulated, anaerobic organism. Necrotic pustules develop at the site of inoculation of the organism.

Dermatophilosis- Disease is caused by *Dermatuphilus congolensis* which is a facultative anaerobic actinomycete. The condition is sometimes referred to asrain scald" as it often looks like raindrops have just fallen on the skin. The condition is initially seen as pustules that are often over-looked. However, the pustules quickly come to together to form large oval crusts as the longer hairs become stuck together in the scab .It has worldwide distribution; and is recorded in cattle, goat, sheep, horse, camel, deer and rabbit . Transmission of disease in man occurs by direct contact with infected animal. Skin infection in man is localized. The most common locations of the lesions in cattle are rump, topline, udder and teats as well as the belly.Lesions in humans occur as discrete, cutaneous pustule on the hands, fore arms or legs .

Chlamydiosis: It is a highly infectious global disease caused by *Chlamydohphila psittaci*, which is an intracellular organism. The infection has been reported in cattle, sheep, horse, goat, pig, buffalo, and birds .Human exposure to infective aerosols or dust-infected bird droppings and nasal discharges and sheep fetuses and membranes can result infection .The persons working in domestic poultry processing plants are at a greater risk of contracting the disease. The disease may vary from being inapparent, as also recognized by the Centre for Disease Control and Prevention (CDC) to fatal in untreated patients. Symptoms include fever, gastrointestinal pain, vomition, headache, arthralgia, keratoconjunctivitis and anorexia .Mild attacks of disease may be confused with influenza. The disease is rarely fatal in properly treated patients. Therefore, early diagnosis and awareness are important. In India, the chlamydial infection was diagnosed in a bird handler .

Q fever-It is an important rickettsial disease caused by*Coxiella burnetii.* The organism remains infective in wool and farm dust for very long periods. Disease occurs most often in cattle, goats and sheep. Inflection in man can occur through inhalation, direct contact or bites of ticks .The symptoms usually take the form of fever, anorexia, chills, frontal headache, myalgia, weakness, cough, chest pain, pneumonia and severe sweating. In more severe cases, pericarditis, endocarditis, and hepatitis is observed. In a study of workers in an abattoir in Edinburgh, 21.1 % showed antibodies to phase 2 antigen of *C. burnetii*. Australian studies in the last two decades have shown evidence of infection in abattoir workers .

Cowpox- This viral zoonosis is caused by **cowpox virus** (DNA virus) and man acquires infection by direct contact with diseased cattle. Cowpox is an acute viral disease characterized by typical vesicular eruptions on the skin and mucous membrane. In man, erythema, vesicle, pustule and scab formation is noticed.Lesions occur on the hands, arms and face and; often accompanied by lymphadenitis and fever. The disease is self-limited. The typical cowpox lesions were observed on the hand of a butcher.

Pseudo cowpox- Also known as “**Milker’s nodules”** is caused by **Pseudocowpox virus** (DNA). Infection occurs by occupational exposure. Papule occurs on the finger, hand or arm disease is mild and self limiting .

Buffalo pox:It is caused by **Buffalo pox virus (DNA)** which belongs to family Poxviridae. Disease was noted first time among water buffaloes in India in 1934. Buffalo pox isalso reported from Indonesia, Pakistan, Egypt and Russia. The virus is transmitted to man by direct contact with infected food animals .Multiple pocks lesions occur on fingers, hands, wrists and fore arms. Occasionally, lymphadenopathy may occur .

Camel pox- It is a viral zoonosis caused by Camel pox virus (DNA) of familyPoxviridae. The disease is reported from several countries of Africa, Asia and Middle East .Human infection can occur by direct contact with diseased camel.Pox lesions occur mainly on the hands of animal handler .

Contagious ecthyma: - It is an occupational disease caused byOrfvirus (DNA) of family **Poxviridae.** Disease is reported in camel, goat and sheep. The virus can enter through abraded /injured skin .Man gets infection by direct contact with diseased animal. Most cases are encountered in adults especially men. Disease occurs frequently in abattoir workers, butchers and meat handlers. Papule, vesicle and pustule occur mainly on the finger, hand, wrist, fore arm and sometimes on the face .The lesions heal in 15-30 days,;and occasionally, ocular lesions may occur .

Foot and mouth disease- It is a economically important infectious disease caused by FMD virus (RNA) of Family **Picornaviridae ;** and is reported in cattle, buffalo, camel, goat, sheep, pig and deer [2]. Transmission of infection occurs by close contact of abraded skin with diseases animals or their discharges. Animal hides can be a source of virus for a long period of time. It is a mild disease in man and vesicles occur on the finger, palm of hand, sole of feet or oral cavity [2].

Newcastle disease - It is a fatal disease chicken, caused by Newcastle disease virus (RNA) of Family **Picornaviridae.**Persons working in poultry slaughter house get the infection. Transmission occurs by handling diseased poultry or carcass. The affected person shows redness, watery discharge from eye, edema of eye lids, conjunctivitis and subconjunctival haemorrhage . The conjunctivitis due to Newcastle disease virus was diagnosed in a poultry handler from India .

Rift Valley fever - The disease was first described in 1931 from Kenya and is caused by Rift Valley fever virus (RNA) of Family **Bunyaviridae.** Man gets infection by direct contact with diseased animals or infected tissues .Bite of mosquitoes can also produce disease in man and animals. Disease is recorded in cattle, camel, goat and sheep. Clinical signs in man include fever, headache myalgia, nausea, vomition, retinitis, jaundice and encephalitis .

Swine influenza- Also known as swine flu or pig flu is a highly infectious emerging viral zoonosis of global significance and is caused by a new strain of influenza A virus subtype H1N1, a RNA virus of the family **Orthomyxoviridae.** These viruses infect a variety of species including humans, swine, equine species, aquatic mammals and birds.Evidence exists demonstrating transmission of influenza viruses from humans to pigs as well as from pigs to humans .Influenza A virus transmission between humans and swine often results in the emergence of new strains with the potential of spreading among both populations .Persons, who work with diseased pigs (swine abattoir employees, pork producers, veterinarians), are at risk of catching swine flu .Clinical signs in man include fever, chills, sore throat, cough, severe headache, running nose, weakness, besides vomition and diarrhoea .Severe fatal infections are recorded in adults between the ages of 30 to 50 years. Continuous exposure of the abattoir workers to animals suffering from subclinical influenza virus infections provided an environment conducive to the occurrence of interspecies infections and the emer-gence of new potentially pathogenic viral variants .Pregnant women are at greater risks of acquiring swine flu. Control measures include use of face mask and gloves at pig farm, personal hygiene, good biosecurity measures and vaccination of pigs

Louping ill: - It is a **viral disease** of cattle, goat, horse, pig, sheep, deer and man; and is caused by Louping ill virus (RNA) of Family**Togaviridae.** Infection can occur by inhalation, ingestion or needle prick.Infection in sheep is transmitted by ticks. Cuts and abrasions frequently predispose the workers to infection .Disease is encountered in **abattoir worker,** and **slaughter man.** The affected person exhibits signs of fever, headache, flu-like symptoms, conjunctivitis and meningoencephalitis .In Scottish abattoir workers in 1966, an incidence of 8.3 % of louping ill infection was reported.

Bird flu- It is a viral zoonosis . Infection can be contracted with sick or dead birds and also during slaughtering, feathering and dressing of diseased birds. Disease is caused by influenza type A virus (H5N1) belonging to family**Orthomyxoviridae (RNA).**Patient shows high fever, chest pain, respiratory distress and hoarse voice. Biosecurity measures at poultry farm and use of protective clothing can help in control of disease .

Dermatophytosis (Ringworm) - This is most commonly encountered mycotic disease of humans and animals. It is caused by dermatophytes of Genus *Microsporum*and*Trichophyton.* Disease is diagnosed in cattle, buffalo, deer, horse, pig, goat, sheep and rabbit. Human gets infection by having **direct contact with diseased animals**.Lesions on the skin show redness, scaling, vesicle, irritation and itching . If hair is involved, hair shaft become fragile and break off a short distance above the skin leaving short stub in a bald circular patch. Hands, arms and forearms are mostly affected .Lesions occur on human head or body, and on most species, are circular or annular because of central healing. Usually scaling or hair loss or breakage, occasional itching occurs. Sometimes erythema, induration, crusting, or suppuration occurs. Maceration, abrasions, and cuts of the skin constitute the predisposing factors . Several cases of ringworm due to *Trichophyton verrucosum* were diagnosed in abattoir workers.

Aspergillosis: Commonly known as brooder pneumonia is a highly fatal fungal disease and is principally caused by *Aspergillus fumigatus*. Infection is mainly acquired by inhalation. Sick person exhibits signs of low grade fever, productive cough breathlessness and haemoptysis. Pulmonary aspergillosis in a poultry worker who was occupationally exposed to fungal spores of *A.flavus* also recorded .Use of face mask in poultry processing industry can prevent *Asperillus* infection

Scabies- It is an **ectoparasitic zoonosis** caused by*Sarcoptes scabiei.* Disease is world wide in distribution and is reported in cattle, buffalo, camel, sheep, goat and pig . Transmission occurs by direct contact with scabietic animal . Lesions occur on the hand and arms and rarely on parts of body. It consists of papules and vesicles and may be intensely pruritic.

During the evisceration step, the spread of zoonotic bacteria out of carcass surface is mainly due to the presence of carrier animal and, therefore, in the slaughterhouses, good hygienic practices and equipment cleanliness are relevant both to prevent and to reduce the level of carcass contamination. The slaughterhouse is an environment that is conducive to the transmission of zoonotic diseases to humans, for in few other locations are so many animals brought into close contact with so many people. As discussed above, slaughterhouse workers directly contact animal tissues and so it would be expected that they would be exposed to these agents regularly. Exposure to these agents may also be indirect (environment and processing equipments). There should be adequate disease reporting systems and sufficient collaboration and communication between human health and veterinary specialists. Vaccination, standard and additional precautions (also known as universal precautions), hand-washing, education and training and the use of personal protective equipment where appropriate are the main control strategies for the prevention of occupationally-related infections in the abattoir employees. As animal infections are often a sentinel for human infection, the veterinarians should report the incident of zoonotic disease to the health authority

Synonyms

Tuberculosis	❖ Pearl disease, scrofula, kings evil, Pthisis
Anthrax	❖ Splenic fever, Milzbrand, wool sorters disease
Avian psittacosis and ornithosis	❖ Chlamydiosis, parrot fever
Botulism	❖ Lumziekte, limberneck, western duck sickness
BovineSponjiform encephalomylitis	❖ BSE, Mad cow disease, prion disease
Brucellosis	❖ Bangs disease, Contagious abortion, Undulant fever
Colibaccillosis	❖ White scour, joint ill, naval ill
Congo fever	❖ Crimean Congo haemorrhagic fever, central Asian haemorrhagic fever
Contagiouspustular dermatitis	❖ CPD, Contagious ecthyma, ORF, sour mouth
Cysticercus bovis	❖ Measly beef
Cysticercus cellulose	❖ Measly pork
Fascioliasis	❖ Liver fluke
Foot and mouth disease	❖ Apthous fever
Glanders	❖ Farcy
Hydatidosis	❖ Disease of soiled hands, Echinococcosis
Hypoderma bovis	❖ Warble fly

Hypoderma lineatum	❖ Heel fly causes grubby gullet
Leptospirosis	❖ Weils disease, Canicola fever, Haemorrhagic jaundice
Listeriosis	❖ Circling disease
Lucilia sericata	❖ Sheep maggot fly , fly blown sheep
Ovine encephalomyelitis	❖ Louping ill
Pneumonic pasturellosis	❖ Shipping fever, snuffles, rhinitis in rabbits
Pox disease	❖ Variolae
Pseudopox	❖ Milker’s nodule, Paravaccinia
Pseudorabies	❖ Mad itch, infectious bulbar paralysis, Aujeszkys disease
Q fever	❖ Query fever, abattoir fever, balken influenza, balken grippe
Rabies	❖ Hydrophobia, Lyssa
Rift valley fever	❖ Enzootic hepatitis
Ring worm	❖ Taenia capitis, kerion, favus
Sarcocyst	❖ Miescher’s tubes
Sarcophaga carnaria	❖ Grey meat fly
Scrapie	❖ RIDA, tremblante Du mouton
Swine erysipelas	❖ Diamond skin disease
Tetanus	❖ Lock jaw
Tularemia	❖ Rabbit fever, Deerfly fever, O’hara disease
Tyroglyphus farinae	❖ Flour mite

High- and low-risk animal waste

Material can be divided into high- and low-risk animal waste

Examples of high- risk animal waste

1. Animal that died or killed on the farm but not slaughtered for human consumption
2. Animal killed in context of disease control
3. By products including blood originating from animal which have been condemned on meat inspection
4. Carcass that have not passed meat inspection including those with unacceptable levels of residues
5. Contaminated meat
6. Parts of carcass except hides, skins, hooves, feathers, wool, horn, hair, blood and similar products of animals slaughter in the normal way which are not presented for post-mortem inspection

Examples of low-risk animal waste

Low risk animal waste are products other than high-risk animal waste, including hides, skins, hooves, feathers, wool, horn, hair, blood and similar products when used in the manufacture of feeding stuffs.

Animal waste must be collected transported in suitable containers or vehicle which prevents leakage. The containers or vehicle must be adequately covered and all must be maintained in clean condition. The container should be labeled for "*not for human consumption*".

The high-risk material should be processed in a high-risk processing plant, which has been approved by government authorities or disposed by burning or burial where transport to the nearest high-risk material processing plant might expose a risk to environment or quantity and distance to be covered does not justify collecting the waste. Burial must be deep enough to prevent carnivorous animal from digging it up and must be in suitable ground to prevent contamination of water tables or any other environment problems. Before burial the waste must be sprinkled with an approved disinfectant.

Low-risk material must be processed in a high-risk or low-risk processing plant, in a pet food plant or plant preparing pharmaceutical or technical products, or disposed by burning o burial

DISPOSAL OF UNSOUND MEAT

Unsound meat means a meat including offal those are condemned in the slaughterhouse as unfit for human consumption, dead animals arriving at the slaughterhouse or fallen animals found in the villages**.**

The unsound meat or condemned meat must be disposed hygienically to avoid spread of disease in the area and increasing environmental pollution. The condemned meat can be converted into a fertilizer or a feed supplement like meat meal, bone meal or carcass meal. The animals dying of deadly disease like anthrax are subjected incineration or ashing at a very high temperature so that the spores are completely destroyed.

Various disposal methods are being used throughout the world depending upon the legislations which are followed in different countries. In developing countries like India the most widely used disposal methods are the traditional methods like: burying, burning, incineration, rendering and composting.

There are some environmental, biosecurity, social and economic issues associated with these methods. Environmental constraints associated with these disposal methods are like: contamination of air, soil and water particularly due to persistency of some infections like TSE (Transmissible Spongiform Encephalitis) and other prion infections. Social concerns with these traditional disposal methods are: odour and fly menace and contamination of drinking water. Similarly the economic constraints are associated with the alarming increase in the costs of raw materials like: kerosene, diesel and wood for burning. Issues are also related with the labor cost, availability of land and transportation of mortalities to site of disposal.

Systems of disposal can be

- Burial
- Incineration
- Rendering

The process of converting dead animals or condemned meat in to value added products like carcass meal are referred as rendering. Actually rendering in earlier days was the process of recovery of fat from dead animals, but now a day the definition also covers the production of carcass meal.

Disposal methods

In order to protect the health of herds and farm personnel, avoid air, soil and water contamination and avoid problems with both the agricultural and non-agricultural neighbors, biologically and environmentally safe methods of dead animal disposal must be employed on animals.Each of these methods has its unique flaws.

- Burial of dead birds in a pit can lead to ground water contamination.
- Incineration is expensive and can potentially pollute the air.
- Rendering dead birds into by-product meal is constrained by transportation cost and
- restrictions on filling is subject to land availability and limitations on diseased carcass movement.
- These different methods in general have other disadvantages as well like cost involvements, labour intensiveness, production of environmental pollutants and obnoxious odour .
- Non traditional and novel methods of disposal are a good option for eco friendly disposal .

Disposal by burial:

Burial is the most common and perhaps least expensive method of dead animal disposal.A pit is dug, into which carcasses are placed. Deep burial is generally recommended. The practice of covering dead animals with lime retards decomposition and is not recommended. Dead animals should never be buried is areas where leaching can occur.The different factors like soil type, permeability, water table depth and rainfall significantly affect the movement of pathogens fromdisposed carcass through soil to groundwater . The use of hydrated lime $Ca(OH)2$) in burial has been effectively reduced the survival of pathogens .

Excessive use of lime both during the construction and subsequent operation of burial sites may reduce the growth of all micro-organisms and hence slow the process of decomposition. Despite the low incidence of drinking water contamination due to burial of carcasses, some infectious material such as anthrax spores or prions can be seen within the soil after carcass decomposition. This may lead to ingesting of contaminated soil and the infectious agents by animals and hence may lead to development of neurodegenerative disease (e.g. BSE or scrapie) in the case of prions or the reintroduction of anthrax.

Burial sites should be located away from livestock fields and at sufficient depth so that the potential for transfer of infectious agents back to the surface is very low. Problems with burial as a method of dead animal disposal include odor and accessibility of scavengers to "dead pits" that are not properly covered. There is also the possibility of significant ground and surface water contamination, for which producers may be held liable.

Incineration: Incineration is burning of animal carcasses at **high temperatures (>850℃)** to inorganic ash. The process destroys all infective agents. Ash represents only 1-5% of initial carcass volume ,although this will depend on the incinerator type, process, fuel and animal species. The principal concern with incineration of carcasses is the gaseous emissions key pollutants represent 60.2% of the total air emissions .Adoption of optimum techniques like use of afterburners has significantly achieved reductions in harmful gases like **polycyclic aromatic hydrocarbon** . Other health concerns with incineration include the release of dioxins and furans released due to incomplete combustion and can settle in areas around carcass incinerators and could enter the food chain through grazing animals or through human consumption of contaminated crops.From a human and animal health perspective, the high temperature of incineration also completely destroys zoonotic and animal pathogens, including resilient spore-forming bacteria such as *Bacillus anthracis* (anthrax) .

Incinerators completely eliminate carcasses and destroy pathogens. In general, incinerators are expensive to buy and to operate. Certain type of incinerators may generate air pollution and objectionable odors. Complete combustion of carcass for safe disposal is achieved when sufficient labour, air and fuel is provided Human health risks associated with on-farm burning (apart from physical burns and direct smoke inhalation) include the emission of dioxins from incomplete carcass combustion. Dioxins and furans are carcinogens and can negatively affect human reproduction, development and immune systems

Disposal by composting: Composting process involves the layering of carcasses between strata of carbon-rich substrate such as straw, sawdust or rice hulks with a final covering of carbon-rich substrate over the entire pile . Larger carcasses are typically placed in single layers while poultry can be multi-layered; and the compost piles are subsequently aerated or turned . Depending on carcass weights, the waste material may decompose at rates as high as 1-2 kg /day into a useful product that can be used as a soil amendment.

The process essentially occurs in two phases – a primary, thermophilic phase (temperatures up to 70°C generated for a number of weeks) and a secondary, mesophilic phase (typically 30-40°C) for a number of months

The action of thermophilic, aerobic bacteria converts nitrogen rich (e.g. Dead animals) and carboniferous (e.g. Straw, Sawdust, etc.) materials into humic acids, bacterial biomass and organic residue (compost).The temperatures generated during the thermophilic phase of carcass composting has been shown to effectively reduce numbers of bacteria, viruses, protozoa and helminthes

During the composting process, heat, CO_2 and water are generated as by products. The resulting product is free from harmful pathogens, is nutrient rich and can be used as fertilizer. The poultry and swine industries have adopted composting as the method of choice for ridding farms of dead animals.

In the typical system, carcasses are placed in a bin containing sawdust creating an ideal environment for the growth of the aforementioned bacteria. The optimal carbon to nitrogen ratio for bacteria is approximately 30 : 1.

Bacteria action rapidly heatthe compost piles to temperatures as high as 160°Fand within several weeks carcasses are reduced, leaving only brittle bones which are easily crumbled. Turning the compost pile by moving to a new bin (i.e. Secondary bin) after 2 weeks helps to maintain high temperatures and promote even further decomposition.

Carcass utilisation plant

Rendering

Rendering is the process of crushing carcasses into small uniform size, heating the particles and separating out the fat, proteinaceous material and water to make useful products like meat and bone meals and tallow. There are two types of technologies for utilisation of carcass they are dry rendering and wet rendering. Continuous low temperature dry rendering is used for producing better quality fats.

In case of large animals like cattle, buffaloes, there is a significant value of dead animals. The value is in the items such as Hide, meat meal, bone meal, tallow hoof and horn.

The separation of hide, horn and hoof are undertaken and the balance carcass is conveniently discarded.The discarded items also have value and the same can be processed.

The carcass rendering plant will be handling feedstock devoid of hide, horn, hoof and rumen contents. The material contains protein, fat and water indifferent proportions and rendering is a process to recover proteins and fat while eliminating water. If rendering is not done the material gets putrefied.

The yield of the final product meat meal, bone meal and tallow depends largely on the weight of the animal and level of nutrition.

WET RENDERING	**DRY RENDERING**
In wet rendering the material is cooked in added water,	In dry rendering the material is cooked in its own water.
Temperature of about 110^0c and pressure of 25 psi, for 40 minutes.	Temperature of about 140^0c and pressure of 45 psi, for 20 minutes Better sterilisation of material at higher temperature
The cooker is vertical type	The cooker is horizontal type
The recovery of fat is better than dry rendering. To produce good grade tallow, viscera must be cut and washed.	The recovery of fat is not better than wet rendering. Tallow is dark coloured. More tallow is lost in dry rendered meat
Yield of carcass meal is less than dry rendering as 25 % of meat is lost in the gravy.	Yield of carcass meal is more than wet rendering

Continuous low temperature dry rendering is used for producing better quality fats. This system uses heating, separation and cooling on a continuous basis, and is usually regarded as the ideal process. The process involves mincing of raw material, melting by live steam injection at **90^0 C**, continuous separation of solids from the liquid fat in a decanter centrifuge, further heating, centrifugation to remove the fines; and cooling in a plate heat exchanger to below solidification point.

A prime consideration in the rendering of fat is the prevention of the breakdown of fat into fatty acids and glycerides by the action of the enzyme lipase which is active at $40\text{-}60^0$ C. Above 60^0 C lipase is inactivated. The continuous low temperature system utilises this action and at the same time minimizes undesirable chemical activity, burning, oxidation and off flavours. This process requires less floor area and less labour than dry batch rendering.

Carcass received
↓
Removal of hide, horn and hoof, evisceration and removal of dung and intestinal contents in sheep, goat, cattle and buffaloes
↓
Manual breaking of carcass in pieces
↓
Feeding pre-breaker to reduce size to about 60 mm
↓
Loading pieces in to dry renderer
↓
Cooking at 100^0C for 20 minutes

<table>
<tr><td colspan="2" align="center">Sterilisation at 140^0C, 45 PIS for 20 min.
Drying at 100^0C for 60 min.</td></tr>
<tr><td colspan="2" align="center">↓</td></tr>
<tr><td colspan="2" align="center">Cooked material emptied into percolating tank</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td align="center">Cracklings</td><td align="center">Fat</td></tr>
<tr><td align="center">↓</td><td align="center">↓</td></tr>
<tr><td align="center">Centrifuged</td><td align="center">Fat removed</td></tr>
<tr><td align="center">↓</td><td align="center">↓</td></tr>
<tr><td align="center">Wet material pressed and moisture reduced and then Dry material is pulvarised (powdered)</td><td align="center">Fat settling tank</td></tr>
<tr><td align="center">↓</td><td align="center">↓</td></tr>
<tr><td align="center">Meat cum bone meal</td><td align="center">Technical fat</td></tr>
</table>

The disadvantage of Continuous low temperature dry rendering is that this system cannot apply pressure hence it cannot sterilize by pressure-cooking and cannot hydrolyse hair and wool.

Rendering Process – 4 hours run
Yield of carcass meal – 30-35 %

Percent yield and composition of poultry by-products meal

Constituent	Feather meal	Blood meal	Offal meal	Mixed by-products meal
Yield	33	18	55	33
Moisture	7	9	10	8
Crude protein	85	86	52	66
Crude fat	3	1	24	18
Ash	4	3	14	1.8

By-products available in poultry industry

- **Poultry dressing plant waste (on live weight)**
- Feathers 6%
- Blood 3.5%
- Offal
- Head 3%
- Feet 4%
- Inedible viscera 9%
- (Intestine, lung, pancreas spleen etc.)
- **Hatchery waste** (infertile eggs, dead in germs, dead embryos, eggshell, un-hatched chicks, un-usable chicks etc.
- **Egg processing plant waste** (unsound eggs and egg shell)
- **Poultry manure** (used up litter and wet droppings from cage houses)
- Dead birds

Applications of poultry industry waste

Feather	Livestock feed, as bedding, fertilizer, ornamental use and in sports	
Blood **(8% moisture)**	Livestock feed, as fertiliser, fish bait	
Offal	Livestock feed, pet food, fish food	
Mixed poultry by - products meal **(Moisture 8%)**	Poultry feed	At 5-7 % level
Hatchery waste Yield 25 % Protein 35% Fat 40% Calcium 0.05% Phosphorous 1%	Poultry feed	At 3% level

Poultry manure	Fertilizer	
Moisture 15-18 % Protein 25-30% Uric acid 6-8 % Ash 15-25 % Calcium 3-6 % Phosphorous 1.5-2%	In poultry feed (layer ration)	Up to 15%

BIOHAZARDOUS WASTE

Solid waste contaminated with infectious agents known to cause human illness and not contaminated with radioactive materials or hazardous chemicals.

In a research laboratory, this would include solid waste generated from any work with human or non-human primate blood, tissue, or cells and microbiological agents that may cause human illness. This also includes animal carcasses infected with infectious agents known to cause human illness. The National Institute of Health has created a list of microbiological agents that is categorized into 4 major risk groups.

Agents that fall within Risk Groups 2,3 and 4 are considered biohazardous and should be treated as such. Agents that fall within Risk Group 1 are considered as biowaste and do not fall under the biohazardous waste requirements although certain precautions still need to be followed.

Classification of Biohazardous Agents by Risk Group (RG)

Risk Group 1 (RG1) Agents that are not associated with disease in healthy adult humans.

Risk Group 2 (RG2) Agents that are associated with human disease which is rarely serious and for which preventative or therapeutic interventions are often available.

Risk Group 3 (RG3) Agents that are associated with serious or lethal human disease for which preventive or therapeutic interventions may be available (high individual risk but low community risk).

Risk Group 4 (RG 4) Agents that are likely to cause serious or lethal human disease for which preventive or therapeutic interventions are not usually available) (high individual risk and high community risk)

<u>Types of Biohazardous Waste</u>

Biohazardous waste includes the following materials:

1.Human blood and blood products: All human blood, blood products (such as serum, plasma, and other blood components) in liquid or semi-liquid form. Items contaminated with blood that, if compressed, would release blood in a liquid or semi-liquid form, or items caked with dried blood capable of being released during handling. Other body fluids or tissues containing visible blood.

2.Human Body Fluids: Human body fluids in a liquid or semi-liquid state, including: semen, vaginal secretions, cerebral spinal fluid, synovial fluid, pleural fluid, pericardial fluid, peritoneal fluid, amniotic fluid, and saliva from dental procedures. Also includes any other human body fluids visibly contaminated with blood, and all body fluids in situations where it is difficult or impossible to differentiate between body fluids.

3.Microbiological Wastes: Laboratory wastes containing or contaminated with concentrated forms of infectious agents. Such waste includes discarded specimen cultures, stocks of etiologic agents, discarded live and attenuated viruses, blood or body fluids known to contain infectious pathogens, wastes from the production of biologicals and serums, disposable culture dishes, and devices used to transfer, inoculate and mix cultures (BSL-1 through BSL-4 etiologic agents as designated by NIH Guidelines/BSC).

4.Pathological waste: All human tissues, organs, and body parts, including waste biopsy materials, tissues, and anatomical parts from surgery, procedures, or autopsy. Any unfixed human tissue, except skin.

5.Animal waste: All animal carcasses, body parts, and any bedding material from animals known to be infected with, or that have been inoculated with human pathogenic microorganisms infectious to humans.

6.Sharps waste: The wastes must be treated, packaged, labeled, and transported the disposal area.

TOXIC RESIDUES IN MEAT

Antibiotic Residues - A Global Health Hazard

Use of Antibiotic that might result in deposition of residues in meat, milk and eggs must not be permitted in food intended for human consumption. If use of antibiotics is necessary as in prevention and treatment of animal diseases, a withholding period must be observed until the residues are negligible or no longer detected. The use of antibiotics to bring about improved performance in growth and feed efficiency, to synchronize or control of reproductive cycle and breeding performance also often lead to harmful residual effects. Concern over antibiotic residues in food of animal origin occurs in two times; one which produces potential threat to direct toxicity in human, second is whether the low levels of antibiotic exposure would result in alteration of microflora, cause disease and the possible development of resistant strains which cause failure of antibiotic therapy in clinical situations.

A withdrawal period is established to safeguard human from exposure of antibiotic added food**. The withdrawal time is the time required for the residue of toxicological concern to reach safe concentration as defined by tolerance.** It is the interval from the time an animal is removed from medication until permitted time of slaughter. Heavy responsibility is placed on the veterinarian and livestock producer to observe the period for a withdrawal of a drug prior to slaughter to assure that illegal concentration of drug residue in meat, milk and egg do not occur.

Use of food additives may improve feed efficiency 17% in beef cattle, 10% in lambs, 15% in poultry and 15% in swine. But their indiscriminate use will produce toxicity in consumers. WHO and FAO establish tolerances for a drug, pesticide or other chemical in the relevant tissues of food producing animals. The tolerance is the tissue concentration below, which a marker residue for the drug or chemical must fall in the target tissue before that animal edible tissues are considered safe for human consumption. Tolerances are established based on extensive toxicological studies of potential hazards of consumption to human.

Antibiotics as Growth Promoter

The antibiotics nowadays used for improved performance in growth especially in broilers and fatteners. They may produce improved growth rate because of thinning of mucous membrane of the gut, facilitating better absorption, altering gut motility to enhance better assimilation, producing favorable conditions to beneficial microbes in the gut of animal by destroying harmful bacteria. Antibiotics also favour growth by decreasing degree of activity of the immune system, reduced waste of nutrients and reduce toxin formation. In most of the cases only young growing animals and poultry are responsive to antibiotic mediated growth promotion.

Antibiotics in Therapeutics

Indiscriminating use of antibiotics in all cases of pyrexia, inflammation, wounds and viral diseases have widespread residual effects on edible tissues. The use of antibiotics only in specific conditions is justified because the roll of microbial agents is mainly to kill the rapidly dividing invading cells. Antibiotics in Prophylaxis Animals and poultry are receiving sub therapeutic levels of antibiotics to prevent possible infection. But the antibiotics are specific to their spectrum of activity only in the active multiplying stage of bacteria. But it will not provide overall protection. Only in certain cases like dry cow therapy and surgical procedures are wanting of antibiotic prophylaxis.

Miscellaneous use of Antibiotics

Antimicrobials are used either directly or indirectly during the production processing and storage of milk and milk products. Direct contamination of milk may occur from air and water during processing, storage and transportation. Besides feed given to animals is also source of indirect contamination. Man will be the ultimate consumer of these antibiotic residues. There are some causes of miscellaneous use like lack of awareness, lack of extension activities, inadequate literature supplied by manufacturers, lack of safer drugs and exploitation of more production and profit from animals. FDA prohibits the extra label use of chloramphenicol, furazolidone, nitrofurazone, sulphonamide drugs, and flouroquinolones in lactating animals.

Techniques used for Detection and Analysis of Drug Residues

- ELISA
- HPLC
- Liquid chromatography
- Gas chromatography
- Paper chromatography.

Pathological Effects produced by Antibiotic Residues in Food

- Transfer of antibiotic resistant bacteria to the human.
- Immunopathological effects
- Autoimmunity
- Carcinogenicity (Sulphamethazine, Oxytetracycline, Furazolidone)
- Mutagenicity
- Nephropathy (Gentamicin)
- Hepatotoxicity
- Reproductive disorders
- Bone marrow toxicity (Chloramphenicol)
- Allergy (Penicillin)

Residues Prevention

- The first step in residue prevention is to make individuals and organizations aware of the problem through education by veterinary personnel, organizations, and literatures and governmental agencies.
- Rapid screening procedures for the analysis of antibiotic residues and instant grading and prohibition of food containing antibiotics more than MRL.
- Processing of milk help for the inactivation of antibiotics. Refrigeration causes disappearance of penicillin. In pasteurization most of antibiotics will loose activity.
- Use of activated charcoal, resin and UV irradiation also help for antibiotic inactivation. - Irrational use of antibiotics in field veterinary practices should be avoided.

- Development of simple and economic field test to identify drug residue in edible animal products. Ethno-veterinary practices may be promoted.
- Nation wide monitoring and periodic surveillance of microbial residue in edible tissues and milk.

CHEMICAL RESIDUES IN MEAT AND MEAT PRODUCTS

Meat and meat products are important for nutrition and the human diet, but are also one of the major routes of human intake of contaminants. Contaminating substances may enter the food chain at many different stages. Through various constituents like fertilizer ingredients and contaminants, irrigation water, contaminants and pesticides can enter food crops through plant roots. Contaminants in forages and other feeds can be transmitted to animal products. Veterinary drugs can leave residues in animal products. Environmental chemicals such as heavy metals (e.g., lead and mercury) from many sources have sometimes been found as food contaminants. Thus range of contaminants/toxicants found in food is varied but can be broadly subdivided into:

- Food processing by products;
- Food additives, processing aids and those chemicals that may migrate from packaging.
- Agricultural and veterinary chemicals used in primary production;
- Natural chemicals found in plants, fungi or bacteria associated with plants
- Environmental contaminants, including heavy metals and organic contaminants.

Food processing toxicants During the processing of foods, products may be produced that, if present in large amounts, could potentially adversely affect health. For example, cooking certain meats at high temperatures creates chemicals that are not present in uncooked meats. A few of these chemicals may increase cancer risk, such as polycyclic aromatic hydrocarbons and heterocyclic amines.

Chemicals used in processing Food additives and processing aids are used in the manufacture of a wide range of meat products. Food additives may be added to achieve a technological function, such as preservation or colouring, and are present in the final food, whereas processing aids fulfil a technological function during processing, but are not present in the final food. The use of food additives and processing aids is regulated in the Code by maximum permitted use\ levels or according to Good Manufacturing Practice (GMP)

Veterinary Drugs and Growth Promoters Residues Veterinary pharmaceutical drugs have been used for a long time in animal production as therapeutic agents to control infectious diseases or as prophylactic agents to prevent outbreaks of diseases and control parasitic infections .Meanwhile, growth - promoting agents like the anabolic agents are added to improve the feed conversion efficiency by increasing the lean - to - fat ratio, while antimicrobial agents are added to make more nutrients available to the animal and not to the gut bacteria.

Hormones are used for therapeutic and growth modifying purposes in animals. They are important because they may be associated with neoplasia. The beta agonists are used to produce less fatty meat from cattle as they have the activities of neurotransmitter and of hormones and as such have both physiological and metabolic activities. They are important in residue analysis because they have major metabolic effects by repartitioning energy from fatto lean meat production. There are about 20 beta-agonists e.g. cematerol, clenbuterol, salbutamol, mabuterol which are added to cattle feed to reduce the fat content of the carcass liked by the consumers. For this effect, the dose required is several times greater than therapeutic. At this concentration residues accumulate in the edible tissues of cattle like kidney or liver.

Environmental Contaminants There are a wide variety of environmental contaminants. The main concern is that they may be present in the feeds consumed by farm animals and thus contaminate the resulting meats.

Pesticides Some well known contaminants are dioxins, organophosphorous, and organochlorine pesticides. These contaminants are quite extensive worldwide, making their control very difficult. The term **"persistent organic pollutants" (POPs)** was defined by the United Nations Environment Programme as those persistent chemical substances that can accumulate in foods and cause adverse effects to consumers. Most of the organo chlorine pesticides were banned during the 1970s and 1980s, but they are persistent and stable and may remain in the environment for many years, constituting a risk of long - term exposure. These substances tend to be accumulated in the fatty tissue of living organisms.

Current maximum residue limits in the EU for the organo chloride pesticides that can be present in animal products are within 0.02 and 1 mg/kg of fat, while in the United States they are established between 0.1 and 7 mg/kg fat .

The use of pesticides have no doubt helped immensely in the success of **India's green revolution,** but these are also the agents polluting water, air and land and posing serious threat to public health, by entering in to food chain. However, since 1993, after imposing ban on some of these pesticides in India, their levels have declined and residues were below the recommended ADI.

Some recommendations as given below which need attention to overcome the worsening situation.

1. Adoption of Integrated Crop Management System (ICMS)
2. Encouragement of organic farming.
3. Application of more biodegradable pesticides
4. Strict ban on use of organo-chlorine pesticides
5. Treatment of pasture/food crops much grazing or harvest.
6. Adoption of genetically modified insect pest resistant seed for commercial cultivation.
7. Agrochemical companies should provide pesticide guide (environment data sheet, dose rate efficacy of individual pesticide etc.) to the farmers.

Heavy Metals: Heavy metals such as **lead, iron, cadmium, mercury, chromium, copper and nickel, together with the metalloid, arsenic** in their many compounds in nature all enter man's food chain from the soil or from water in the first instance. Since man has always been exposed to certain low levels of many of the chemical contaminants known today, a population would have to be subject to exposure at well above average levels to detect an increased incidence of a health effect. The levels of heavy metal residues which have been found in food in recent times have only been detected by sophisticated developments in analytical methodology. Whereas ten, or even five years ago, the detectability and sensitivity of an analytical method was expressed in terms of milligrams, or micrograms, today, the nanogram and picogram are invoked with increasing frequency .

Mycotoxins Problems from molds and mycotoxins have considerable worldwide significance in terms of public health, agriculture, and economics. Examples of molds affecting health include,

a.Aflatoxin metabolic products of the molds Aspergillus flavus and A. parasiticus, may occur in food as a result of mold growth in a number of susceptible commodities including peanuts and corn. Other domestic nuts and grains are susceptible but less prone to contamination with aflatoxins. Because aflatoxins are known carcinogens to laboratory animals and presumably man, the presence of aflatoxins in foods should be restricted to the minimum levels practically attainable using modern processing techniques.

b.Ochratoxin A is a naturally-occurring nephrotoxic fungal metabolite produced by certain species of the genera Aspergillus and Penicillium. It is mainly a contaminant of cereals (corn, barley, wheat, and oats) and has been found in edible animal tissues as well as in human blood sera and milk. Studies indicate that this toxin is carcinogenic in mice and rats. It is not completely destroyed during the processing and cooking of food, therefore the implication for risk to human health and safety must be considered.

There are extensive regulatory and non-regulatory measures in place to ensure that chemicals used or present in meat and meat products present a very low public health and safety risk. The regulations and control measures currently in place along the meat primary production chain have resulted in minimal public health and safety concerns regarding the use or presence of chemicals in meat and meat products. The extensive monitoring of chemical residues in meat over many years has demonstrated a high level of compliance with the regulations. Continuation of the current management practices, particularly monitoring programs for chemicals along the primary production chain, will ensure that the meat industry continues to maintain a high standard of public health and safety. Thus, by observing proper scientific guidelines and precautions we can minimize the harmful effects of antibiotic residues.

Veterinary drugs residues

The residues of veterinary drugs or its metabolites in meat and other foods of animal origin may cause adverse toxic effects on consumers' health. In fact, the European Food Safety Authority has recently issued an opinion on the effect of hormones residues in meat and reflected that epidemiological data provided evidence for an association between some forms of hormone-dependent cancers and red meat consumption. Furthermore, recent intoxications by consumption of lamb and bovine meat containing residues of clenbuterol resulted in 50 intoxicated people with symptoms described as gross tremors of the extremities, tachycardia, nausea, headaches and dizziness. Other important effects mainly due to the presence of residual antibiotics consist in allergic reactions or the selection of a resistant bacteria that could be transferred to humans through the food chain. In addition, the consumption of trace levels of antimicrobial residues in foods from animal origin may have consequences on the indigenous human intestinal microflora which constitutes an essential component of human physiology. This flora acts as a barrier against colonization of the gastrointestinal tract by pathogenic bacteria and has an important role for food digestion. So, the ingestion of trace levels of antimicrobials in foods must take into account potentially harmful effects on the human gut flora.In view of all these circumstances, foods of animal origin must be monitored for the presence of veterinary drug residues.

Effects of veterinary drugs on meat quality

Most of the drugs used as growth promoters may exert more or less important effects on meat quality, usually towards poorer eating quality. The meat tends to be tougher because there is an increase in connective tissue production and also a higher rate of collagen cross-linking as well as an increase in the insoluble fraction of the intramuscular collagen. Another factor, which is important from the point of view of meat tenderness, consists in the inhibitory action that these substances may exert against muscle proteases, enzymes responsible for protein breakdown in postmortem meat. For instance, myofibrillar protein fragmentation has been reported to be decreased in agonists-treated animals probably due to calpains inhibition by β-agonists. On the contrary, other endogenous enzymes like the hormone-sensitive lipase appear to be activated, increasing the lipolysis rate and the breakdown of triacylglycerols. The final result of the altered lipid metabolism is a sensible reduction in the amount of fat although this reduction in fat is associated to a lower sensory quality (poor juiciness and flavour). As mentioned above, some toxic effects in humans have been reported after consumption of lamb and bovine meat containing residues of clenbuterol. In other cases, like thiouracyls, the result is a substantial retention of water which is rapidly lost during cooking, giving a meat with lower juiciness.

Control of veterinary drugs residues in meat

The use of substances having hormonal or thyreostatic action as well as β-agonists is controlled by official inspection and analytical services following Commission Directive 96/23/EC on measures to monitor certain substances and residues in live animals and animal products. This Directive gave procedures for European Union Member States to set up national monitoring programmes and sampling procedures. This Directive contributed to a sensible reduction in the number of growth promoting reported cases. However, laboratories in charge of residues control usually face a large number of samples with a great varieties of residues to search in short periods of time making it rather difficult. The availability of simple and useful screening techniques is really necessary for an effective control.

Main veterinary drugs and substances with anabolic effect are listed in Table 1 and Table 2. In these tables, two groups of substances may be differentiated: those unauthorised substances having anabolic effect belonging to group A and those veterinary drugs of group B, some of them having established maximum residue limits (MRL). Analytical methodology, including criteria for identification and confirmation, for the monitoring of compliance was also given in the Commission Decisions 93/256/EEC and 93/257/EEC. The Commission Decision 2002/657/EC, which is in force since 1 September 2002, implemented Council Directive 96/23/EC by providing rules for the analytical methods to be used in testing of official samples and specific common criteria for the interpretation of analytical results of official control laboratories for such samples. For instance, substances in group A (Table) would require 4 identification points while those in group B (Table) only require a minimum of 3. The guidelines given in the new Directive imply new concepts like the decision limit (CCα) which means the limit at and above which it can be concluded with an error probability of α that a sample is non-compliant, and the detection capability (CCβ) that means the smallest content of the substance that may be detected, identified and/or quantified in a sample with an error probability of β.

Recently, the EC Quality of Life Programme supported European collaborative projects in the area of antimicrobials and hormone residues analysis (Boenke, 2002). These projects consisted in the development and validation of screening and confirmatory analytical methods and sensors for a cost-effective and time efficient control of synthetic glucocorticoids, nitrofurans, coccidiostatics, β-lactam residues and androgen residues in live and postmortem animals.

Lists of substances having anabolic effect belonging to group A according to Council Directive 96/23/EC

Substance	*Main representative*
Group A: substances having anabolic effect	
1. Stilbenes	(Diethylestilbestrol)
2. Antithyroid agents	(Thiouracyls)
3. Steroids	
Androgens	(Trenbolone acetate)
Gestagens	(Melengestrol acetate)
Estrogens	(17-β estradiol)
4. Resorcycilic acid lactones	(Zeranol)
5. β-agonists	(Clenbuterol)
6. Other compounds	(Nitrofurans)

Lists of veterinary drugs belonging to group B according to Council Directive 96/23/EC

Group B: veterinary drugs

1. Antibacterial substances

Sulphonamides and quinolones

2. Other veterinary drugs

(a) Antihelmintics

(b) Anticoccidials, including nitroimidazoles

(c) Carbamates and pyrethroids

(d) Sedatives

(e) Non-steroideal anti-inflammatory drugs

(f) Other pharmacologically active substances (dexamethasone)

Causes and concerns of pesticide residues

The principal sources of pesticide residues in crops, food animals, soil, water and almost all food commodities are given below

1. Carry-over from insecticide application to soil or to growing crops
2. Leaching of pesticides (herbicides) or insecticides into ground water
3. Drift of the pesticides from adjacent field
4. Translocation of soil applied pesticide into growing crops
5. Disposal of pesticides in streams, rivers and lakes
6. Effluents of pesticide industry in rivers and streams, and into soil which may be translocated in crops.

Pesticides along with certain environmental chemicals are known to cause endocrine disruption by mimicking or antagonising natural hormones in the body and it has been postulated that their long-term, low-dose exposure are increasingly linked to human health effects such as immunosuppression, hormone disruption, diminished intelligence, reproductive abnormalities and cancer. A cytogenitic study conducted on males associated with the spraying of pesticides in grape garden revealed a significant increase in chromatid breaks and gaps in chromosomes in the peripheral blood cells. However, the potential toxicity of residues still remained a matter of controversy. Although it is believed that adipose tissue acts as a protective reservoir, p,p'-DDT residues in the adiposetissue can be mobilized during starvation, reach tissues such as brain and produce symptoms of neurotoxicity. Similarly, kinetics of Hexachlorobenzene (HCB) and p,p'-DDE in rats before, during and after partial starvation and noted that food restriction produced a drastic mobilization of the residues stored in the adipose tissue resulting in symptoms of neurotoxicity. Furthermore, the redistribution was reversible and did not result in loss/excretion of the residues leading to their re-accumulation in the adipose tissue.

Majority of the pesticides used for agriculturaland domestic pest control purpose degrade rapidly in the environment, a few are trans-located from soil to plant tissues and then to animals where they have specific affinity for adipose tissue and ultimately lead to contamination of the livestock products. Since livestock food constitutes significant part of human diet and human beings are placed at the top of most food chains, it is not surprising that high levels of these compounds have been found in human adipose tissue and milk fat. Most of these pesticides are eliminated from the mammalian system more or less rapidly. To facilitate the clearance of such poorly excretable lipophilic pesticides, most organisms have a battery of enzymes that are specialized for the conversion of lipophilic materials to hydrophilic metabolites which can readily be eliminated by excretion in body wastes. The rate of elimination is a major factor governing the severity and duration of any toxic effects that may occur. Some products of pesticide metabolism are more toxic than the parent compound and are responsible for many of the undesirable toxic effects, both chronic and acute .

Observation of withdrawal period of antibiotics before slaughter of animals will help in reducing the risk of antibiotic residues in meat to a great extent. The withdrawal period of some of the common antibiotics used in veterinary practice is shown in Table 2.

Table 2-Withdrawal time of commonly used antibiotic drugs

Drugs	Pre-slaughter withdrawal time (days)
Ampicillin	6
Procaine penicillin G	5
Dihydro streptomycin sulphate	30
Erythromycin	14
Oxytetracycline hydrochloride	3
Sulphadimethoxime	7
Sulphamethoxy pyridazine	16
Sulphamethazine	10
Tylosin	8

Hazard Analysis Critical Control Point (HACCP)

HACCP has become synonymous with food safety. It is a worldwide-recognized systematic and preventive approach that addresses biological, chemical and physical hazards through anticipation and prevention, rather than through end-product inspection and testing.

The HACCP system for managing food safety concerns grew from two major developments. The first breakthrough was associated with W.E. Deming, whose theories of quality management are widely regarded as a major factor in turning around the quality of Japanese products in the 1950s. Dr Deming and others developed total quality management (TQM) systems which emphasized a total systems approach to manufacturing that could improve quality while lowering costs.

The second major breakthrough was the development of the HACCP concept itself. The HACCP concept was pioneered in the 1960s by the Pillsbury Company, the United States Army and the United States National Aeronautics and Space Administration (NASA) as a collaborative development for the production of safe foods for the United States space programme.

Hazard Analysis and Critical Control Point (HACCP) is a process control system designed to identify and prevent microbial and other hazards in food production. It includes steps designed to prevent problems before they occur and to correct deviations as soon as they are detected. Such preventive control system with documentation and verification are widely recognized by scientific authorities and international organizations as the most effective approach available for producing safe food.

HACCP involves a system approach to identification of hazard, assessment of chances of occurrence of hazards during each phase, raw material procurement, manufacturing, distribution, usage of food products, and in defining the measures for hazard control. In doing so, the many drawbacks prevalent in the inspection approach are provided and HACCP overcomes shortcomings of reliance only on microbial testing. HACCP enables the producers, processors, distributors, exporters, etc, of food products to utilize technical resources efficiently and in a cost effective manner in assuring food safety. Food inspection too would be more systematic and therefore hassle-free. It would no doubt involve deployment of some additional finances initially but this would be more than compensated in the long run through consistently better quality and hence better prices and returns.

DEFINITIONS

Control (verb): To take all necessary actions to ensure and maintain compliance with criteria established in the HACCP plan.

Control (noun): The state wherein correct procedures are being followed and criteria are being met.

Control measure: Any action and activity that can be used to prevent or eliminate a food safety hazard or reduce it to an acceptable level.

Corrective action: Any action to be taken when the results of monitoring at the CCP indicate a loss of control.
Critical Control Point (CCP): A step at which control can be applied and is essential to prevent or eliminate a food safety hazard or reduce it to an acceptable level.
Critical limit: A criterion which separates acceptability from unacceptability.
Deviation: Failure to meet a critical limit.
Flow diagram: A systematic representation of the sequence of steps or operations used in the production or manufacture of a particular food item.
HACCP: A system which identifies, evaluates, and controls hazards which are significant for food safety.
HACCP plan: A document prepared in accordance with the principles of HACCP to ensure control of hazards which are significant for food safety in the segment of the food chain under consideration.
Hazard: A biological, chemical or physical agent in, or condition of, food with the potential to cause an adverse health effect.
Hazard analysis: The process of collecting and evaluating information on hazards and conditions leading to their presence to decide which are significant for food safety and therefore should be addressed in the HACCP plan.
Monitor: The act of conducting a planned sequence of observations or measurements of control parameters to assess whether a CCP is under control.
Step: A point, procedure, operation or stage in the food chain including raw materials, from primary production to final consumption.
Validation: Obtaining evidence that the elements of the HACCP plan are effective.
Verification: The application of methods, procedures, tests and other evaluations, in addition to monitoring to determine compliance with the HACCP plan.

PRINCIPLES OF THE HACCP SYSTEM

The HACCP system consists of the following seven principles:
Principle 1 : Conduct a hazard analysis.
Principle 2 : Determine the Critical Control Points (CCPs).
Principle 3: Establish critical limit(s).
Principle 4: Establish a system to monitor control of the CCP.
Principle 5: Establish the corrective action to be taken when monitoring indicates that a particular CCP is not under control.
Principle 6: Establish procedures for verification to confirm that the HACCP system is working effectively.
Principle 7: Establish documentation concerning all procedures and records appropriate to these principles and their application.

APPLICATION OF THE HACCP PRINCIPLES

1. Assemble HACCP team

The food operation should assure that the appropriate product specific knowledge and expertise is available for the development of an effective HACCP plan. Optimally, this may be accomplished by assembling a multidisciplinary team. Where such expertise is not available on site, expert advice should be obtained from other Sources. The scope of the HACCP plan should be identified. The scope should describe which segment of the food chain is involved and the general classes of hazards to be addressed (e.g. does it cover all classes of hazards or only selected classes).

2. Describe product

A full description of the product should be drawn up, including relevant safety information such as: composition, physical/chemical structure (including Aw, pH, etc.), packaging, durability and storage conditions and method of distribution.

3. Identify intended use

The intended use should be based on the expected uses of the product by the end user or consumer. In specific cases, vulnerable groups of the population, e.g. institutional feeding, may have to be considered.

4. Construct flow diagram

The flow diagram should be constructed by the HACCP team. The flow diagram should cover all steps in the operation. When applying HACCP to a given operation, consideration should be given to steps preceding and following the specified operation.

5. On-site verification of flow diagram

The HACCP team should confirm the processing operation against the flow diagram during all stages and hours of operation and amend the flow diagram where appropriate.

6. List all potential hazards associated with each step, conduct a hazard analysis, and consider any measures to control identified hazards (see Principle 1)

The HACCP team should list all hazards that may be reasonably expected to occur at each step from primary production, processing, manufacture, and distribution until the point of consumption.

7. Determine Critical Control Points (see Principle 2) :

Since the publication of the decision tree by Codex, its use has been implemented many times for training purposes. In many instances, while this tree has been useful to explain the logic and depth of understanding needed to determine CCPs, it is not specific to all food operations, e.g. slaughter, and therefore it should be used in conjunction with professional judgement, and modified in some cases.

8. Establish critical limits for each CCP (see Principle 3)

Optical limits must be specified and validated if possible for each Critical Control Point. In some cases more than one critical limit will be elaborated at a particular step. Criteria often used include measurements of temperature, time, moisture level, pH, Aw, available chlorine, and sensory parameters such as visual appearance and texture.

9. Establish a monitoring system for each CCP (see Principle 4)

Monitoring is the scheduled measurement or observation of a CCP relative to its critical limits. The monitoring procedures must be able to detect toss of control at the CCP. Further, monitoring should ideally provide this Information in time to make adjustments to ensure control of the process to prevent violating the critical limits. Where possible, process adjustments should be made when monitoring results indicate a trend towards loss of control at a CCP. The adjustments should be taken before a deviation occurs. Data derived from monitoring must be evaluated by a designated person with knowledge and authority to carryout corrective actions when indicated. If monitoring is not continuous, then the amount or frequency of monitoring must be sufficient to guarantee the CCP d in control. Most monitoring procedures for CCPs will need to be done rapidly because they relate to on-line processes and there will not be time for lengthy analytical testing. Physical and chemical measurements are often preferred to microbiological testing because they may be done rapidly and can often indicate the microbiological control of the product. All records and documents associated with monitoring CCPs must be signed by the person(s) doing the monitoring and by a responsible reviewing official(s) of the company.

10. Establish corrective actions (see Principle 5)

Specific corrective actions must be developed for each CCP in the HACCP system in order to deal with deviations when they occur.

The actions must ensure that the CCP has been brought under control, Actions taken must also include proper disposition of the affected product. Deviation and product disposition procedures must be documented in the HACCP record keeping.

11. Establish verification procedures (see Principle 6)

Establish procedures for verification. Verification and auditing methods, procedures and tests, including random sampling and analysis, can be used to determine if the HACCP system is working correctly The frequency of verification should be sufficient to confirm that the HACCP system is working effectively. Examples of verification activities include:

- Review of the HACCP system and its records;
- Review of deviations and product dispositions;
- Confirmation that CCPs are kept under control.

Where possible, validation activities should include actions to confirm the efficacy of all elements of the HACCP plan.

12. Establish documentation and record keeping (see Principle 7)

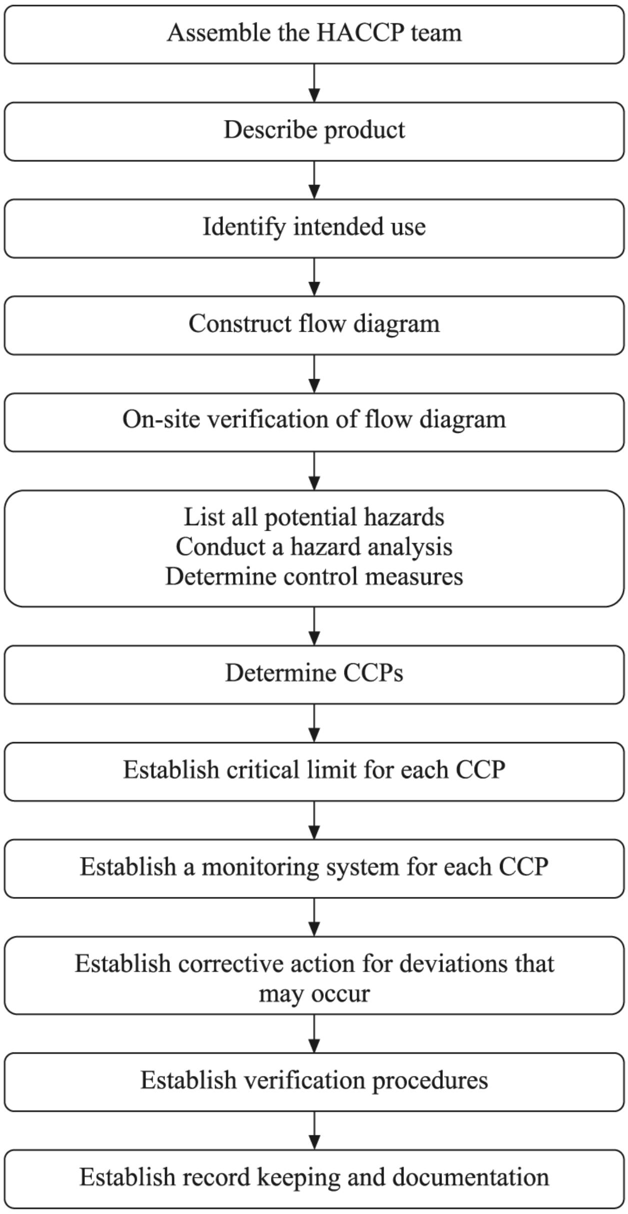
Efficient and accurate record keeping is essential to the application of an HACCP system. HACCP procedures should be documented. Documentation and record keeping should be appropriate to the nature and size of the operation.

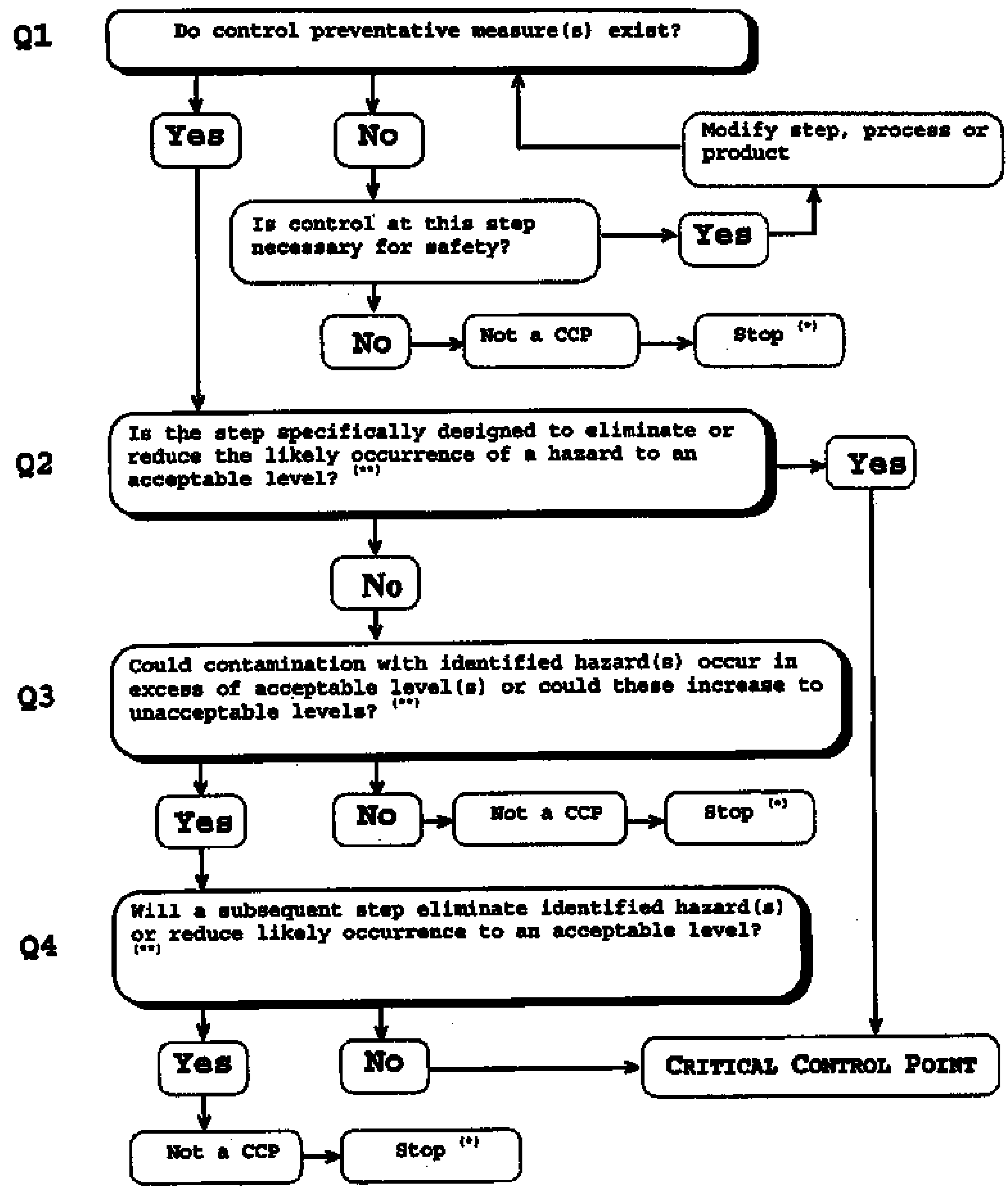
Documentation examples are:

- Hazard analysis;
- CCP determination;
- Critical limit determination.

Record examples are:

- CCP monitoring activities;
- Deviations and associated corrective actions;
- Modifications to the HACCP system.


Assemble the HACCP team
Describe product
Identify intended use
Construct flow diagram
On-site verification of flow diagram
List all potential hazards
Conduct a hazard analysis
Determine control measures
Determine CCPs
Establish critical limit for each CCP
Establish a monitoring system for each CCP
Establish corrective action for deviations that may occur
Establish verification procedures
Establish record keeping and documentation



(*) Proceed to the next identified hazard in the described process.

(**) Acceptable and unacceptable levels need to be defined within the overall objectives in identifying the CCPs of HACCP plan.

POULTRY

The flow diagram for the slaughtering process for poultry with identified critical control points is given below,

Flow diagram for the slaughtering process for poultry with identified Critical Control Points (CCP)

1. Holding
2. Bleeding ----CCP1

3. Scalding -----CCP1
4. Defeathering---CCP2
5. Evisceration------CCP2
6. Washing
7. Trussing ----CCP1
8. Chilling -----CCP1

Six Critical Control Points (CCPs) have been identified in the slaughter process of poultry. Among the six CCPs four can be categorized as CCP 1 (bleeding, scalding, trussing and chilling) and the other two can be categorized as CCP 2 (defeathering and evisceration).

The corrective action to be adopted to keep the identified CCPs in the slaughtering process of cattle and buffalo under control is as follows,

1) Bleeding (CCP 1)

- The major source of contamination is the blood oozing from the cut blood vessels which accidentally contaminates the carcass.
- Knives should be sterilized after bleeding operation.
- The stagnant blood on the floor should be washed away.

2) Scalding (CCP 1)

- The major cause for the contamination associated with scalding is the wash water. Major contributing factor to microbial load of the wash water is the microorganisms inhabiting the feathers, skin and droppings of the poultry from where it is transmitted to the wash water.
- Keen care should be taken to maintain the birds clean before slaughter by providing a thorough wash prior to slaughter and remove the droppings prior to scalding.
- Wash water must be changed frequently enough to ensure that microbial quality of water is safe.
- Temperature of the water and length of exposure of the poultry carcasses to scalding are two important factors that are to be controlled.
- Butchers should be trained on scientific methods of scalding.

3) Defeathering (CCP 2)

- The major source of contamination associated with defeathering is the feathers of the poultry.
- The defeathering machine should be thoroughly washed prior to commencement of operation and in between each operation and after completion of operations.
- Butchers should be trained on scientific methods of defeathering.

4) Evisceration (CCP 2)

- The contents of the gut are the major source of contamination during the process of evisceration.
- Butchers should be trained in wholesome meat production to enhance their skills to prevent the escape of gut contents.
- If any accidental contamination with gut contents occurs, it should be trimmed.

Bacterial reductions are possible up to one log10 APC (aerobic plate count) by washing of carcasses before they are sent to chill room. The removal of bacteria will be greater if carcasses are washed at each processing point. Therefore after evisceration the carcass should be washed with potable water and should be sprayed with lactic acid or acetic acid to reduce the microbial load and thereby extend the shelf life of meat.

5) Trussing (CCP 1)

- The hands of the operators are the major source of contamination during trussing operations.
- Use of disposable gloves will reduce the contamination.
- Visible contamination, if any, should be trimmed.

Among the various sources of contamination identified within the premises of the slaughterhouse, the hands of the operators and the water used for slaughter operations were the major sources of contamination of the carcass. Thus it is imperative to educate the butchers in the maintenance of personal hygiene, clean habits and the water used for slaughter and washing operations should be of potable quality.

6) Chilling (CCP 1)

Time and temperature are the two major factors that must be controlled for efficient storage of carcasses and meat in chiller. Hence the optimum time and temperature required to preserve the meat under both chilled conditions should be maintained.

RISK ANALYSIS

Risk analysis is widely recognised as the fundamental methodology underlying the development of food safety standards. As recognised in the 1995 consultation, risk analysis is composed of three separate but integrated elements, namely risk assessment, risk management and risk communication. That consultation recognised risk communication as an interactive process of exchange of information and opinion on risk among risk assessors, risk managers, and other interested parties. Risk management is defined within Codex as the process of weighing policy alternatives in the light of the results of risk assessment and, if required, selecting and implementing appropriate control options, including regulatory measures. The outcome of the risk management process, as undertaken by Committees within the Codex Alimentarius system, is the development of standards, guidelines and other recommendations for food safety, m the national situation it is likely that different risk management decisions could be made according to different criteria and different ranges of risk management options. The overall objective of Codex is to ensure consumer protection and to facilitate international trade.

Risk managers, in developing approaches to managing risk, utilise the risk characterisation that results from the risk assessment process. An important principle that was recognised by the 1995 consultation was the functional separation of risk assessment from risk management.

The significant world-wide increase in foodborne illness that has been recognized in recent years, especially arising from enteric organisms, suggests the need for more effective control using internationally agreed risk management methods.

The goal of food risk management

The primary goal of the management of risks associated with food is to protect public health by controlling such risks as effectively as possible through the selection and implementation of appropriate measures.

Definitions of key risk management terms

Food Safety : It is the condition which ensures that food will not cause harm to the consumer when prepared and/or eaten according to their intended use. Safety is one of four basic groups of features along with nutrition, the organoleptic, and commercial make up the food quality.

Food Safety Management System" means the adoption Good Manufacturing Practices, Good Hygienic Practices, Hazard Analysis and Critical Control Point and such other practices as may be specified by regulation, for the food business;

Risk management: The process of weighing policy alternatives in the light of the results of risk assessment and, if required, selecting and implementing appropriate control options, including regulatory measures.

> This definition of risk management, which has been proposed for inclusion in the Codex Procedural Manual (4), includes consideration of all the elements (listed below) that may be included in the risk management process (i.e. risk evaluation, risk management option assessment, implementation of management decision, and monitoring and review). However, in a practical context, it may not be necessary to include all the elements. For example, risk management decisions at the national level are likely to use all of the elements of this definition, whereas the risk management activities of Codex do not generally include implementation, monitoring and review.

Risk assessment policy: Guidelines for value judgement and policy choices which may need to be applied at specific decision points in the risk assessment process.

> Risk assessment policy setting is a risk management responsibility, which should be carried out in full collaboration with risk assessors, and which serves to protect the scientific integrity of the risk assessment. The guidelines should be documented so as to ensure consistency and transparency. Examples of risk assessment policy setting are establishing the population(s) at risk, establishing criteria for ranking of hazards, and guidelines for application of safety factors.

Risk profile: A description of the food safety problem and its context.

> Risk profiling is the process of describing a food safety problem and its context, in order to identify those elements of the hazard or risk relevant to various risk management decisions. The risk profile would include identifying aspects of hazards relevant to prioritising and setting the risk assessment policy and aspects of the risk relevant to the choice of safety standards and management options.

> A typical risk profile might include the following: a brief description of the situation, product or commodity involved; the values expected to be placed at risk, (e.g. human health, economic concerns); potential consequences; consumer perception of the risks; and the distribution of risks and benefits.

ELEMENTS OF RISK MANAGEMENT

A. Risk evaluation

- Identification of a food safety problem.
- Establishment of a risk profile.
- Ranking of the hazard for risk assessment and risk management priority.
- Establishment of risk assessment policy for conduct of risk assessment.
- Commissioning of risk assessment.

· Consideration of risk assessment result.

B. Risk management option assessment

- Identification of available management options.
- Selection of preferred management option, including consideration of an appropriate safety standard.*

* "Safety standard" here refers to the level of acceptable risk, which is adopted by risk managers or is implicit in the chosen risk management option. Examples include "zero-risk" standards (such as are usually implicit in de minimis and ADI levels), "balancing" standards (such as cost-benefit, cost-effectiveness, and ALARA), "threshold" standards (where a non-zero level of risk is stipulated as acceptable), or "procedural" standards (where the acceptable risk level is determined by an agreed process, such as a negotiation or referendum).

C. Implementation of management decision

D. Monitoring and review

- Assessment of effectiveness of measures taken.
- Review risk management and/or assessment as necessary.

The outcome of the risk evaluation process should be combined with the evaluation of available risk management options in order to reach a decision on management of the risk. In arriving at this decision, human health protection should be the primary consideration, with other factors (e.g. economic costs, benefits, technical feasibility, risk perceptions, etc.) being considered as appropriate. Implementation of the management decision should be followed by monitoring both the effectiveness of the control measure and its impact on risk to the exposed consumer population, to ensure that the food safety objective is being met.

It is important that all interested parties** who are likely to be affected by risk management decisions have an opportunity for input into the risk management process. These groups may include (but should not be limited to) consumer organizations, representatives of the food industry and trade, education and research institutions, and regulatory bodies. A consultative process can be implemented in many ways, ranging from public meetings to opportunities to comment on public documents. Inputs from interested parties can be introduced and considered at every stage of the risk management policy formulation process, including evaluation and review.

** These interested parties are commonly referred to as "stakeholders" in a number of countries.

GENERAL PRINCIPLES OF FOOD SAFETY RISK MANAGEMENT

Principle 1: Risk management should follow a structured approach.

The elements of a structured approach to risk management are Risk Evaluation, Risk Management Option Assessment, Implementation of Management Decision, and Monitoring and Review. In certain circumstances, not all of these elements will be included in risk management activities (e.g. standard setting by Codex, with implementation of control measures by national governments).

Principle 2: Protection of human health should be the primary consideration in risk management decisions.

Decisions on acceptable levels of risk should be determined primarily by human health considerations, and arbitrary or unjustified differences in the risk levels should be avoided. Consideration of other factors (e.g. economic costs, benefits, technical feasibility, and societal preferences) may be appropriate in some risk management contexts, particularly in the determination of measures to be taken. These considerations should not be arbitrary and should be made explicit.

Principle 3: Risk management decisions and practices should be transparent.

Risk management should include the identification and systematic documentation of all elements of the risk management process including decision-making, so that the rationale is transparent to all interested parties.

Principle 4: Determination of risk assessment policy should be included as a specific component of risk management.

Risk assessment policy sets the guidelines for value judgements and policy choices which may need to be applied at specific decision points in the risk assessment process, and preferably should be determined in advance of risk assessment, in collaboration with risk assessors.

Principle 5: Risk management should ensure the scientific integrity of the risk assessment process by maintaining the functional separation of risk management and risk assessment.

Functional separation of risk management and risk assessment serves to ensure the scientific integrity of the risk assessment process and reduce any conflict of interest between risk assessment and risk management. However, it is recognised that risk analysis is an iterative process, and interactions between risk managers and risk assessors are essential for practical application.

Principle 6: Risk management decisions should take into account the uncertainty in the output of the risk assessment.

The risk estimate should, wherever possible, include a numerical expression of uncertainty, and this must be conveyed to risk managers in a readily understandable form so that the full implications of the range of uncertainty can be included in decision-making. For example, if the risk estimate is highly uncertain the risk management decision might be more conservative.

Principle 7: Risk management should include clear, interactive communication with consumers and other interested parties in all aspects of the process.

On-going reciprocal communication among all interested parties is an integral part of the risk management process. Risk communication is more than the dissemination of information, and a major function is the process by which information and opinion essential to effective risk management is incorporated into the decision.

Principle 8: Risk management should be a continuing process that takes into account all newly generated data in the evaluation and review of risk management decisions.

Subsequent to the application of a risk management decision, periodic evaluation of the decision should be made to determine its effectiveness in meeting food safety objectives. Monitoring and other activities will likely be necessary to carry out the review effectively.

SANITARY AND PHYTOSANITARY MEASURES

Definitions

- Sanitary = health in general (sometimes animal health)
- Phytosanitary = plant health
- Sanitary and phytosanitary (SPS) = food safety and animal and plant health

SPS measures include import restrictions or maximum residue levels. The agreement says governments' SPS measures should be based on:
-recognized international standards, particularly those of the "three sisters" — FAO/WHO Codex Alimentarius Commission, the World Organization for Animal Health (OIE), the International Plant Protection Convention (IPPC)
-science, including scientific assessment of risk
-a temporary precautionary principle in the absence of international standards or scientific evidence
The SPS Committee's job is to monitor how countries are applying the SPS Agreement and to discuss issues that arise from that.
The SPS Agreement tries to sort out genuine cases from those that are potentially excuses for protectionism. It does this by saying that measures either have to be based on scientific evidence of risk, or on recognized international standards (see above). If a country applies

international standards, it is less likely to be challenged legally in the WTO than if it sets its own standards. But countries are free to set their own standards based on science.
•SPS issues are becoming more important as tariff barriers fall
•Food producers, particularly in developing countries, are becoming increasingly concerned that their attempts to export to richer markets are being hampered by SPS measures. At issue are genuine risks versus protectionism in disguise
•Individual private sector exporters tend to assume that the real motive for importing countries' SPS measures is really protectionism — to protect producers rather than consumers, livestock or crops.
•But there are genuine needs to have SPS measures.
What are sanitary and phytosanitary measures?
For the purposes of the SPS Agreement, sanitary and phytosanitary measures are defined as any measures applied:
- to protect human or animal life from risks arising from additives, contaminants, toxins or disease-causing organisms in their food;
- to protect human life from plant- or animal-carried diseases;
- to protect animal or plant life from pests, diseases, or disease-causing organisms;
- to prevent or limit other damage to a country from the entry, establishment or spread of pests.

These include sanitary and phytosanitary measures taken to protect the health of fish and wild fauna, as well as of forests and wild flora.
Measures for environmental protection (other than as defined above), to protect consumer interests, or for the welfare of animals are not covered by the SPS Agreement. These concerns, however, are addressed by other WTO agreements.

CHAPTER 22

NATIONAL AND INTERNATIONAL AGENCIES RELATED TO FOOD SAFETY

Food safety challenges
Focus on Food Safety ?
- Increasing population and demand for food
- Greater public demand for health protection
- Increasing volume and diversity of trade in foods
- Sophisticated detection and management of hazards
- Changing human/ animal interactions
- Changing hazards e.g. resistant microbes

List of main international standards

- International standards organisation (ISO)
- Codex Alimentarius Commission (CAC)
- World Health Organisation (WHO)
- World Trade Organisation (WTO)
- World Organisation For Animal Health (WOAH)
- European Union (EU) Standards
- Food and Drugs Administration (FDA), USA

- International Plant Protection Organisation (IPPO)
- Convention on BioDiversity (CBD)
- International commission on microbiological specification for foods (ICMSF)

Codex Alimentarius Commission (Codex)

- An intergovernmental body
- Formed in 1963
- By a joint venture of the FAO and the WHO
- Coordinates food standards at the international level

Objective of Codex

- Develop standards for foodProtecting the health of the consumers
- Ensuring fair practices in the food trade
- Developed standards: Based on a sound, scientific risk assessment for protecting public health
- Member States : Used as national legislation.
- World Trade Organization (WTO)- Emphasized on the importance of Codex standards

Structure of the CAC

The Commission

- The supreme decision making body
- Provides a forum for discussion and debate on all major food standards/safety issues of interest and concern to Codex Member States

- The Executive Committee
- The subsidiary bodies

Codex committees

- General Subject Committees (10)
- Commodity specific committees (14)

Codex coordinating committees

- The Commission: Meets every year alternately in Rome and in Geneva

General Subject Committees

- Committee on General Principles (CCGP) hosted by France
- Committee on Food Labelling (CCFL) hosted by Canada
- Committee on Methods of Analysis and Sampling (CCMAS) hosted by Hungary
- Committee on Food Hygiene (CCFH) hosted by USA
- Committee on Pesticide Residues (CCPR) hosted by China

- Committee on Food Additives (CCFA) by China
- Committee on Contaminants in Foods (CCCF) hosted by Netherlands
- Committee on Import and Export Inspection and Certification Systems (CCFICS) hosted by Australia
- Committee on Nutrition and Foods for Special Dietary Use (CCNFSDU) hosted by Germany
- Committee on Residues of Veterinary Drugs in Food (CCRVDF) hosted by USA

Steps in Codex Standards Development

Step 1 Decision of Commission to take up standard

- Member request
- Critical review
- Executive Committee Decision
- Member consensus
- Decides which subsidiary body to take work

Step 2 The Secretariat prepares proposed draft standard

- Takes help of Joint (FAO/WHO) committees JECFA, JEMRA, JMPR, etc

Step 3 The proposed draft standard sent to Members and interested international organizations for comment
Step 4 Subsidiary body or body concerned considers comments and to amends the draft standard
Step 5 Executive Committee for critical review and to the Commission with a view to its adoption as a draft standard.
Step 6 Sent to all Members and interested international organizations for comment
Step 7 The comments considered by subsidiary body for amendments
Step 8 Executive Committee for critical review and to the Commission with a view to its adoption as Codex standard.

World Health Organization (WHO)

- Established in 1948
- WHO has a specific mandate for the protection of public health
- Also plays important role in food safety
- Advisory role
- Assisting Member States in development of measures along the entire food production chain.
- Risk-based
- Sustainable
- Science-based measures
- Prevent unacceptable levels of microbiological agents and chemicals in food.
- Strengthening surveillance systems of foodborne diseases:
    - FERG: Foodborne Disease Burden Epidemiology Reference Group
    - Established in 2006

- An advisory body to WHO on global epidemiology of foodborne diseases

Risk assessment: Several Expert Committees -Assessing the risks of chemicals in food

- JECFA: The Joint FAO/WHO Expert Committee on Food Additives
- JMPR: The Joint FAO/WHO Meeting on Pesticide Residues. Microbial risk assessment
- JEMRA: Joint WHO/FAO expert committee for microbial risk assessment
- Risk communication and International cooperation
- INFOSAN: International Food Safety Authority Network In collaboration with the FAO
- Established in 2004
- Sharing of information between the global networks of national food safety authorities

Food and Agriculture Organization (FAO)

- An intergovernmental organization
- FAO has 194 Member Nations
- Headquartered in Rome, Italy
- Core mandate of FAO
- Food security i.e. to ensure that all people have regular access to enough high-quality food to lead active, healthy life is at the.
- Sustainable management and utilization of natural resources
- Land, water, air, climate and genetic resources for the benefit of present and future generations.
- The FAO and WHO joint efforts : Vital role in deciding global strategies and policies for food safety

World Organization for Animal Health (OIE)

- The Office International des Epizooties (OIE)
- Created on January 25th 1924 to fight animal diseases at global level.
- OIE is the intergovernmental organization
- The responsible for improving animal health worldwide
- World Organization for Animal Health: May 2003 -But kept its historical acronym OIE.
- Recognized as a reference organization by the World Trade Organization (WTO)
    - Trade related disease
    - Animal food safety issues/zoonotic pathogens

International Organization for Standardization (ISO)

- Independent, non-governmental international organization
- Membership of 163 national standards bodies
- BIS is an Indian Member body
- Experts from the member bodies
- Share knowledge
- Develop International Standards

- Voluntary, consensus-based, market relevant
- Provide solutions to global challenges
- Ensure quality, safety, efficiency
- Instrumental in facilitating international trade

International Standards Organisation

Popular name for international organisation for standards (IOS), a voluntary, non-treaty federation of standards setting bodies of some 130 countries. Founded in 1946-47 in Geneva as a UN agency, it promotes development of standardization and related activities to facilitate international trade in goods and services and corporation on economic, intellectual scientific and technological aspects. ISO covers standardization in all fields including computers and data communication.

Salient Features of International Standards Organisation (ISO)

- Most important international standards setting nongovernmental organisation
- World federation of 130 countries national standards bodies
- Develops consensus based international standards
- Develops standard through 200 technical committees split in to 650 subcommittees and 200 working groups
- ISO 9000 series is the general quality certification of ISO

ISO 9000

ISO 9000 is a generic name given to a family of standards developed to provide a framework around which a quality management system can effectively be implemented.

ISO 9001

Model for design/development production, installation and servicing

ISO 9002

Model for quality assurance in production and installation

ISO 9003

Model for quality assurance in final inspection & testing this standard is used for external quality assurance purpose.

ISO 9004

This standard provides guidance on the technical, administrative and human factors and services at all stages of the quality loop from detection of need to the customer’s satisfaction.

ISO 22000

ISO 22000: 2005 specifies requirements for a **food safety management** system where an organisation in the food chain needs to demonstrate its ability to control food safety hazards in order to ensure that food is safe at the time of human consumption.

ISO 14000

The ISO 14000 **environmental management** standards exist to help organisations (a) minimise how their operations (process etc), negatively affect the environment (ie cause adverse changes to air , water or land (b) comply with applicable laws, regulations and other environmentally oriented requirements and (c) continually improve in the above.

International Commission on Microbiological Specifications for Foods (ICMSF)

- The ICMSF is a non-governmental, non-profit, scientific advisory body.Established in 1962 .Working under the auspices of the International Union of Microbiological Societies (IUMS). Through the IUMS, the Commission is formally linked to the WHO and FAO, and thus Codex.

ICMSF's primary goal

- To be a leading source for independent and impartial scientific concepts in microbial food safety and quality management
- ICMSF membership : 15 food microbiologists from 13 countries
- Members: Selected based on their technical expertise, not as national delegates
- Voluntary and without honoraria
- Partners - ILSI, FAO, WHO, IUFoST, IAFP
- The recommendations of ICMSF have no official statusConcepts adopted by governmental agencies and industry
- Reduce the incidence of microbiological food-borne illness, Food spoilage
- Facilitate global trade
- Profound and global impact on the field of food microbiology
- Test methods for microorganisms
- Sampling plans
- Microbiological criteria
- HACCP
- Risk assessment and risk management

World Trade Organization (WTO)

- An intergovernmental organization
- Established: 1 January 1995
- Location: Geneva, Switzerland
- Replaced the General Agreement on Tariffs and Trade (GATT) (1948)
- Membership: 164 countries on 29 July 2016

Functions:

• Regulates international trade
• Forum for trade negotiations
• Handling trade disputes
• Monitoring national trade policies
• Technical assistance and training for developing countries
• Cooperation with other international organizations

SPS Agreement

- The Agreement on the Application of Sanitary and Phytosanitary Measures
- Entered into force on 1 January 1995.
- It concerns the application of food safety and animal and plant health regulations
- Basic rules for food safety and animal and plant health standards .Allows countries to set their own standards.
- But must be based on science
- Should be applied only to the extent necessary to protect human, animal or plant life or health
- Should not arbitrarily or unjustifiably discriminate between countries where identical or similar conditions prevail
- Member countries encouraged to use international standards guidelines andrecommendations where they exist
- Members may use higher standards if there is scientific justification.
- They can also set higher standards based on appropriate assessment of risks
- The agreement still allows countries to use different standards and different methods of inspecting products

International Life Sciences Institute (ILSI)

- A non-profit , Non-government science organization
- Founded in 1978
- Headquartered in Washington, DC
- Members are primarily food and beverage, agricultural, chemical, and pharmaceutical companies

Mission

- Improve public health and well-being
- Engaging academic, government, and industry scientists in a neutral forum
- To advance scientific understanding inNutrition, food safety, risk assessment, and the environment.

NATIONAL AGENCIES
Bureau of Indian Standards (BIS)

- National Standards Body promoting and nurturing standards movement
- BIS came into existence on 01 April 1987

- It took over the functions of the erstwhile Indian Standards Institution (ISI)
- Various activities performed by BIS are
    - Standards formulation
    - Certification, testing & calibration services
    - International harmonization with ISO

The Agricultural and Processed Food Products Export Development Authority (APEDA)

Established under the Agricultural and Processed Food Products Export Development Authority Act, 1986.

- Mandated with export promotion and development of the following scheduled products:

- Fruits, vegetables and their products

- Meat and meat products
- Poultry and poultry products
- Dairy products
- Confectionery, biscuits and bakery products
- Honey, jaggery and sugar products
- Cocoa and its products, chocolates of all kinds
- Alcoholic and non-alcoholic beverages
    - Cereal and cereal products
    - Groundnuts, peanuts and walnuts
    - Pickles, papads and chutneys
    - Guar gum
    - Floriculture and floriculture products
    - Herbal and medicinal plants

Export Inspection Council (EIC)

- EIC set up under the Export (Quality Control and Inspection) Act, 1963
- To ensure sound development of export trade of India - Quality Control and Inspection and for matters connected thereof.
- EIC is an advisory body with following activities
- Notify commodities which will be subject to quality control and/ or inspection prior to export
- Establish standards of quality for such notified commodities
- Specify the type of quality control and / or inspection to be applied to such commodities.

The Food Safety and Standards Authority of India (FSSAI)

- Established on 5 September 2008 , under the Food Safety and Standards Act, 2006
- under the ministry of health & family welfare,govt. of India
- Headed by a chairman appointed by central govt. of India and 22 members
- headquarters at new Delhi
- 6 regional offices located in Delhi, Guwahati, Mumbai, Kolkata, cochin, Chennai
- 14 referral laboratories , 72 State/UT laboratories , 112 laboratories are NABL accredited private laboratories notified by FSSAI

Highlights of the Food Safety and Standard Act, 2006-Various central Acts like Prevention of Food Adulteration Act,1954,Fruit Products Order , 1955, Meat Food Products Order,1973, Vegetable Oil Products (Control) Order, 1947,Edible Oils Packaging (Regulation)Order 1988, Solvent Extracted Oil, De- Oiled Meal and Edible Flour (Control) Order, 1967, Milk and Milk Products Order, 1992 etc will be repealed after commencement of FSS Act, 2006.

The Act also aims to establish a single reference point for all matters relating to food safety and standards, by moving from multi- level, multi- departmental control to a single line of command. To this effect, the Act establishes an independent statutory Authority – the Food Safety and Standards Authority of India with head office at Delhi. Food Safety and Standards Authority of India (FSSAI) and the State Food Safety Authorities shall enforce various provisions of the Acted from modern biotechnology. Its including guide lines for management of official import and export inspection and certification system for food.

FSSAI has been mandated by the FSS Act, 2006 for performing the following functions:

- To provide scientific advice and technical support to Central Government and State Governments in the matters of ***framing the policy and rules*** in areas which have a direct or indirect bearing of food safety and nutrition.
- ***Collect and collate data*** regarding food consumption, incidence and prevalence of biological risk, contaminants in food, residues of various, contaminants in foods products, identification of emerging risks and introduction of rapid alert system.
- ***Creating an information network*** across the country so that the public, consumers, Panchayats etc receive rapid, reliable and objective information about food safety and issues of concern.
- *Provide training programmes* for persons who are involved or intend to get involved in food businesses.
- Contribute to the *development of international technical standards* for food, sanitary and phyto-sanitary standards.

- ***Promote general awareness about food safety and food standards*.**

The act in the chapter 3 schedule IV, part I propose essential measure that needs to be followed even by the petty food manufacturer or operator for ensuring the food safety. the measures include from location to type of construction, ceiling, flooring, water facilities, machinery, utensils to be used, waste disposal, pest control and personal hygiene of the food handlers, etc.

INTERNATIONAL STANDARDS FOR FOOD SAFETY

World Organisation for Animal Health (Office International des Épizooties) OIE

The World Organisation for Animal Health (OIE) is an independent intergovernmental organization founded in 1924, and having 172 member countries in January 2008. OIE's mandate is to improve animal health worldwide. The organization achieves this mandate through its six primary objectives, which include ensuring transparency in the global animal disease situation, permanent update of disease prevention and control methods, provision of international solidarity in the control of animal diseases, publication of international animal health standards, and improvement of the legal framework and resources of veterinary services. For several years, the OIE has also had a strong focus on improving animal production food safety and animal welfare. OIE's headquarters are in Paris (France), and there are nine regional offices in the five regions. The OIE now has two regional animal health centres operating in collaboration with FAO, based in Bamako and Beirut, and is planning to establish other centres that will serve as regional centres of expertise.

In order to fulfil the mandate to ensure transparency in the global animal disease situation, the OIE manages the World Animal Health Information System (WAHIS), based on the commitment of member countries to notify the main animal diseases, including zoonoses, to the OIE. In 2004, OIE member countries approved the creation of a single list of diseases notifiable to the OIE to replace the former lists A and B. The content of the list is based on a decision tree which is part of the Terrestrial Animal Health Code. Currently about one-hundred diseases are listed; thirteen of these are poultry diseases, among which are highly pathogenic avian influenza (HPAI), Newcastle disease, Marek's disease, infectious bursal disease and avian infectious laryngotracheitis.

As the international standard-setting body for animal health, the OIE has defined standards on notification, trade aspects and surveillance of the listed diseases, including the poultry diseases. The aim of the Terrestrial Animal Health Code is to ensure the sanitary safety of international trade in terrestrial animals and their products, by detailing the health measures to be used by the veterinary services of importing and exporting countries. The measures are also meant to avoid the transfer of pathogenic or zoonotic agents without imposing unjustified trade restrictions.

The OIE is in a continuous process of updating its disease standards, while taking into account the latest scientific information on the diseases. For example, the chapter on avian influenza in the Terrestrial Animal Health Code was updated in 2004. The new chapter has several significant changes compared to the previous one, such as differentiating between low and highly pathogenic avian influenza and defining HPAI as an infection of poultry. The chapter gives trade recommendations for poultry and poultry products like fresh meat, meat products, eggs, feathers and down.

World Trade Organization

There are a number of ways of looking at the WTO. It's an organization for liberalizing trade. It's a forum for governments to negotiate trade agreements. It's a place for them to settle trade disputes. It operates a system of trade rules.

Essentially, the WTO is a place where member governments go, to try to sort out the trade problems they face with each other. The first step is to talk. The WTO was born out of negotiations, and everything the WTO does is the result of negotiations. The bulk of the WTO's current work comes from the 1986-94 negotiations called the Uruguay Round and earlier negotiations under the General Agreement on Tariffs and Trade (GATT). The WTO is currently the host to new negotiations, under the "Doha Development Agenda" launched in 2001.

Where countries have faced trade barriers and wanted them lowered, the negotiations have helped to liberalize trade. But the WTO is not just about liberalizing trade, and in some circumstances its rules support maintaining trade barriers — for example to protect consumers or prevent the spread of disease.

At its heart are the WTO agreements, negotiated and signed by the bulk of the world's trading nations. These documents provide the legal ground-rules for international commerce. They are essentially contracts, binding governments to keep their trade policies within agreed limits. Although negotiated and signed by governments, the goal is to help producers of goods and services, exporters, and importers conduct their business, while allowing governments to meet social and environmental objectives.

The system's overriding purpose is to help trade flow as freely as possible — so long as there are no undesirable side-effects — because this is important for economic development and well-being. That partly means removing obstacles. It also means ensuring that individuals, companies and governments know what the trade rules are around the world, and giving them the confidence that there will be no sudden changes of policy. In other words, the rules have to be "transparent" and predictable.

This is a third important side to the WTO's work. Trade relations often involve conflicting interests. Agreements, including those painstakingly negotiated in the WTO system, often need interpreting. The most harmonious way to settle these differences is through some neutral procedure based on an agreed legal foundation. That is the purpose behind the dispute settlement process written into the WTO agreements. The WTO began life on 1 January 1995, but its trading system is half a century older. Since 1948, the General Agreement on Tariffs and Trade (GATT) had provided the rules for the system. (The second WTO ministerial meeting, held in Geneva in May 1998, included a celebration of the 50th anniversary of the system.)

It did not take long for the General Agreement to give birth to an unofficial, de facto international organization, also known informally as GATT. Over the years GATT evolved through several rounds of negotiations. The last and largest GATT round, was the Uruguay

Round which lasted from 1986 to 1994 and led to the WTO's creation. Whereas GATT had mainly dealt with trade in goods, the WTO and its agreements now cover trade in services, and in traded inventions, creations and designs (intellectual property).

CODEX ALIMENTARIUS

It is about safe, good food for everyone - everywhere.International food trade has existed for thousands of years but until not too long ago food was mainly produced, sold and consumed locally. Over the last century the amount of food traded internationally has grown exponentially, and a quantity and variety of food never before possible travels the globe today.

The Codex Alimentarius international food standards, guidelines and codes of practice contribute to the safety, quality and fairness of this international food trade. Consumers can trust the safety and quality of the food products they buy and importers can trust that the food they ordered will be in accordance with their specifications. Public concerns about food safety issues are often placing Codex at the centre of global debates. Biotechnology, pesticides, food additives and contaminants are some of the issues discussed in Codex meetings. Codex standards are based on the best available science assisted by independent international risk assessment bodies or ad-hoc consultations organized by FAO and WHO. While being recommendations for voluntary application by members, Codex standards serve in many cases as a basis for national legislation. The reference made to Codex food safety standards in the World Trade Organizations' Agreement on Sanitary and Phytosanitary measures (SPS Agreement) means that Codex has far reaching implications for resolving trade disputes. WTO members that wish to apply stricter food safety measures than those set by Codex may be required to justify these measures scientifically.

Codex members cover 99% of the worlds population. More and more developing countries are taking an active part in the Codex process - in many cases assisted by the Codex Trust Fund, which strives to finance - and train - participants from such countries to enable efficient participation. Being an active member of Codex helps countries to compete in sophisticated world markets - and to improve food safety for their own population. At the same time exporters know what importers demand, and importers are protected from substandard shipments.

International governmental and non-governmental organizations can become accredited Codex observers to provide expert information, advice and assistance to the Commission. Since its beginnings in 1963, the Codex system has evolved in an open, transparent and inclusive way to meet emerging challenges. International food trade is a 200 billion dollar a year industry, with billions of tonnes of food produced, marketed and transported. There is a lot at stake for protecting consumers' health and ensuring fair practices in the food trade. All information on Codex is public and free.

The CAC Statements of Principle Concerning The Role of Science in the Codex Decision-making Process and the Extent to Which Other Factors Are Taken Into Account* state that standards, guidelines and other recommendations of Codex shall be based on the principle of sound scientific advice and evidence, and where appropriate. Codex will have regard to other legitimate factors relevant to the health protection of consumers and for the promotion of fair practices in food trade. Codex principles of risk management must be guided by these statements, as well as by the provisions of the SPS Agreement. To date, these "other legitimate factors" have not been defined or considered in the risk management context.

The Statements of Principle Concerning the Role of Science in the Codex Decision-making Process and the Extent to Which Other Factors are Taken into Account include the following:

1. The food standards, guidelines and other recommendations of Codex Alimentarius shall be based on the principle of sound scientific analysis and evidence, involving a thorough review of all relevant information, in order that the standards assure the quality and safety of the food supply.

2. When elaborating and deciding upon food standards Codex Alimentarius will have regard, where appropriate, to other legitimate factors relevant for the health protection of consumers and for the promotion of fair practices in food trade.

3. In this regard it is noted that food labelling plays an important role in furthering both of these objectives.

4. When the situation arises that members of Codex agree on the necessary level of protection of public health but hold differing views about other considerations, members may abstain from acceptance of the relevant standard without necessarily preventing the decision by Codex.